



User Manual **Model 686**

Simple AFG



Summary

CONTACTING BNC	6
GENERAL SAFETY SUMMARY	7
TO AVOID FIRE OR PERSONAL INJURY	7
<i>Use Proper Power Cord</i>	<i>7</i>
<i>Ground the Product.....</i>	<i>7</i>
<i>Observe All Terminal Ratings</i>	<i>7</i>
<i>Power Disconnect.....</i>	<i>7</i>
<i>Do Not Operate Without Covers</i>	<i>7</i>
<i>Do Not Operate With Suspected Failures.....</i>	<i>7</i>
<i>Avoid Exposed Circuitry.....</i>	<i>7</i>
<i>Do Not Operate in Wet/Damp Conditions.</i>	<i>7</i>
<i>Do Not Operate in an Explosive Atmosphere.....</i>	<i>7</i>
<i>Keep Product Surfaces Clean and Dry.</i>	<i>7</i>
<i>Provide Proper Ventilation</i>	<i>7</i>
SAFETY REQUIREMENTS.....	8
<i>Safety Symbols</i>	<i>8</i>
ENVIRONMENTAL CONSIDERATIONS	10
PRODUCT END-OF-LIFE HANDLING	10
EQUIPMENT RECYCLING.....	10
PREFACE	11
PACKAGE CONTENTS	11
MODELS.....	12
RECOMMENDED OPTIONS AND ACCESSORIES.....	12
KEY FEATURES.....	13
MECHANICAL CHARACTERISTICS	14
OPERATING REQUIREMENTS.....	15
ENVIRONMENTAL REQUIREMENTS.....	15
POWER SUPPLY REQUIREMENTS	15
CLEANING.....	16
CALIBRATION.....	16
ABNORMAL CONDITIONS	17
INSTALLING YOUR INSTRUMENT.....	18
POWER THE INSTRUMENT ON AND OFF AND LAUNCH AFG APPLICATION.....	18
PROTECT YOUR INSTRUMENT FROM MISUSE	18
OBTAINING THE LATEST VERSION RELEASES.....	19
INSTALL SIMPLE AFG APPLICATION.....	19
USB PEN DRIVE AND RECOVERY PROCEDURE	21
<i>Recovery Procedure</i>	<i>21</i>
INSTRUMENT OVERVIEW	23
FRONT PANEL Model 686-2C	23

FRONT PANEL Model 686-2C	24
<i>Analog Outputs</i>	25
<i>Marker Outputs</i>	25
<i>Trigger Inputs</i>	25
<i>Soft keyboard and rotary knob</i>	26
<i>Numeric Keypad</i>	27
REAR PANEL Model 686-2C	29
REAR PANEL Model 686-4C	30
<i>External Modulation Input Connector</i>	31
<i>Reference Clock Input Connector</i>	31
<i>Reference Clock Output Connector</i>	31
<i>Digital Output connector</i>	31
<i>External Clock Input Connector</i>	31
<i>Sync Clock Output Connector</i>	31
<i>Sync In / Sync Out Connectors</i>	31
<i>Pattern Force Jump In Connector (with 686-FSS Option only)</i>	31
INTRODUCTION	32
SIMPLE AFG APPLICATION	32
SIMPLE AFG TOUCH UI.....	32
USER INTERFACE DESCRIPTION	33
<i>Waveform Parameters Area</i>	34
<i>Graph Area</i>	37
<i>Zoom graph</i>	38
<i>Channel Information</i>	42
<i>Command Toolbar Area</i>	43
INPUT / OUTPUT CHANNELS	46
ANALOG OUTPUT CHANNEL	46
<i>Main Parameters</i>	47
AUXILIARY CHANNELS	52
<i>Marker Outputs</i>	52
<i>Trigger In</i>	52
<i>External Modulation Input Connector</i>	53
<i>Reference Clock Input Connector</i>	54
<i>Reference Clock Output Connector</i>	54
<i>External Clock Input Connector</i>	54
<i>Sync Clock Output Connector</i>	54
<i>Digital Output connector</i>	55
<i>Sync In / Sync Out Connectors</i>	55
PREDEFINED WAVEFORMS	56
<i>List and parameters of predefined waveforms</i>	56
RUN MODE	60
CONTINUOUS	61

<i>Marker Out behaviour in Continuous Run Mode</i>	61
MODULATION	62
<i>Marker Out behaviour in Modulation Run Mode</i>	64
<i>Modulation General Parameter</i>	65
<i>Modulation Types and associated parameters</i>	65
SWEEP	67
<i>Sweep Mode</i>	67
<i>Parameters</i>	68
<i>Sweep Trigger Mode</i>	68
<i>Marker Out behaviour in Sweep Run Mode</i>	69
BURST.....	70
<i>Burst Mode</i>	70
<i>Marker Out behaviour in Burst Run Mode</i>	73
CHANNELS AND DEVICE SETTING	74
CHANNELS SETTINGS.....	74
SETTINGS BUTTON	76
DEVICE SETTINGS TAB	77
CLOCK SETTINGS TAB	78
TRIGGER SETTINGS TAB.....	80
MARKER SETTINGS	82
MAIN COMMAND BUTTON DESCRIPTION.....	84
SAVE AS.....	84
EXPORT CONFIGURATION.....	84
LOAD FROM.....	84
REMOTE CONTROL.....	86
REMOTE DESKTOP CONNECTION	86
GPIB AND USBTMC (WITH – GPIBUSB OPTION ONLY)	87
<i>GPIB control</i>	87
<i>USBTMC control</i>	88
WAVEFORM LIST.....	89
HOW TO IMPORT A WAVEFORM FROM A FILE	90
<i>How to export a waveform to a file</i>	90
<i>How to promote a waveform to a Predefined</i>	91
<i>How to edit a Waveform</i>	91
<i>How to create a new Waveform</i>	91
CHANNEL COUPLING.....	92
CALIBRATION AND DIAGNOSTIC	95
MULTI-INSTRUMENT SYSTEM	96
BEHAVIOR OF A MULTI-INSTRUMENT SYSTEM	100
MASTER MULTI-INSTRUMENT BAR.....	101
SLAVE MULTI-INSTRUMENT TOOLBAR	103

LICENSE.....	104
CERTIFICATIONS.....	105
EMC COMPLIANCE.....	105
SAFETY COMPLIANCE	105
ENVIRONMENTAL COMPLIANCE.....	106

General Safety Summary

Review the following safety precautions to avoid injury and prevent damage to this product or any products connected to it.

To avoid potential hazards, use this product only as specified.

Only qualified personnel should perform service procedures.

To Avoid Fire or Personal Injury

Use Proper Power Cord

Use only the power cord specified for this product and certified for the country of use.

Ground the Product

This product is grounded through the grounding conductor of the power cord. To avoid electric shock, the grounding conductor must be connected to earth ground. Before making connections to the input or output terminals of the product, ensure that the product is properly grounded.

Observe All Terminal Ratings

To avoid fire or shock hazard, observe all ratings and markings on the product. Consult the product manual for further ratings information before making connections to the product.

Power Disconnect

The power cord provides Mains disconnect.

Do Not Operate Without Covers

Do not operate this product with covers or panels removed.

Do Not Operate With Suspected Failures

If you suspect that there is damage to this product, have it inspected by qualified service personnel.

Avoid Exposed Circuitry

Do not touch exposed connections and components when power is present.

Do Not Operate in Wet/Damp Conditions.

Do Not Operate in an Explosive Atmosphere.

Keep Product Surfaces Clean and Dry.

Provide Proper Ventilation

Refer to the manual's installation instructions for details on installing the product so it has proper ventilation.

Safety Requirements

This section contains information and warnings that must be observed to keep the instrument operating in a correct and safe condition. You are required to follow generally accepted safety procedures in addition to the safety precautions specified in this section.

Safety Symbols

Where the following symbols appear on the instrument's front or rear panels, or in this manual, they alert you to important safety considerations.



This symbol is used where caution is required. Refer to the accompanying information or documents in order to protect against personal injury or damage to the instrument.



This symbol warns of a potential risk of shock hazard.



This symbol is used to denote the measurement ground connection.



This symbol is used to denote a frame or chassis connection.



This symbol is used to denote a safety ground connection.



On (Supply). This is the DC power connect/disconnect switch at the back of the instrument.



Off (Supply). This is the DC power connect/disconnect switch at the back of the instrument.



This symbol is used to denote Power. It is located on the front panel and denotes Power On/Off status of the instrument.



This symbol is used to denote Direct Current.



This symbol is used to denote that the device connectors are sensitive to electrostatic discharge.

CAUTION

The **CAUTION** sign indicates a potential hazard. It calls attention to a procedure, practice or condition which, if not followed, could possibly cause damage to equipment. If a **CAUTION** is indicated, do not proceed until its conditions are fully understood and met.

WARNING

The **WARNING** sign indicates a potential hazard. It calls attention to a procedure, practice or condition which, if not followed, could possibly cause bodily injury or death. If a **WARNING** is indicated, do not proceed until its conditions are fully understood and met.

CAT I

Installation (Overvoltage) Category rating per EN 61010-1 safety standard and is applicable for the instrument front panel measuring terminals. CAT I rated terminals must only be connected to source circuits in which measures are taken to limit transient voltages to an appropriately low level.

Environmental considerations

Product End-of-life Handling

Observe the following guidelines when recycling an instrument or component.

Equipment Recycling

Production of this equipment required the extraction and use of natural resources. The equipment may contain substances that could be harmful to the environment or human health if improperly handled at the product's end of life. In order to avoid release of such substances into the environment and to reduce the use of natural resources, we encourage you to recycle this product in an appropriate system that will ensure that most of the materials are reused or recycled appropriately.



The symbol shown to the left indicates that this product complies with the European Union's requirements according to Directive 2002/96/EC on waste electrical and electronic equipment (WEEE).

Preface

This manual describes the installation and operation of Model 686 using the Simple AFG software. Basic operations and concepts are presented in this manual.

The easiest touch screen display interface allows to create waveforms scenarios, only in few screen touches.

The Model 686 series offers premium signal integrity thanks to the 14 bits resolution DAC. For single ended models the output signal can reach up to 5 Volts pk-pk into 50 Ohm and a total voltage window of 10 Volts into 50 Ohm with a hardware offset of ± 2.5 Volts into 50 Ohm.

For differential models the output signal can reach up to 2 Volts pk-pk into 100 Ohm with a common mode voltage of ± 2 Volts into 50 Ohm.

The Model 686 can generate signals up to 10 GHz. The software architecture provides the possibility to easily generate predefined waveforms and modulated signals or to load an arbitrary waveform from a user defined file.

Package Contents

The standard Model 686 Series package includes the following:

- Model 686 Arbitrary Waveform Generator equipment
- 32 GB USB Pen Drive for Software Recovery procedure
- Power Cord
- Performance/Calibration Certificate
- CE certificate

Models

Item	Description
686-2C-SE	2 Ch – 20GS/s AWG – 5Vpp Single ended outputs – Long memory
686-2CD	2 Ch – 20GS/s AWG – 2Vpp (1Vpp single ended) Differential outputs – Long memory
686-4C-SE	4 Ch – 20GS/s AWG – 5Vpp Single ended outputs – Long memory
686-4CD	4 Ch – 20GS/s AWG – 2Vpp (1Vpp single ended) Differential outputs – Long memory

Recommended Options and Accessories

Item	Description
686-4C-8DIG	O 686-4C-8DIG 8CH Dig license
686-4C-16DIG	O 686-4C-16DIG 16CH Dig license
686-4C-32DIG	O 686-4C-32DIG 32CH Dig license
686-MINI-SAS-HD	A Mini Sas HD cable for digital probe, 8 Differential signal (available only for 4-channels models with long memory)
AT-DTTL8	A LVDS to LVTTTL digital adapter probe (available only for 4-channels models with long memory)
AT-LVDS-SMA8	A CML to SMA digital adapter cable (available only for 4-channels models with long memory)
686-2C-WAR	O 3 years warranty extension for 686-2C
686-4C-WAR	O 3 years warranty extension for 686-4C
RIDER-686-SYNC	A Multi-Instrument synchronization cable for 686 series 0.5m
686-2C-PAT	O Serial Pattern Generator (SPG)
686-4C-PAT	O Serial Pattern Generator (SPG)

¹ All 20GS/s models require the export license.

686-FSS	O	686 Fast Sequence Switch
GP-IB / USB-TMC	O	GPIB and USBTMC Ports for Remote Control
RIDER-RACK	A	Rackmount kit for rider instrument system

O = Options, **A** = Accessories

Key features

The following list describes some of the key features of the Model 686 series

- High resolution, high sampling rate: 14 Bits, 20 GS/s
- Best output frequency vs amplitude trade off: 10 GHz, 10V voltage window.
- 5 V_{pp} single ended outputs or 2 V_{pp} (1 V_{pp} single ended) differential outputs (into 50 Ohm).
- 4 operating modes in the same instruments: Function Generator (AFG), Arbitrary Waveform Generator (AWG).
- Simple touch screen user interface to create complex waveforms scenarios just in few screen touches.
- Large 7 inch, 1024x600 capacitive touch LCD.
- Touchscreen or Keypad data entering.
- Windows 10 operating system.
- USB and LAN interfaces.
- 3U case size with the possibility of rack mounting.

Mechanical Characteristics

Model 686-2C

Net Weight	23.1 lbs
Net Weight with Package	24.3 lbs
Overall Dimensions	Height: 5.31 in Width: 17.5 in Depth: 12.6 in

Model 686-4C

Net Weight	24.3 lbs
Net Weight with Package	25.4 lbs
Overall Dimensions	Height: 5.31 in Width: 17.5 in Depth: 12.6 in

Operating Requirements

CAUTION. To ensure proper cooling, keep sides of the instrument clear of obstructions.

Place the instrument on a cart or bench, observing clearance requirements:

- Top: 0.8 in (20 mm)
- Left and right side: 5.9 in (150 mm)
- Bottom: 0.8 in (20 mm)
- Rear: 3 in (75 mm)

CAUTION. Ensure that the equipment is positioned in a way that the disconnecting device can be readily accessible.

The instrument is intended for indoor use and should be operated in a clean, dry, nonconductive environment. Occasionally a temporary conductivity that is caused by condensation must be expected. This location is a typical office/home environment. Temporary condensation occurs only when the product is out of service.

Environmental requirements

Before using this product, ensure that its operating environment is maintained within these parameters:

Temperature	Operating +5 °C to +40 °C (+41 °F to 104 °F) Non-operating -20 °C to +60 °C (-4 °F to 140 °F)
Humidity	Operating 5% to 80% relative humidity with a maximum wet bulb temperature of 29 °C at or below +40 °C, non-condensing. Non-operating 5% to 95% relative humidity with a maximum wet bulb temperature of 40 °C at or below +60 °C, non- condensing.
Altitude	Operating 3,000 m (9,843 feet) Non-operating 12,000 m (39,370 feet)

Power supply requirements

WARNING. To reduce the risk of fire and shock, ensure that the mains supply voltage fluctuations do not exceed 10% of the operating voltage range.

No manual voltage selection is required because the AC Adapter automatically adapts to line voltage.

Source Voltage and Frequency	100 to 240 VAC $\pm$ 10% @ 45-66 Hz
Power Consumption	• 686-2C Maximum: 100W • 686-4C Maximum: 200W



WARNING - Electrical Shock Hazard

Only use the power cord provided with your instrument

Cleaning

WARNING. To avoid personal injury, power off the instrument and disconnect it from line voltage before performing any other following procedures.

Inspect the arbitrary waveform generator as often as operating conditions require. To clean the exterior surface, perform the following steps:

- Remove loose dust on the outside the instrument with a lint-free cloth. Use care to avoid scratching the front panel display.
- Use a soft cloth dampened with water to clean the instrument. Use a 75% isopropyl alcohol solution as a cleaner.

CAUTION. To avoid damage to the surface of the arbitrary waveform generator, do not use any abrasive or chemical cleaning agents.

Calibration

The recommended calibration interval is one year. Calibration should be performed by qualified personnel only.

Abnormal Conditions

Operate the instrument only as intended by the manufacturer.

If you suspect the instrument's protection has been impaired, disconnect the power cord and secure the instrument against any unintended operation.

The instrument's protection is likely to be impaired if, for example, the instrument shows visible damage or has been subjected to severe transport stresses.

Proper use of the instrument depends on careful reading of all instructions and labels.

WARNING. Any use of the instrument in a manner not specified by the manufacturer may impair the instrument's safety protection.

Installing your instrument

Unpack the instrument and check that you received all items listed in the Package Contents paragraph.

Power the Instrument On and Off and Launch AFG Application

Power On

- Insert the AC power cord into the power receptacle on the rear panel.
- Use the front-panel power button  to power on the instrument.
- Wait until the system shows windows desktop.
- The Simple AFG software will start automatically if at the previous power off the instrument was working in AFG mode.

Alternatively push the AFG icon  to launch the application from the desktop or push the

Switch App button  to switch in AFG mode from another application.

Power Off

- Close the application in use. 
- Press the front-panel power button  to power off the instrument.

Protect Your Instrument from Misuse

Check Input and Output Connectors

When connecting a cable, be sure to distinguish the input connector from the output connectors to avoid making the wrong connection.



CAUTION. Do not short output pins or apply external voltages to Output connectors. The instrument may be damaged.

CAUTION. Do not apply excessive inputs over ± 15 Vpk to Trigger Input connector. The instrument may be damaged.

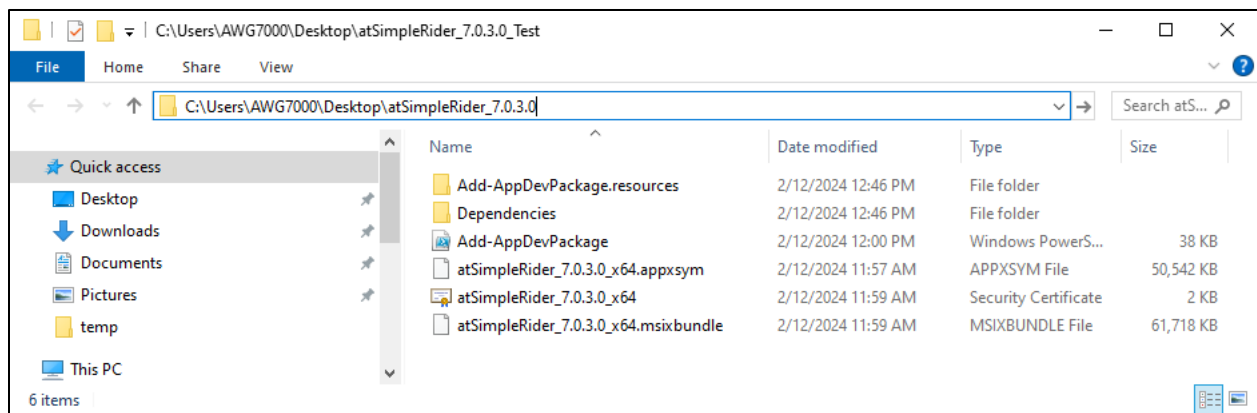
Obtaining the Latest Version Releases

The latest release of the software may not be installed on your instrument. The latest version could be found on Berkeley Nucleonics Corporation website in the support area.

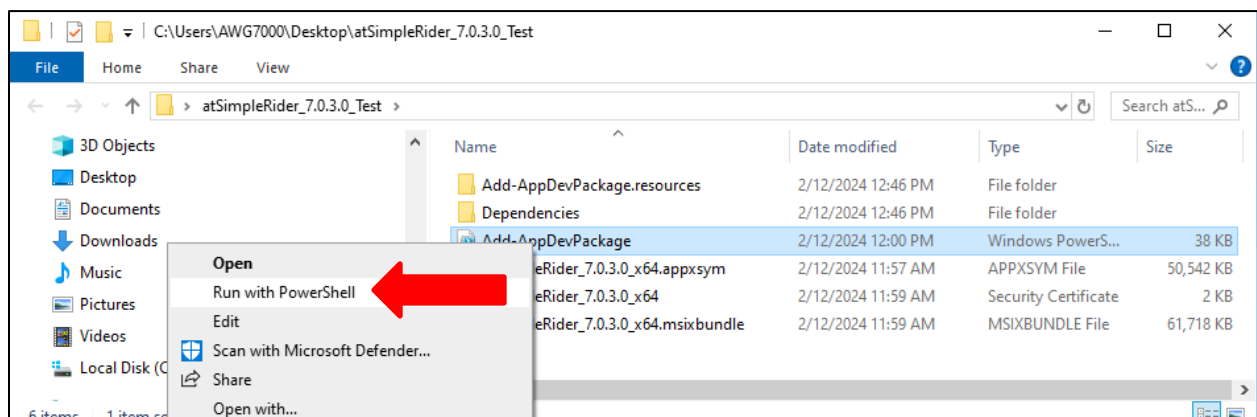
Install Simple AFG Application

If your instrument has already installed another version of the Simple AFG application, DO NOT uninstall it otherwise you will lose all the configurations and projects.

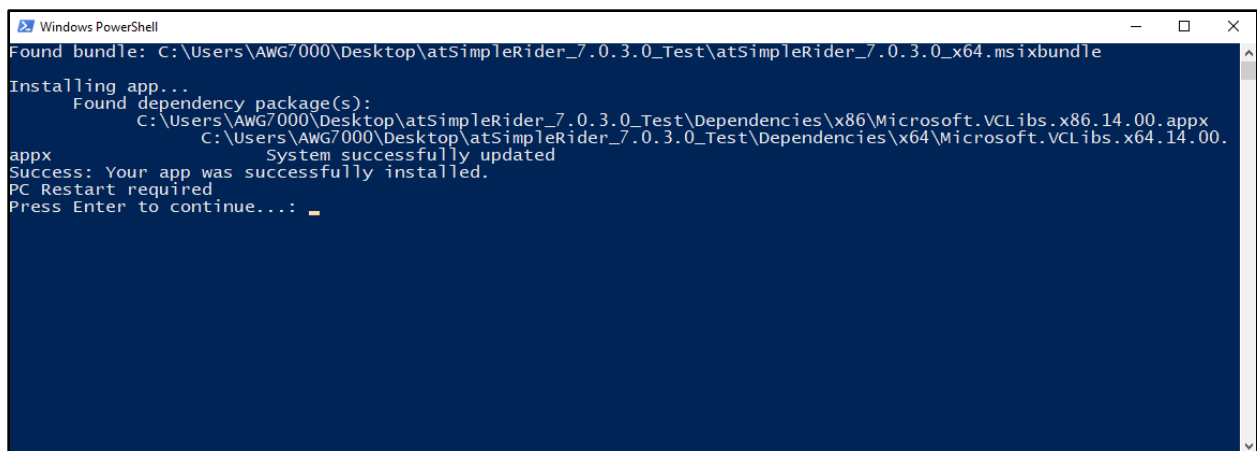
- 1) Download the Simple AFG setup package from Berkeley Nucleonics Corporation website and decompress it to instrument's local disk.



- 2) Right click on the “Add-AppDevPackage” file and select **Run with PowerShell** to start the installation.



- 3) When the application has been installed, press the “Enter” button to continue.



USB Pen Drive and Recovery Procedure

In case of software failure or corrupted applications, it's possible to reinstall the full factory image of the software using the 32 GB Usb Pen Drive included in the package.



Note: In the Recovery USB Pen Drive you'll find the same version of applications, drivers and updates that are installed on the instrument at the time of purchase. To get the latest application release follow the instructions in the paragraph "Obtaining the Latest Version Releases".

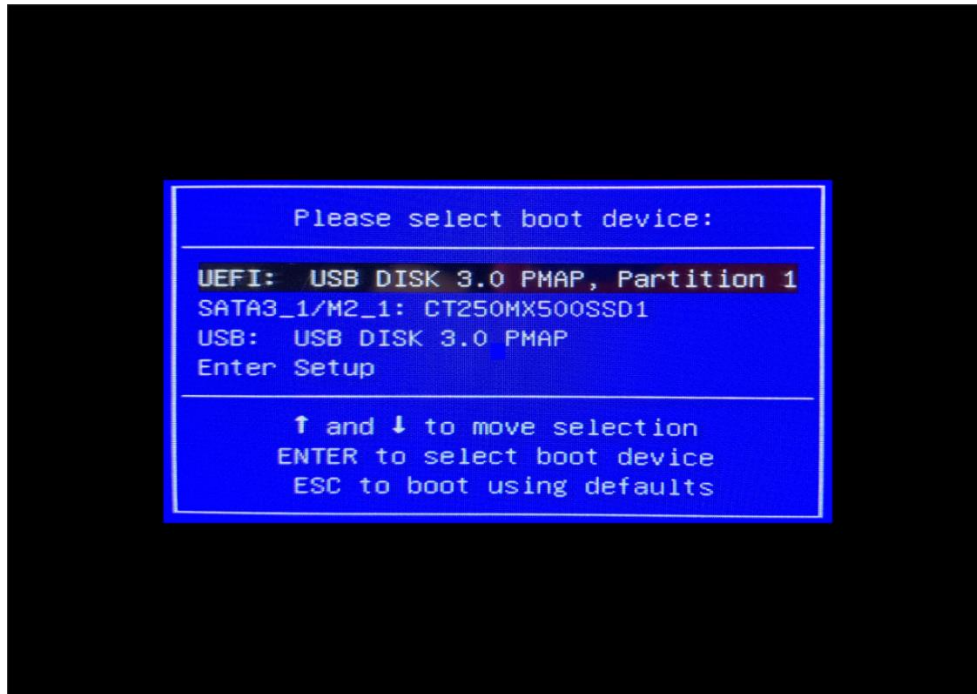
Note: The procedure completely formats the SSD of the instrument so remember to save all important data (Configuration files, Arbitrary User Waveform files...) in an external device because all the current settings will be lost when the Recovery Procedure is launched.

Note: If two SSD are mounted on the instrument remember to leave only the one on which you want to launch the procedure, the other must be removed.

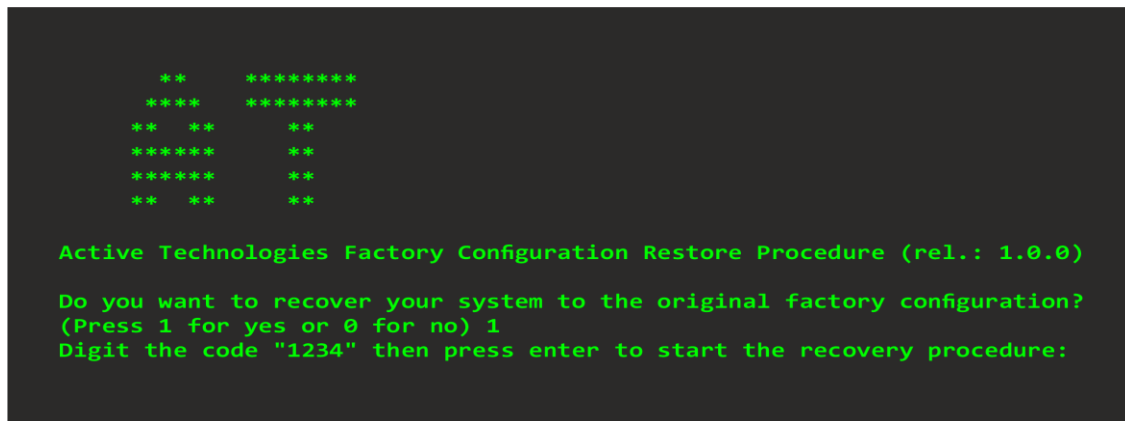
Note: A keyboard is required for the recovery procedure from USB Pen Drive.

Recovery Procedure

- 1) Insert the Recovery USB Pen Drive into a USB port of the instrument. If the instrument is OFF press the power-ON button, otherwise restart the instrument. Check that a keyboard is correctly connected to the instrument.
- 2) Once the instrument is started press repeatedly F11 button on the keyboard during the Boot process and access the Boot menu (see the image below).



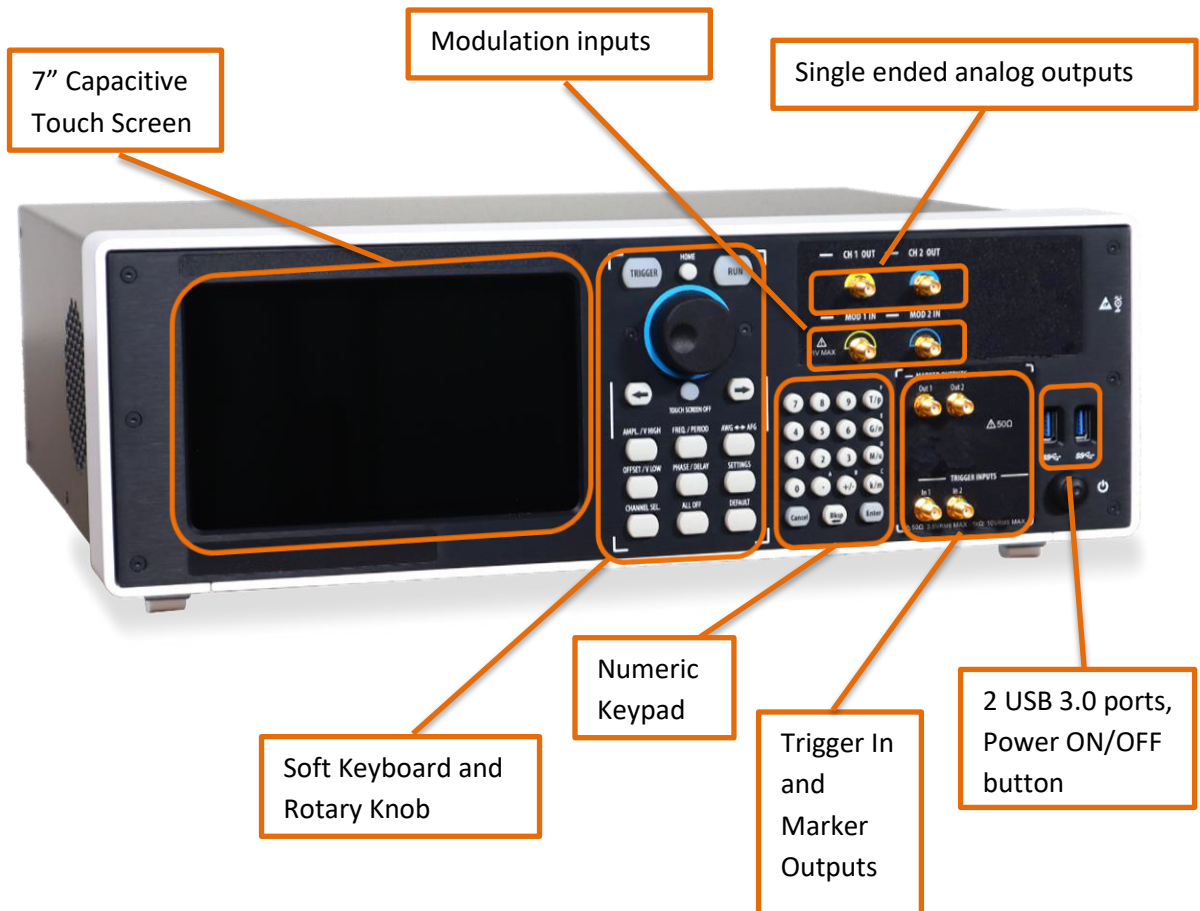
- 3) In the Boot menu select the "UEFI: USB DISK 3.0 PMAP, partition 1" choice and then press Enter.
- 4) Press '1' on the keyboard to start the recovery procedure.



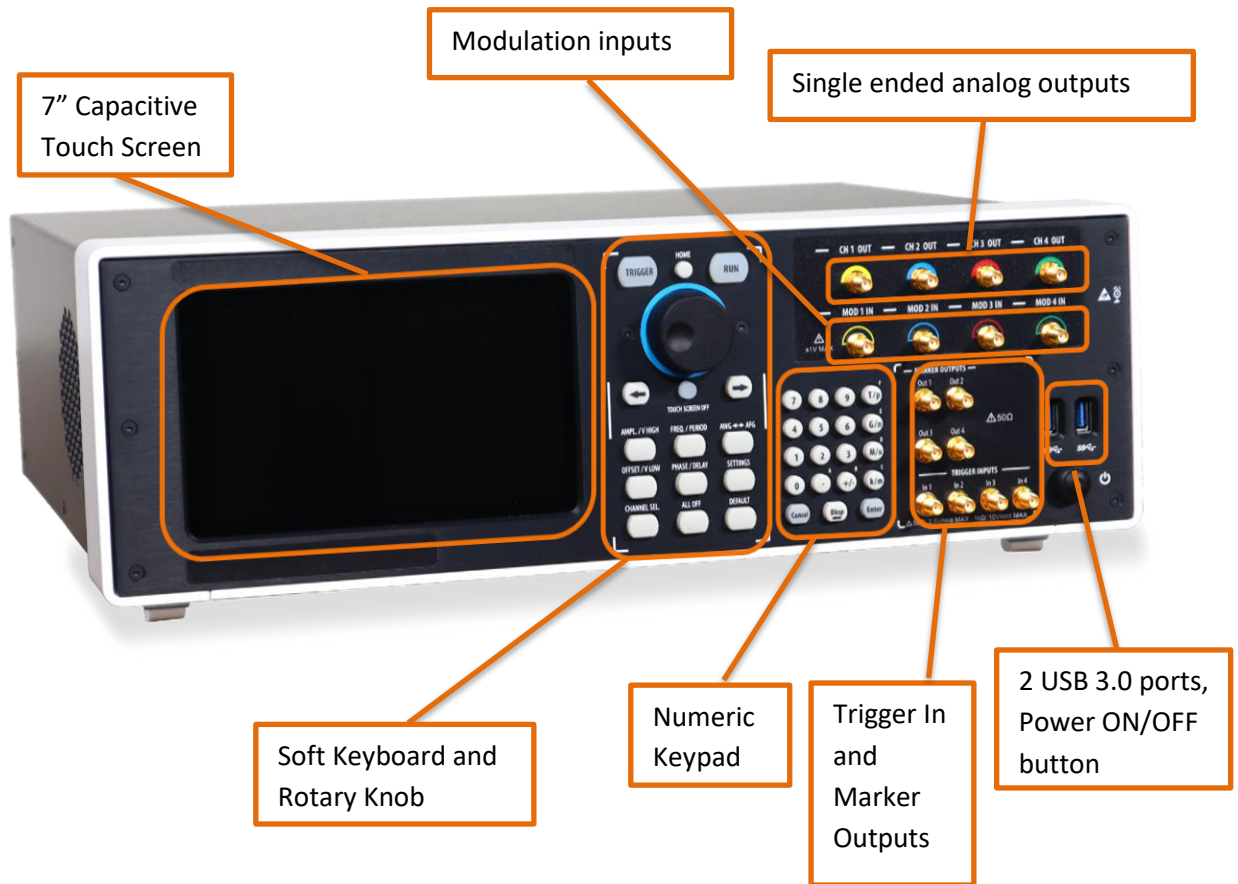
- 5) Enter the code "1234" to confirm the execution of the recovery procedure.
- 6) Once the procedure is complete, press enter to shut down the instrument.
- 7) Remove the Recovery USB Pen Drive and power ON the instrument. Follow the instructions at point 2) to access the Boot menu and select SATA SSD source. Press enter to confirm.

Instrument Overview

Front Panel Model 686-2C



Front Panel Model 686-4C



The front panels of the Model 686-2C models differ from that shown in the picture above because they have twice the SMA analog output connectors because each output channel has two complementary outputs (+ and -).

The Touch screen functionalities and features are described in the Simple AFG Application paragraph.

Analog Outputs

The Model 686 series instrument has 2 or 4 analog output channels, each one is single-ended or differential (depending on the model) and the connector type is a SMA.

Marker Outputs

The Marker Out is a digital output channel that generates a pulse related to the analog waveform. Its impedance is 50 Ohm and the output voltage amplitude ranges from -0.5 V to +1.65 V into 50 Ohm load. The Marker Out generates a digital pulse synchronous with the waveform depending on the Run Mode. To set the Marker Out parameters refer to the Marker Out Settings. The connector type is a standard SMA.

Marker Out Specification	Value
Connector	1 SMA for each pair of channels on the Front Panel
Output impedance	50 Ω
Output level (into 50 Ω)	-0.5 V to 1.65 V

Model	Marker Out Connectors
686-2C	2 SMA on the Front Panel
686-4C	4 SMA on the Front Panel

Trigger Inputs

The Trigger In 1/2/3/4 (Trigger In connectors on the front panel) allow to control the generation by an external signal source. They have a selectable impedance of 1 kOhm or 50 Ohm. In Continuous mode the Trigger Inputs don't have any effect.

Trigger In Specification	Value
Connector	SMA on the Front Panel
Number of connectors	2 in 2-channels models or 4 in 4-channels models
Input impedance	1 k Ω or 50 Ω selectable
Slope/Polarity	Positive or negative selectable

Soft keyboard and rotary knob

Most of the buttons you use with Simple AFG application are virtual ones on the touchscreen, but a few physical buttons control basic functions, such as the setting of amplitude, offset, frequency, etc.

A physical numeric keypad is available on the front-panel and it can be used instead of the virtual numeric pad.

A useful central knob is available for fine-tuning and adjustments during the on the fly set up operation. The rotary knob will change the value in continuous, analog fashion. The push button rotary knob lets you to change the value increment between Coarse and Fine adjustment.

The → key will move the selected digit to the right and the ← key will move the selected digit to left. You can keep pressed the rotating knob and rotate it on the right or on the left to change the Delta increment.



Button	Description
HOME	If you are in a sub-menu page, use this button to return to the main page.
TRIGGER	Use this button to send an internal trigger to the instrument.
RUN	Use this button to start and stop the signal generation. If the button is on and green the instrument is running while if it is off the instrument is stopped. Pushing the button will change the instrument state.
LEFT ARROW	Once the virtual numeric keypad will be opened, use this button to move to the left the digit selection cursor.
RIGHT ARROW	Once the virtual numeric keypad will be opened, use this button to move to the right the digit selection cursor.
TOUCH SCREEN OFF	Use this button to disable the touch screen.
AMPL./V HIGH	Use this button to set the high voltage level or the amplitude of the waveform.
FREQ/PERIOD	Use this button to set the period or the frequency of the waveform.
AWG <-> AFG	Use this button to switch between AFG mode and AWG operating mode.
OFFSET/V LOW	Use this button to set the low voltage level or the offset of the waveform.
PHASE/DELAY	N.A.
SETTINGS	Use this button to open the Settings page
CHANNEL SEL.	Use this button to change the output selection in the user interface
ALL OFF	Use this button to turn off all the outputs.
DEFAULT	Use this button to restore the default settings.

Numeric Keypad

The physical numeric keypad lets you to set the parameter value and their measure unit.

Once a parameter to be edited is selected by using the touch panel or the soft keyboard, each number pressed in the keypad will be displayed in the display. The Bksp key is provided for deleting erroneous key presses. The [+/-] key will toggle the sign of the number being entered and may be pressed after terminating the entry. After the sign and the numeric portion of the desired value have been entered, the pressing of the multiplier button applies the parameter. The Enter button closes the virtual keyboard and will apply the entered value.



When you select a parameter on the user interface, if you press a Unit Measure Range button it will automatically update the available range allowed for that parameter.

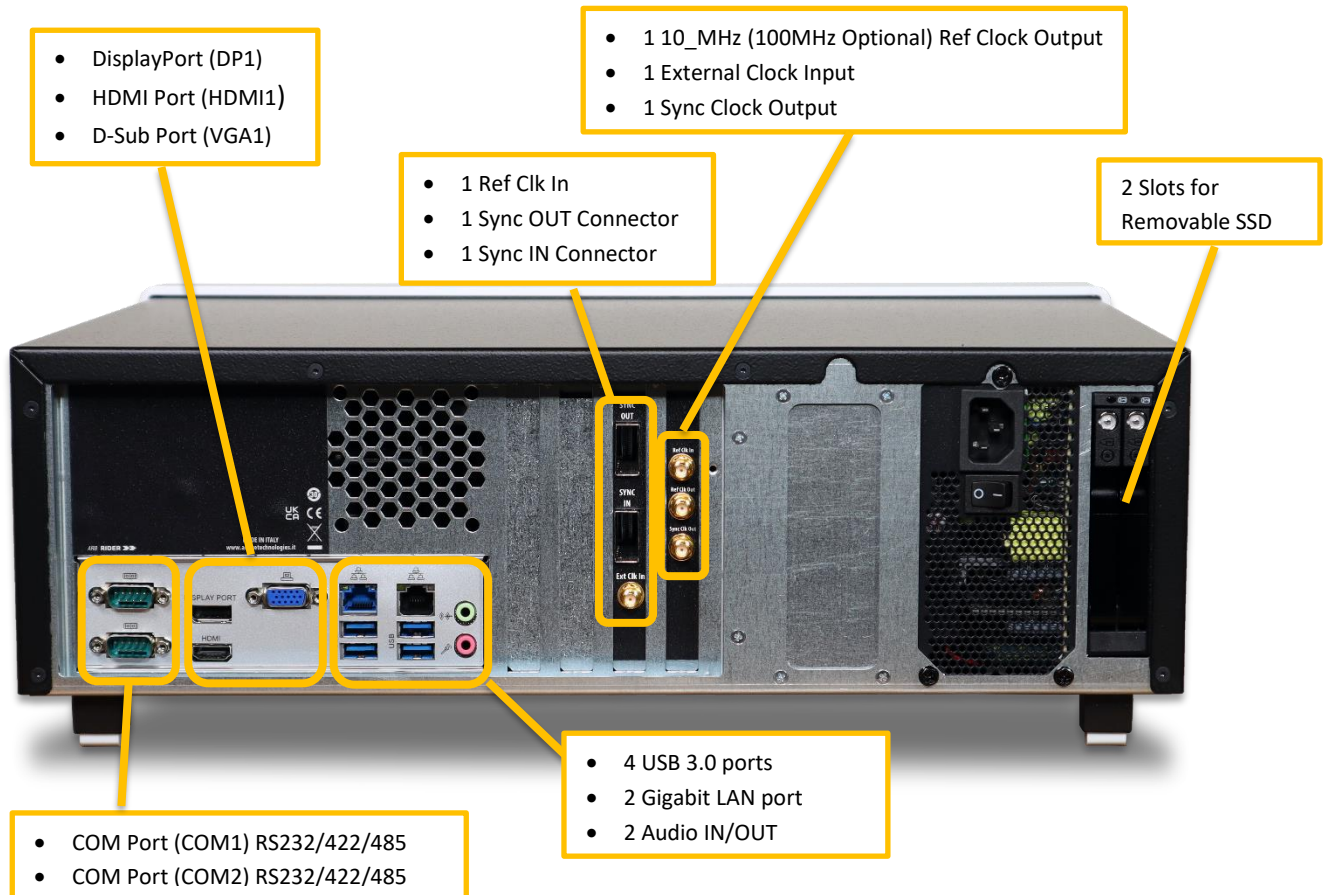
Unit Measure Range Button	Unit Measure Range
T/p	Tera / pico
G/n	Giga / nano
M/u	Mega / micro
k/m	kilo / milli

For example if you select the Frequency parameter and you press k/m the unit measure range will be kHz, if you press M/u it will be MHz, if you press G/n it will be GHz, if you press T/p nothing will happen because that range is not available for the selected parameter.

If both the two unit of measure of a Unit Measure Range button are available for the selected parameter (i.e. Mega and Micro), if you press the range button **M/u**, the range will switch accordingly between Mega and Micro.

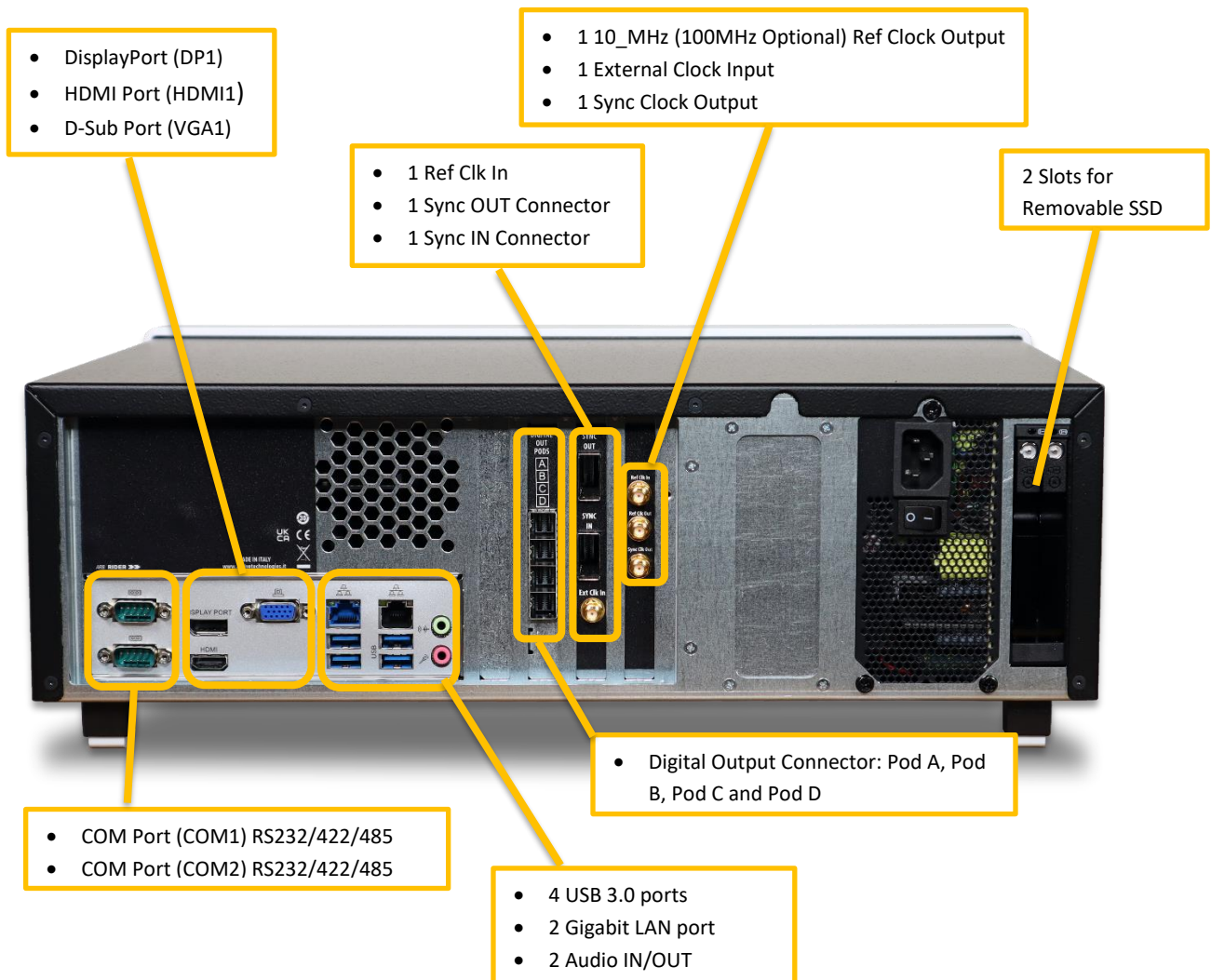
Rear Panel Model 686-2C

The callouts on this image gives the description of the corresponding connectors:



Rear Panel Model 686-4C

The callouts on this image gives the description of the corresponding connectors:



External Modulation Input Connector

For each analog output, the Model 686 series instrument has an input connector (MOD IN) to receive an external analog signal that is used as modulating source. When the selected Run Mode is “Modulation” and the source is “External”, the instrument will use one of these external signals to modulate the carrier waveform of a specific channel.

The connector type is a SMA.

Reference Clock Input Connector

The Model 686 series instrument can use an external clock source to generate the sampling clock frequency. This feature allows to synchronize the generator with an external clock.

The connector type is a SMA.

Reference Clock Output Connector

This connector outputs the internal 10MHz (100MHz Optional) reference clock used to synthesize the DAC sampling clock. If the clock source is internal it produces a signal at 10 MHz (100 MHz Optional), if the source is external it is disabled.

The connector type is a SMA.

Digital Output connector

Important Note: this connector is not used by AFG application.

External Clock Input Connector

This connector input gives the User the possibility to feed directly a sampling clock for the System. This Clock bypasses the internal Generator Clock System of the instrument.

The connector type is a SMA.

Sync Clock Output Connector

This connector outputs a divided clock generated from the sampling clock of the instrument. The resulting frequency can be selected by the User from a list with all possible values.

The connector type is a SMA.

Sync In / Sync Out Connectors

The purpose of those connectors is to connect and synchronize together multiple instruments. Up to 4 instruments can be synchronized.

Pattern Force Jump In Connector (with 686-FSS Option only)

Important Note: this connector is not used by AFG application.

Introduction

The Model 686 series instrument, when used in Arbitrary Function Generator mode, has two or four independent analog channels. Each channel can generate a predefined waveform or a user-defined waveform loaded from a file.

Any characteristic parameter of the selected waveform can be modified at runtime. For example, if a pulse waveform is selected it is possible to define at runtime its amplitude, offset, frequency, duty cycle and the duration of leading and trailing edge.

Simple AFG Application

The Model 686 series instrument includes a 7" capacitive touch screen and an easy touch user interface based on a Microsoft Windows 10 platform.

You can control instrument operations using one or all of the following entering methods:

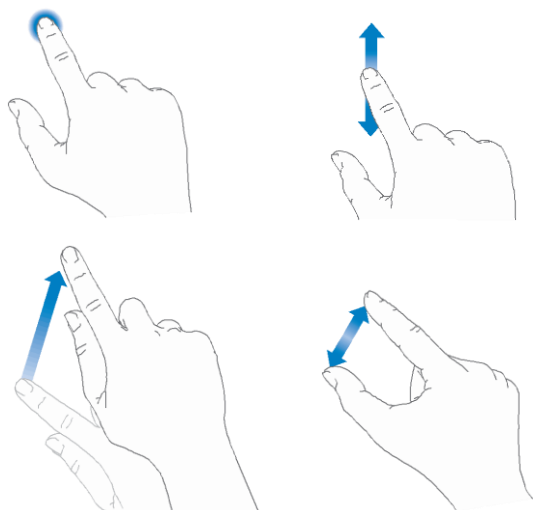
- Touch Screen and Front-panel soft key controls
- Keyboard and mouse

Simple AFG Touch UI

Simple AFG Touch UI is designed for touch to drive simplicity in operating with an Arbitrary Waveform Generator, by using the today's modern technique, used on Tablet or smart phones, available in capacitive touch-screen displays.

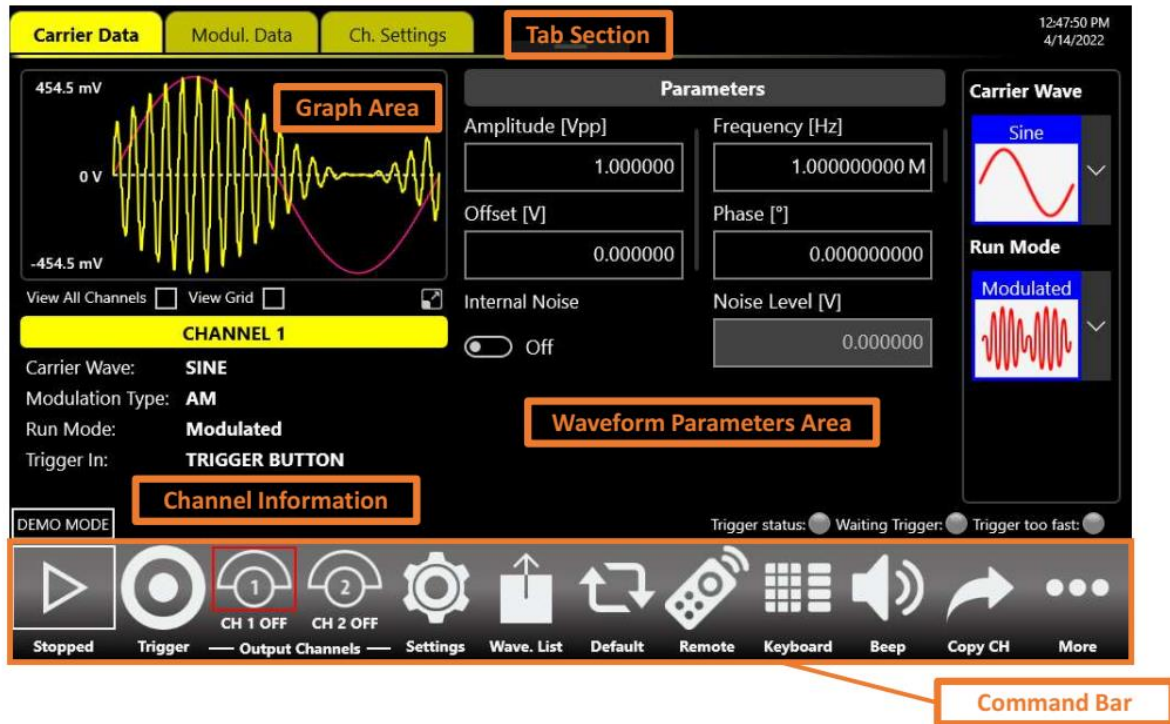
All the important instrument controls and settings are always one touch away:

- swipe down gesture to change the output channel
- swipe left or right to navigate through the sequencer entries
- pinch in-out to zoom the waveform graph
- use the touch-friendly virtual numeric keyboard to modify the parameters and to entry new values on the fly



User Interface Description

The Simple AFG software environment provides an easy access to all instrument functionalities and parameters.



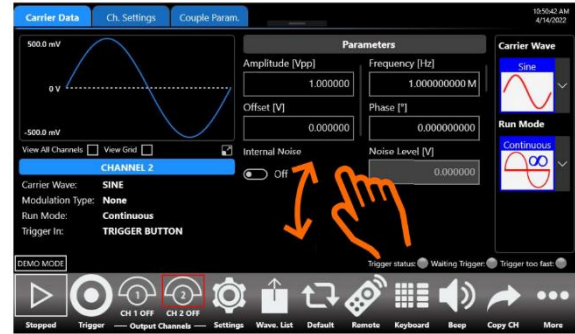
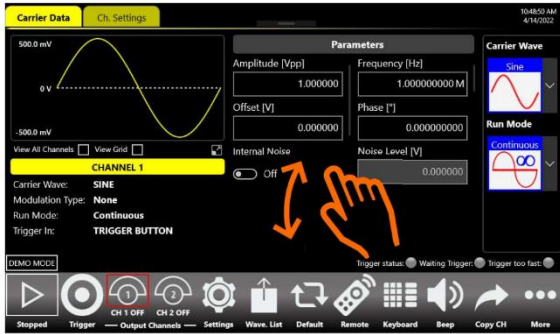
AFG user interface consists of four main elements:

- **Waveform Parameters Area:** it contains all the waveform settings. It is composed by the **Carrier** tab and the **Secondary** tab.
The **Carrier** tab allows to choose the Run Mode and the Waveform type and to set its parameters.
The **Secondary** tab is used to define the **Modulation**, **Sweep** or **Burst** parameters depending on the selected Run Mode.
- **Graph Area:** it shows a qualitative graphical representation of the generated waveform.
- **Channel Information:** it summarizes the channel settings.
- **Command Bar:** in this bar there are elements to control the instrument operations, to modify the instrument settings and to manipulate waveforms.

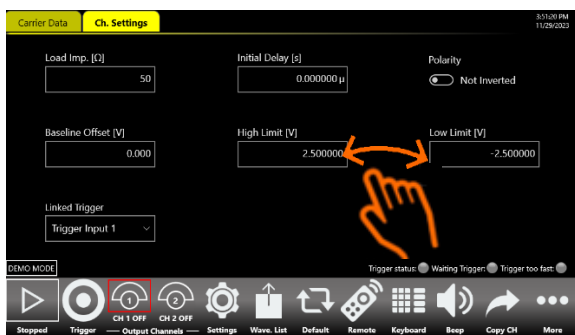
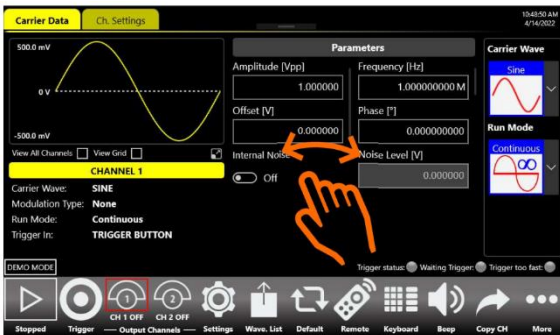
As mentioned, the display is 7" capacitive touch screen display and you can use the gestures like in a mobile phone:



If you use the Swipe Up or Down gesture over the Command Bar Area you can switch between the adjacent channels.



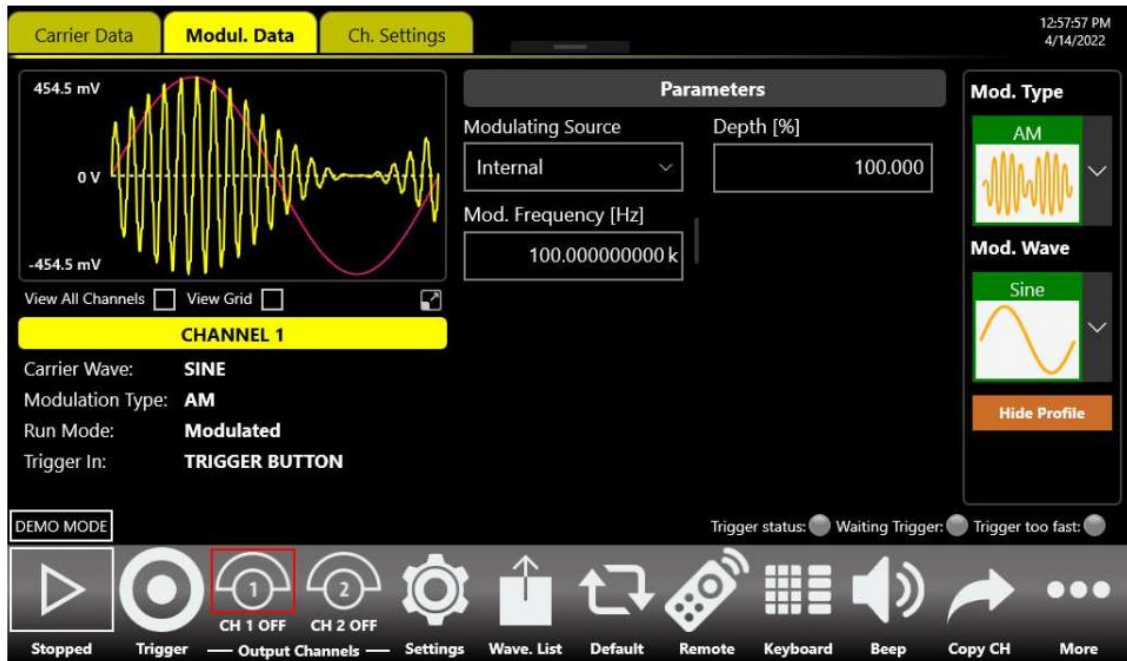
If you use the Swipe Left or Right gesture over the Command Bar Area you can switch between the Tab Section.



Waveform Parameters Area

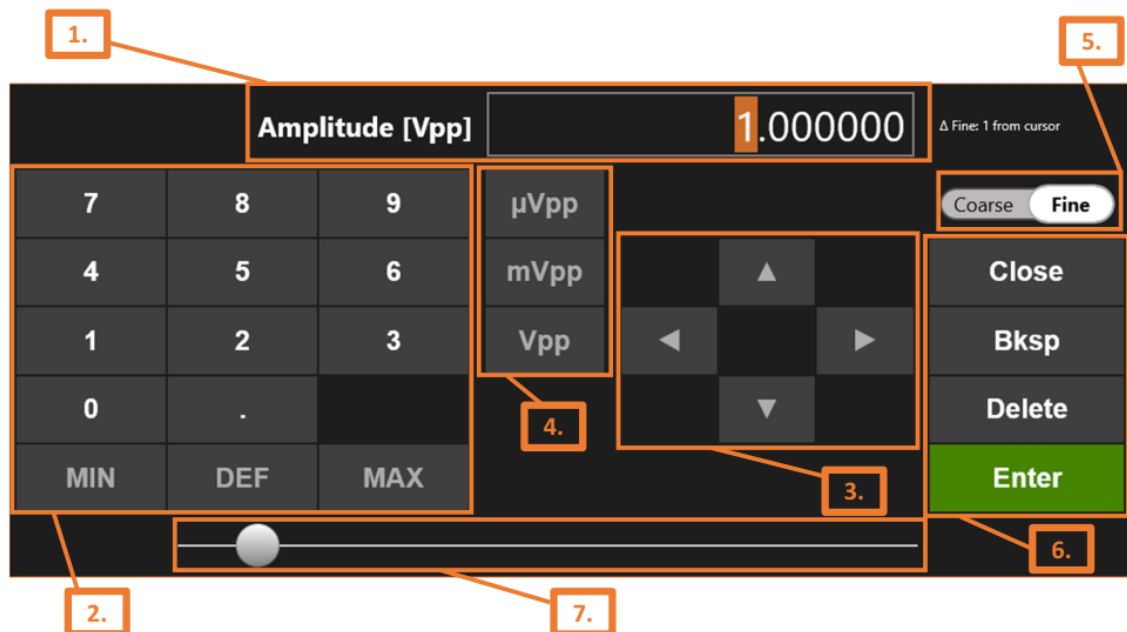
This section is different in base of the tab selected:

- In the **Carrier** tab it is possible to define the Run Mode, the Carrier Waveform and its parameters as explained in the relative chapter.
- If the Run Mode option selected is different from Continuous, will compare another tab right side Carrier Tab, it is possible to define the parameters that describe the **Modulation**, **Sweep** and **Burst**. This tab changes title name and functionality depending on the selected Run Mode. For example, if the Run Mode is Sweep the Secondary tab will take the title name "Sweep" and the tab page will show the Sweep parameters. The same will happen for Modulation and Burst modes. In Continuous Run Mode the Secondary tab is not active.



You can touch the parameter area to open the Virtual numeric keypad, edit the parameter value and its measure unit.

Below there is a description of the keypad items:



1. **Parameter Name and Value:** This area of the virtual keyboard displays the parameter name, value and unit of measure.
2. **Numeric Keypad:** this area contains the keys to edit the number that will be displayed in the area 1. The [+/-] key will toggle the sign of the number being entered and can be pressed at the end of the number editing.

Touch the “MIN” and “MAX” buttons to set the minimum and maximum allowed value for the selected parameter. Use the “DEF” button to set the default value.
3. **Arrows:** The left/right arrows allow to move the cursor or select the different digit position as the arrows on the front panel. The up/down arrows allow to modify the value.
4. **Measurement Unit:** After typing the numeric value these buttons can apply a different multiplier of the measurement unit. When a measurement unit is pressed, the value is applied on the fly.
5. **Coarse / Fine:** the coarse/fine button let you to modify the granularity of the increment. You can increment or decrement the selected parameter using the UP/DOWN arrows button or rotating knob on the front panel.

When Fine is selected, the increment is of 1 unit at the current cursor position.

When Coarse is pressed, the Delta increment is displayed in the parameter area and the parameter value changes in steps of the selected increment.

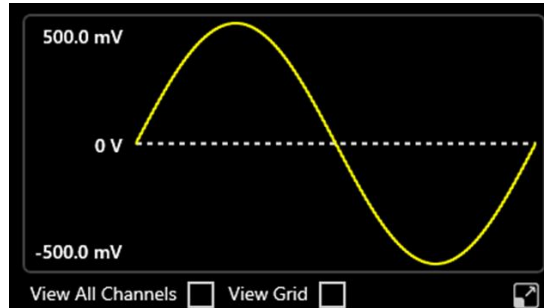
You can keep pressed the knob and rotate it on the right or on the left to change the Delta Coarse increment.
6. **Control Buttons:** The “Close” button closes the virtual keypad without applying any changes on the instrument while the “Enter” button confirms the changes and it applies them on the instrument.

“Bksp” (backspace) button is provided for deleting erroneous key presses, “Delete” button deletes all digit of the textbox.
7. The **horizontal scrollbar** allows to change quickly the selected value. The position specifies the value between the allowed minimum and the maximum.

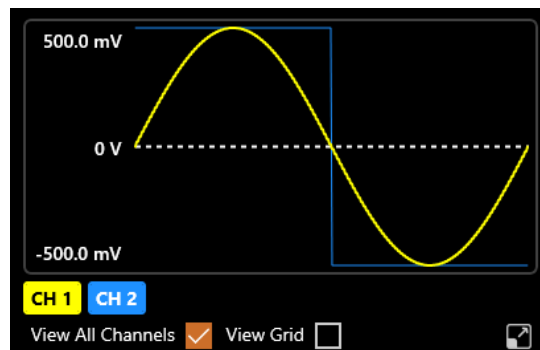
The increment/decrement value entered by the rotary knob or by the scrollbar are applied to the instrument on the fly.

Graph Area

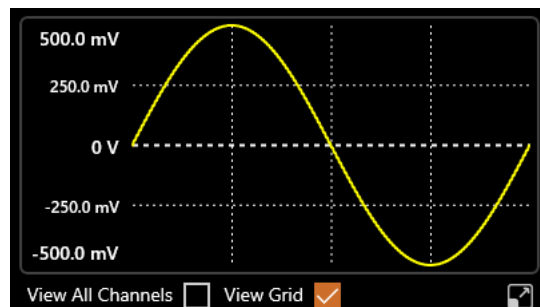
The graph area displays the Output channel waveform with a vertical legend that shows the minimum and maximum voltage levels and the offset.



When **View All Channels** is checked the graph shows all channels graphs overlapped. The vertical scale is referred to the most amplitude from the channels. Furthermore, you can decide to View only the channels that you want tapping on coloured buttons under the graph.

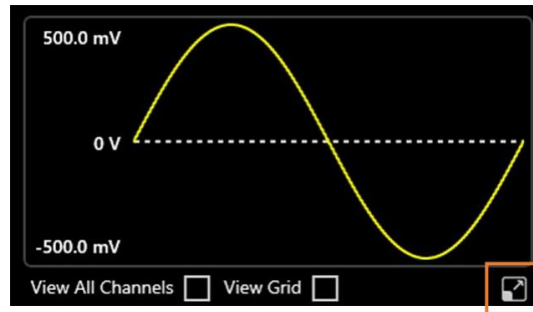


Checking **View Grid**, you can view further horizontal and vertical lines with other values of amplitude.

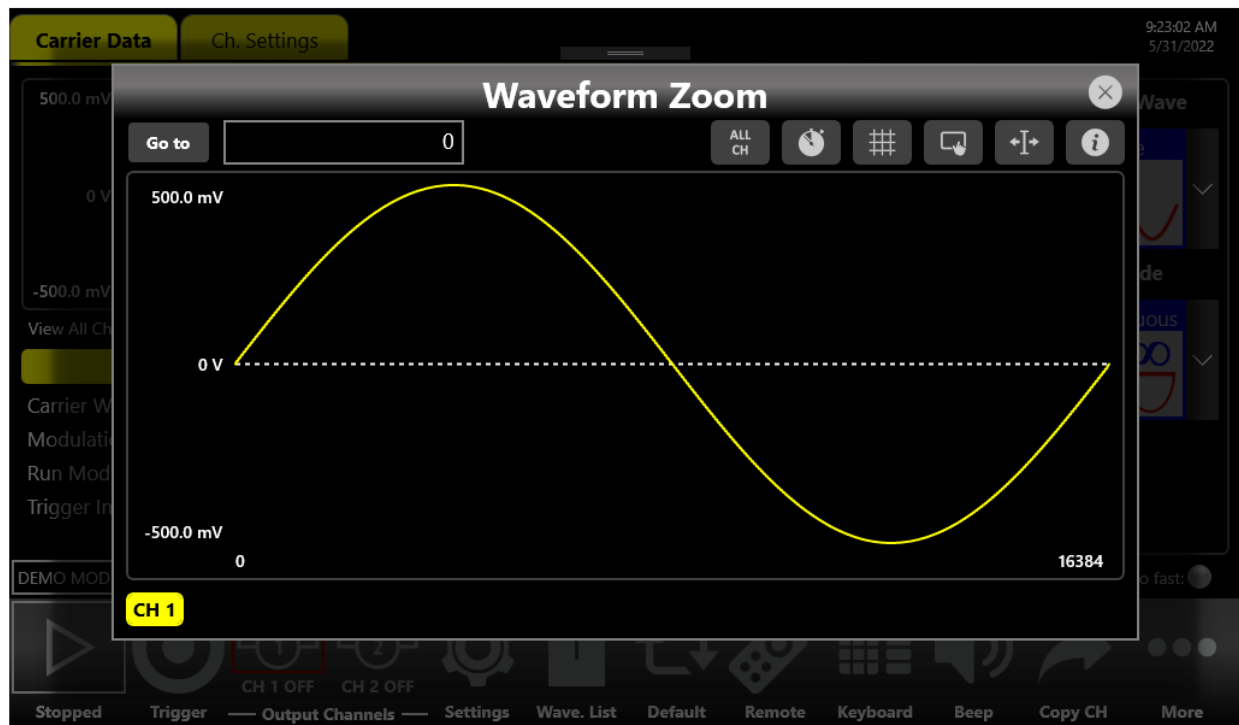


Zoom graph

Tapping on the highlighted button below, you can open another panel with more functions that expands the functionality of the graph.

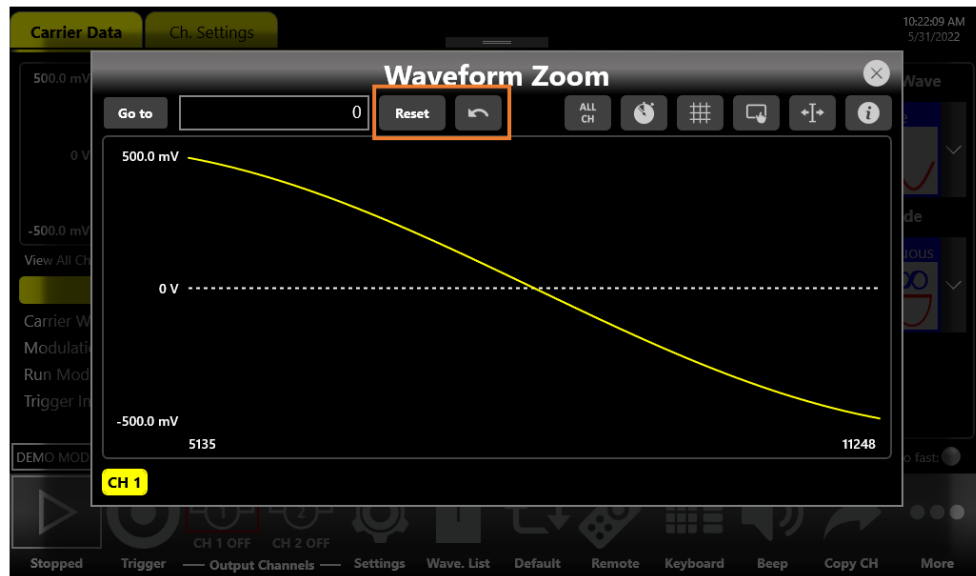


In particular, this new panel give you the possibility to Zoom into area of the graph with different modality.

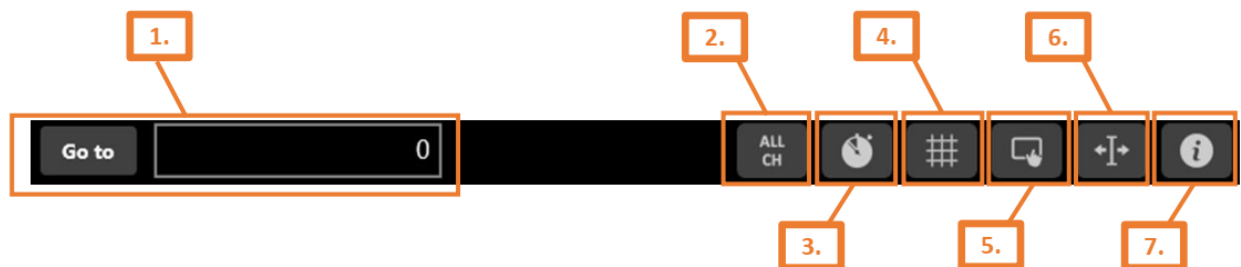


As you can see in the previous figure, the panel is divided in two areas, on top there is the command bar while the rest of the area is employed by the graph. The Waveforms on the graph can be represented with Samples Scale or Times Scale, you can zoom into area using two fingers as multitouch to zoom in and out, one finger to translate window zoom left or right, otherwise you can select a rectangle area if the correct button is enabled. Every time that you Zoom In, will be add to a stack the previous value of

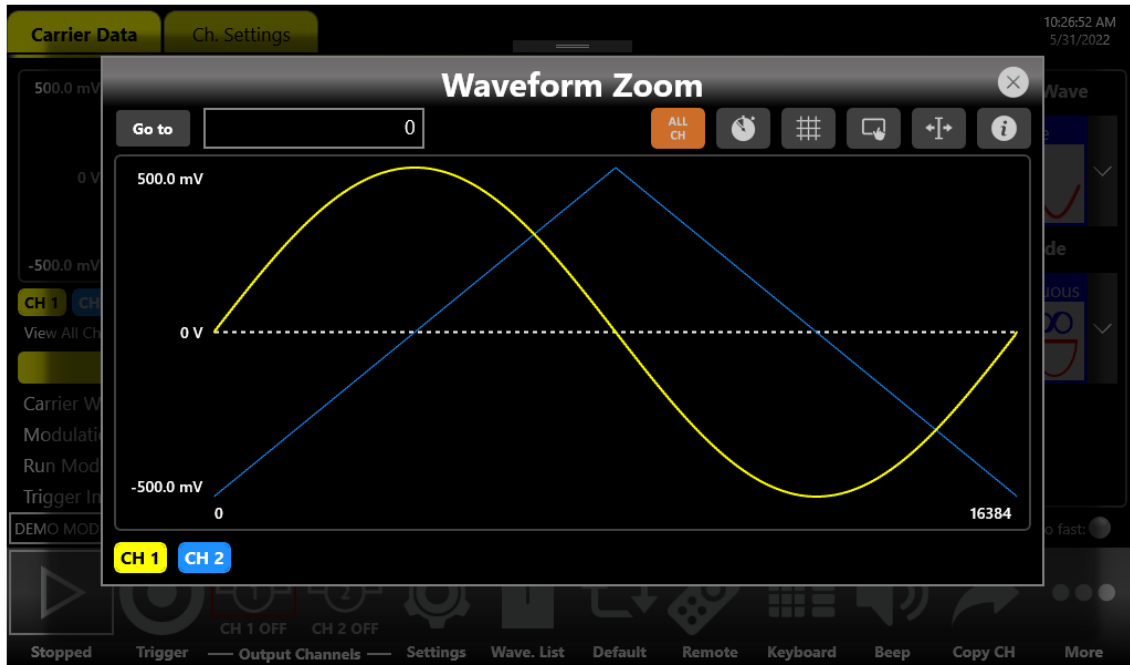
Zoom and will be show two buttons, “Reset” and “Back” in the command bar, these buttons permit you to come back the previous values or set the default value state of the waveform with zoom at 100%.



Below the description of the command bar:

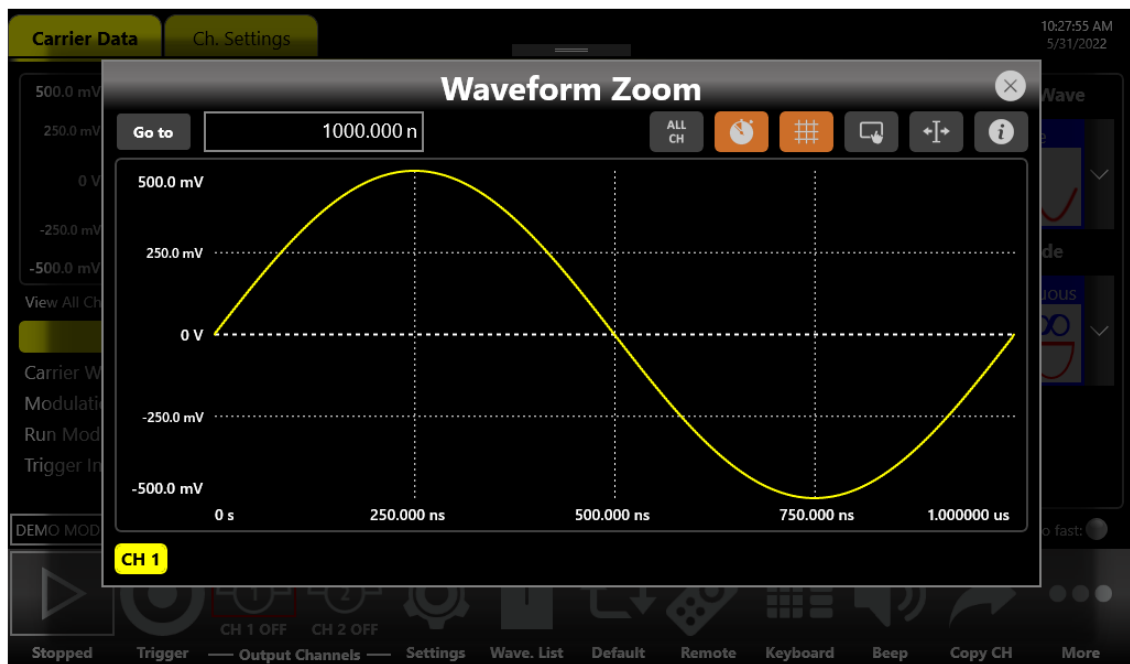


1. **Go To:** If you are in Samples Scale, you can insert in the textbox the sample that you want to go and the graph will move the actual window zoom to that sample, at same way if you are in Times Scale, but you must insert a time value. The range of the Samples Scale is 0 to 16384, while the Time Range depends of the values set on the Channel Data.
2. **ALL CH:** enable or disable the View All Channels.



You can decide to View only the channels that you want tapping on coloured buttons under the graph.

3. **Times Scale:** if enabled the Times Scale is visible, otherwise Samples Scale by default.
4. **View Grid:** if enabled, you can view further horizontal e vertical line with an others value of amplitude and Samples/Times.



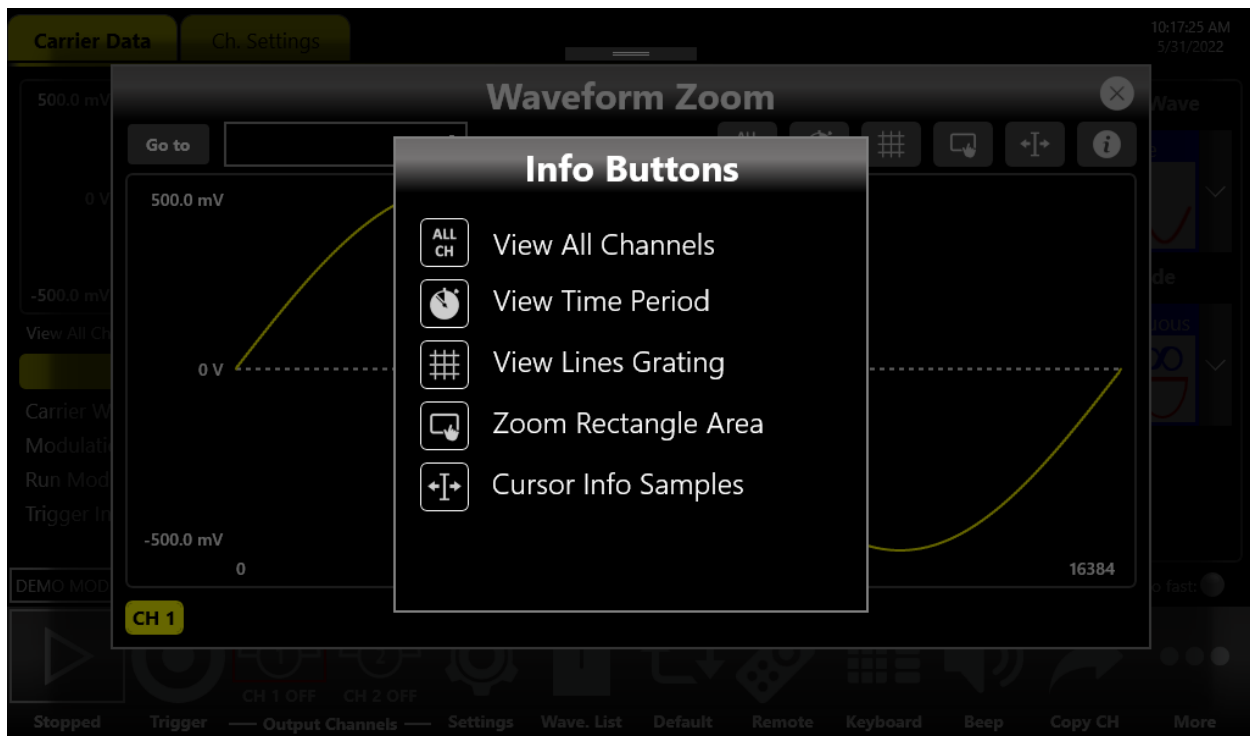
5. **Zoom Rect Area:** if enabled, the multitouch and the translation left/right will be disabled, you can holding down and then release to select an area to Zoom.



6. **Cursor:** enable this button if you want to see a cursor on the graph that visualize the values (Sample/Time and Amplitude) of the waveforms in the position set.



7. **Info Buttons:** tap to open a panel where they are indicated a lite description of the buttons just listed.



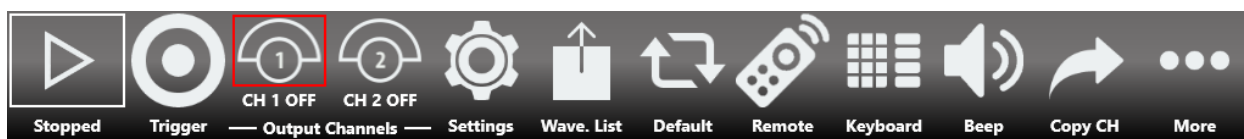
Channel Information

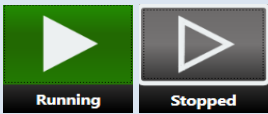
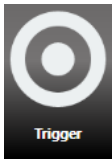
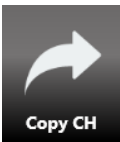
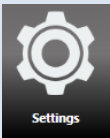
This area displays the channel name and a list of all the main current channel settings: the selected waveform type, the Modulation / Sweep / Burst mode, the Generation mode, the channel status and the Trigger Source.

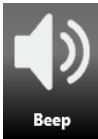
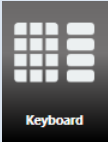
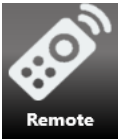
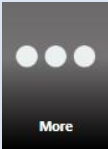
CHANNEL 1	
Carrier Wave:	SINE
Modulation Type:	None
Run Mode:	Continuous
Trigger In:	TRIGGER BUTTON

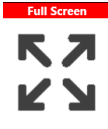
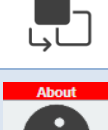
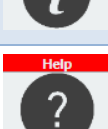
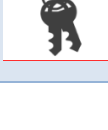
Command Toolbar Area

The command bar contains several touch buttons to control the instrument and its layout changes depending on the model (in the 4-channels models some buttons can be located in the More menu instead of in the Command Bar). Below a detailed description of these bars is provided.



Command Bar Buttons	Description
	Running/Stopped Button – Use this button to set the instrument in Running state (or Ready to receive a Trigger) or in the Stopped state. If the button is green the instrument is running while if it is grey the instrument is stopped. By pushing this button, the instrument state will change.
	Trigger Button – Use this button to send an internal software trigger to the instrument. Independently from the setting this trigger is always received.
	Output Channels Buttons - Press CH 1, CH 2, ..., CH 4 to change the Output Channel page. By pressing and keep pressed the Channel button for a programmable time (ON/OFF waiting time), you will turn it OFF/ON. The <i>ON/OFF waiting time</i> can be set in the Device Settings. When a channel is OFF, it is mechanically disconnected from the output (For more information, please refer to the relative paragraph).
	Copy Ch Button –This button copies all settings of the visualized channel to all other channels. When you press the button a dialogue window appears to Confirm or Cancel the operation. As example on four channel models, you can copy the channel 1 into channel 2, 3, 4 or the channel 2 into channel 1, 3, 4 depending on the current selected one. Note: this button is in the More menu on 4-channels models.
	Settings Button This button opens the output channel Settings and device Settings. (For more information, please refer to the relative section).

	Wave. List Button – This button opens the page where you can manage all imported and predefined waveforms of the current configuration. These waveforms can be used for Carrier Wave, Modulation Law or Sweep Profile. (For more information, please refer to the relative section).
	Default Button – This button restores the default value of all parameters of the instrument.
	Beep Button– This button enables or disables the beep audio signal when the user touches a button.
	Numeric Keyboard Button – This button enables or disables the virtual numeric keyboard.
	Remote Control Button – Use this button to open the SCPI server page. In that page you can enable or disable the SCPI server and see the sequence of commands sent to the instrument and its response.
	More Button –This button gives access to other instrument features. These buttons as explained in the following table.

More Button Menu Items	Description
	Exit Button – Press this button to close the application.
	Full/Float Button – press this button to maximize or reduce the application screen; in this way you can access to Windows OS.
	Load From Button – Use this button to load a configuration file. (For more information, please refer to the relative section).
	Save As Button – Use this button to Save the Current configuration into an existing one or create a new one. (For more information, please refer to the relative section).
	Export Button – Use this button to export the current configuration. (For more information, please refer to the relative section).
	Change Application Button – Use this button to switch from AFG to TrueArb or SPG application.
	About Button – Use this button to check the credits, the software and firmware release number and the instrument serial number.
	Help Button – Use this button to open the User Manual.
	Calibration Button – Use this button to enter the Calibration and Diagnostic page. (For more information, please refer to the relative section).
	Waveform Editor Button – Use this button to open the Waveform Editor software. (For more information, please refer to the Waveform Editor User Manual).
	License Button – Use this button to enter the License setup page. (For more information, please refer to the relative section).

Input / Output Channels

The Model 686 has 2 or 4 independent analog channels. Each channel can be single-ended or differential (depending on the model) and it is available on a SMA connector (or two SMA connectors for the differential output) located in the front instrument panel (**CH1 OUT** and **CH2 OUT** for 2-channels models and **CH1 OUT**, **CH2 OUT**, **CH3 OUT** and **CH4 OUT** for 4-channels models).

The Marker Out is a digital output signal available on a SMA connector located in the front instrument panel (**MARKER OUTPUTS: Out 1** and **Out 2** for 2-channels models and **MARKER OUTPUTS: Out 1**, **Out 2**, **Out 3** and **Out 4** for 4-channels models).

Each marker output signal is correlated to the corresponding analog output signal.

The Trigger In is an input signal available in a SMA connector located in the front instrument panel (**TRIGGER INPUTS: In 1** and **In 2** for 2-channels models and **TRIGGER INPUTS: In 1**, **In 2**, **In 3** and **In 4** for 4-channels models).

The SMA Input connectors for the External Modulation are also located on the front panel of the instrument (**MOD 1 IN** and **MOD 2 IN** for 2-channels models and **MOD 1 IN**, **MOD 2 IN**, **MOD 3 IN** and **MOD 4 IN** for 4-channels models).

Analog Output Channel

The term Output Channel, as used in this user manual, refers to the analog signal provided on the SMA output connector located in the front instrument panel.

For some parameters the Simple AFG application supports different input formats. For example, it is possible to specify the frequency of a waveform or alternatively it is possible to specify the corresponding time period. It is possible to switch between two different parameter formats by just touching on the label of the parameter.



Main Parameters

The set of available parameters depends on the selected waveform.

The Simple AFG application provides the control of the vertical (voltage) parameters of the Output channel in the format **Amplitude [Vpp] / Offset[V]** or in the format **Voltage High[V] / Voltage Low[V]** or in the format **Amplitude RMS[V] / Offset[V]** or in the format **Amplitude DBM[dBm] / Offset[V]**.

The output signal levels displayed by the Simple AFG UI text are calculated for the specified source and load impedances that by default are 50 Ohm. To change the expected load please refer to the Channel Settings.

Amplitude

The amplitude can be represented in three different formats that can be selected by opening the menu beside the amplitude label:

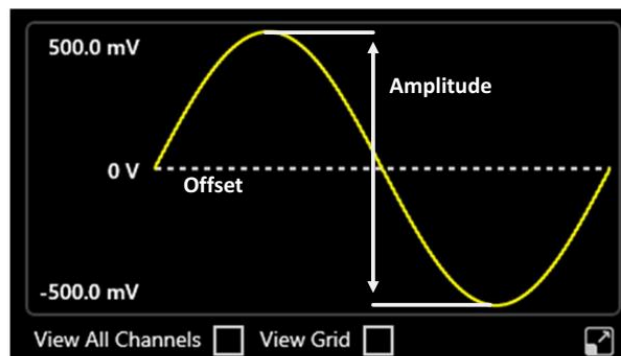
- **Vpp (Peak to Peak Voltage):** it is the difference between the highest and lowest level of the waveform.
- **Vrms (Root Mean Square Voltage):** it is the rms value of the waveform.
- **dBm:** it is the power transferred to the load expressed in dBmW. Its value takes into account the Load impedance parameter specified in Channel Settings.

Note: The Vrms and the dBm values set in the textbox are referred only to the waveform amplitude. They don't take into consideration the Offset of the waveform.

These parameters are available for all function excepts the DC level that is identified only by the Offset parameter.

Offset[V]

It defines the voltage of $(V_{max} + V_{min}) / 2$ expressed in Volts where V_{max} is the maximum level of the waveform and V_{min} is the minimum level of the waveform

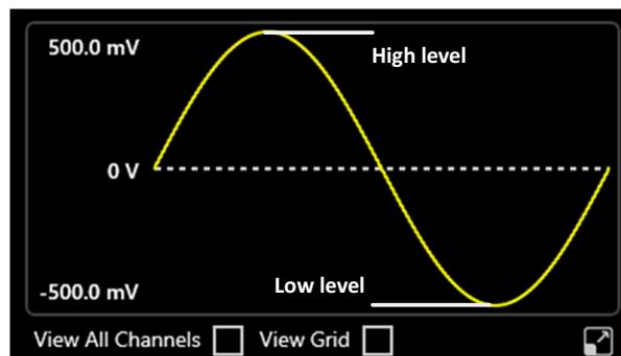


High Level [V]

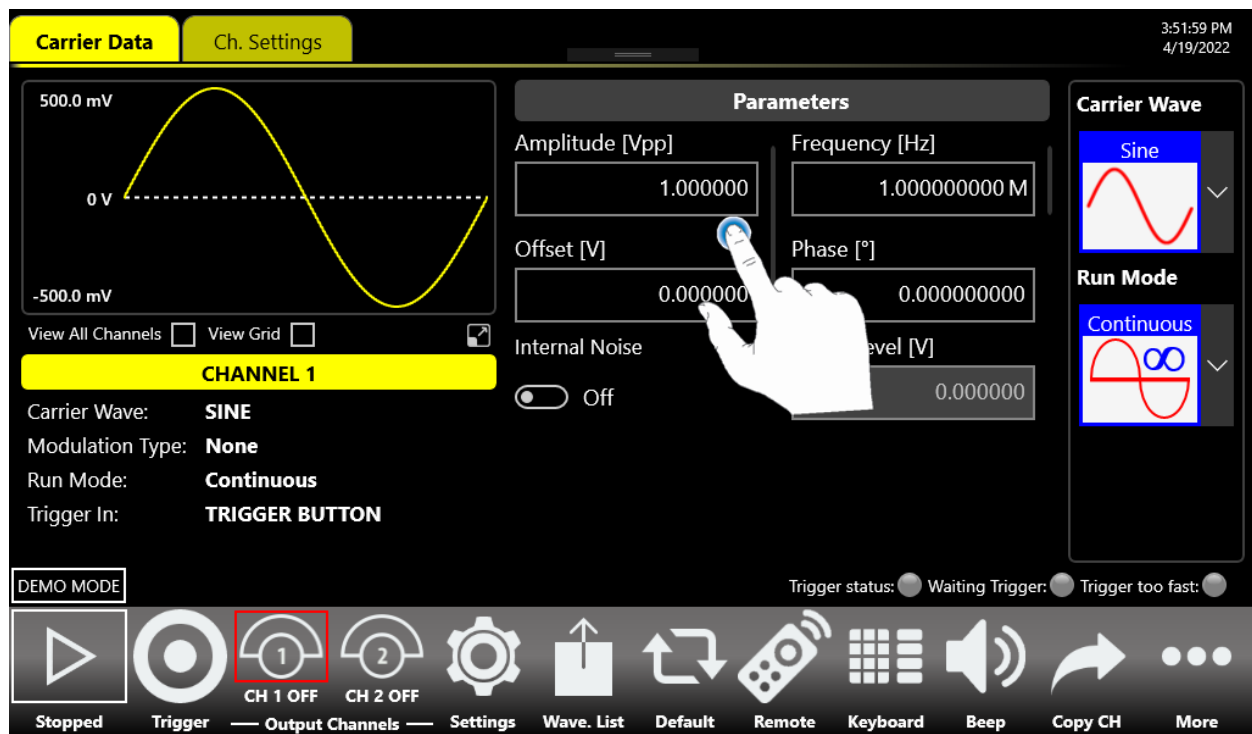
It defines the maximum level of the waveform expressed in Volts

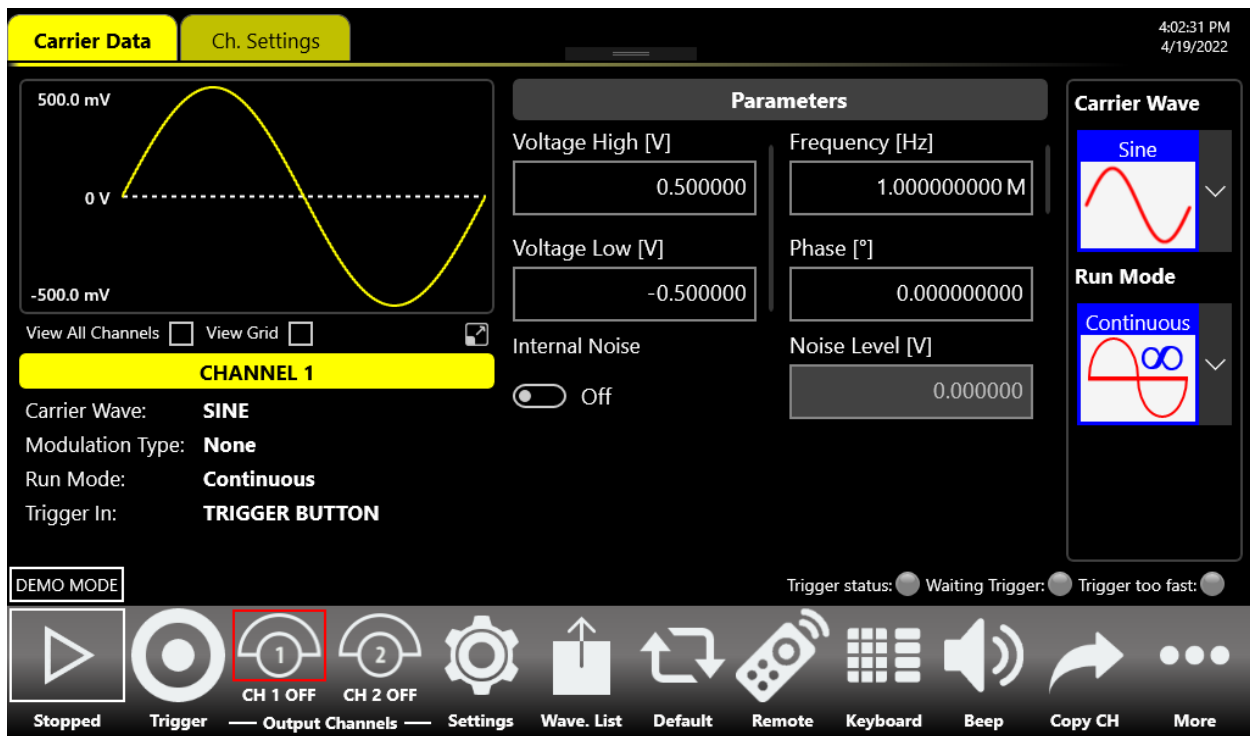
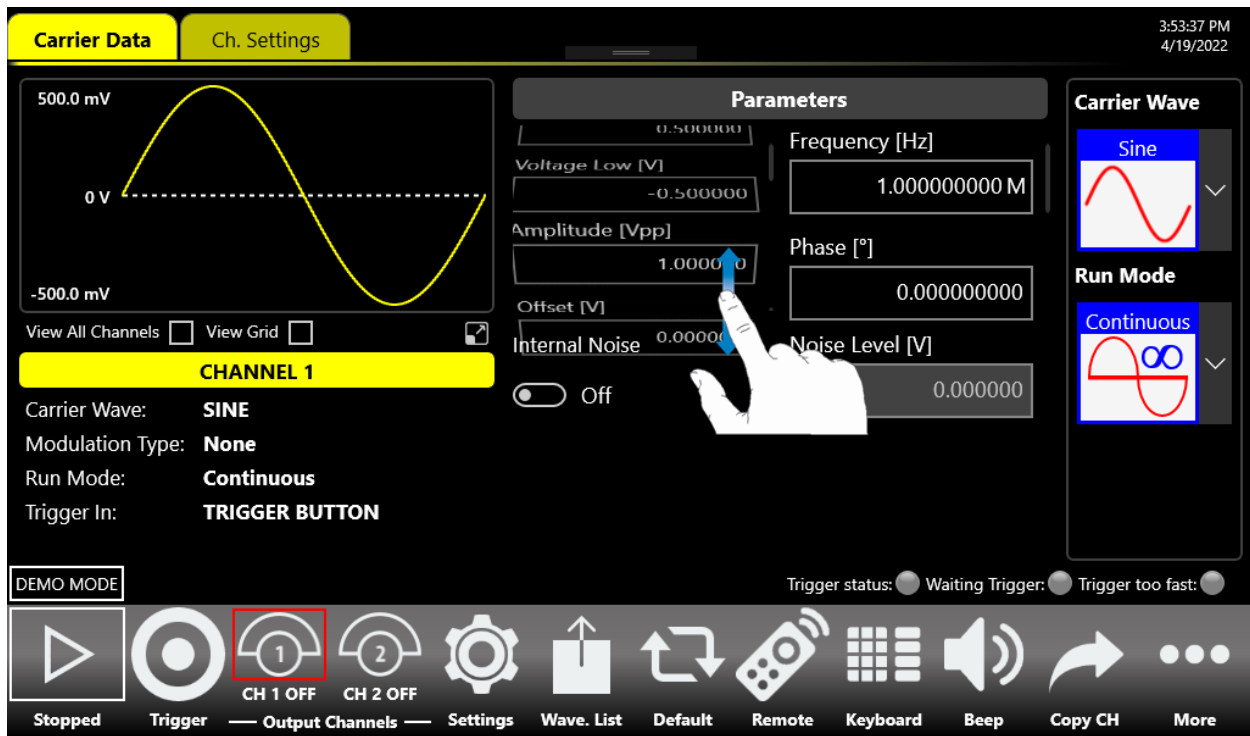
Low Level [V]

It defines the minimum level of the waveform expressed in Volts



The amplitude and the offset can be represented in four different formats, as listed before, that can be selected by rotate the 3D cube as below:





Frequency [Hz] / Period [s]

This parameter defines the frequency or the period of the generated waveform. This parameter is available for all the functions except DC Level and Noise.

In sweep run mode it is replaced by the Phase[deg] or Start Freq./Stop Freq. [Hz] parameters.

Rotating the parameter (in the same way as amplitude parameter) the format switches between Frequency and Period.

Phase [deg]

It controls the initial phase of the waveform. This control is available for all function except DC Level and Noise.

Noise Level [V]

Use this setting to add noise to the output signal. The voltage shown in the textbox defines the peak voltage of the noise level. The range is 0V to 2.5 Vpk (i.e. 0V to 5Vpp) for single-ended models and 0V to 500 mVpk for differential models. The Noise is generated by hardware using a pseudo random algorithm.

Note: When you set the Noise amplitude please pay attention that Signal + Noise does not exceed the amplitude of 5Vpp (single-ended models) or 1Vpp (differential models).

Internal Noise

Use this control to enable or disable the noise added to the output signal.

Symmetry [%]

This parameter is defined only for the Ramp function. It represents the percentage of the cycle in which the ramp function is rising.

Width [s] / Duty Cycle [%]

It defines the duration of the High-level part of the Pulse function. The width is defined as Full Width at Half Maximum (FWHM) that means the time from the medium of the rising edge to the medium of the falling edge. The duty cycle is the percentage value of the width compared to the period.

Rotating the parameter (in the same way as amplitude parameter), you can change the format between the Width (absolute) and Duty Cycle.

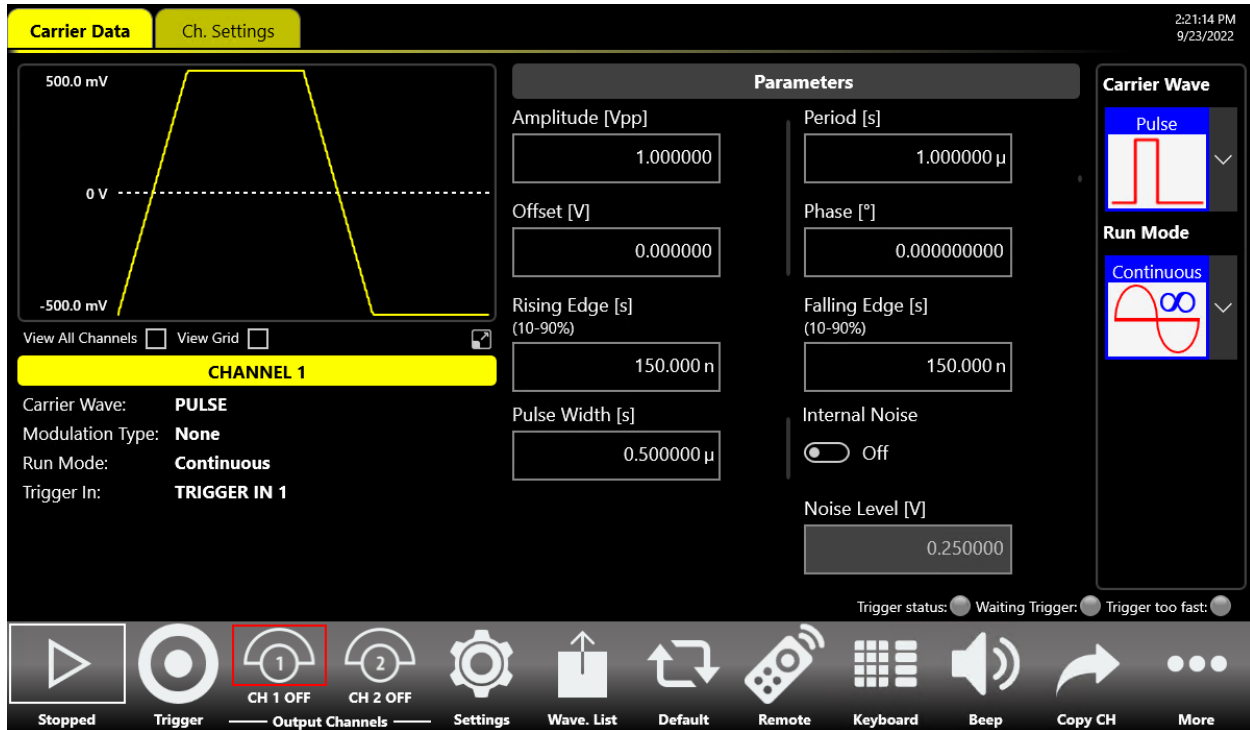
Rising – Falling Edge [s]

In the Pulse function, it defines the transition time between Low level and High level for the Rising (or Leading) Edge and between High level and Low level for the Falling (or Trailing) Edge.

Note: the values refer to the rise and fall time between the 10% and the 90% of the pulse amplitude. The 0% to 100% transitions will be longer than the set values anyway the graph represents the entered value as the 0% to 100% transition time.

Note: Using the Pulse Waveform the following constraints must be met:

- $Rising\ Edge + Falling\ Edge < Period$
- $Pulse\ Width < Period$



Auxiliary Channels

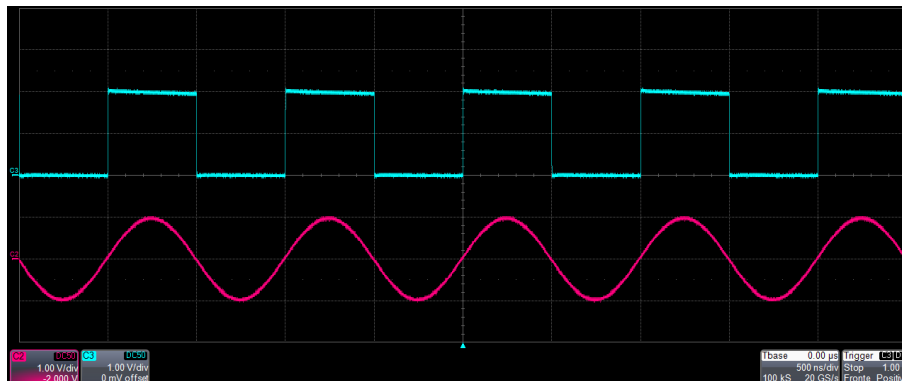
Marker Outputs

The Marker Out generates a digital pulse synchronous with the waveform or with the modulating function depending on the Run Mode.

Its impedance is 50 Ohm and the output voltage amplitude ranges from -0.5 V to 1.65 V into 50 Ohm load.

To set the Marker Out parameters refer to the Channel Settings paragraph.

Marker Out Specification	Value
Connector	1 SMA per channel on the Front Panel
Output impedance	50 Ω
Output level (into 50 Ω)	-0.5 V to 1.65 V



Marker out (blue, top) synchronous with the analog signal (red, bottom)

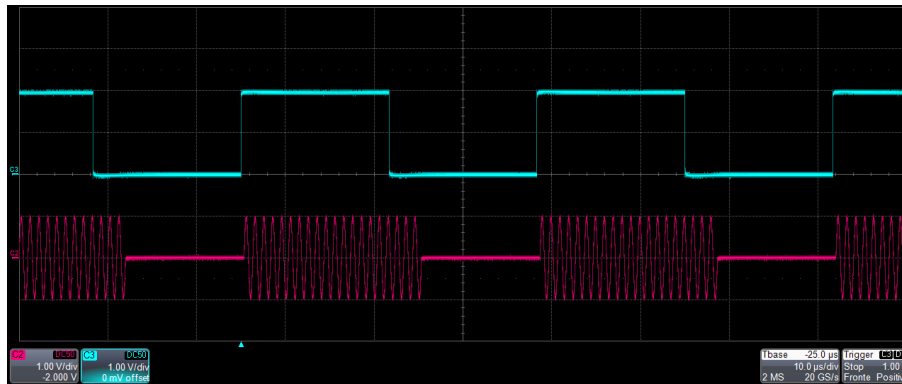
Trigger In

The Trigger In signals (SMA connectors located on the front panel) allow to control the signal generation when a channel is in Burst Run Mode or in Sweep Run Mode.

Refer to the Device Settings paragraph to know how to define the trigger parameters or the Run Mode.

In Continuous and Modulation Run Modes the Trigger In signals have no effect.

Trigger In Specification	Value
Connector	SMA on the Front Panel
Number of connectors	2 in 2-channels models or 4 in 4-channels models
Input impedance	1 k Ω or 50 Ohm selectable
Slope/Edge	Positive or negative selectable



Trigger In signal (blue, top) that starts a burst of sine waveform (red, bottom)

External Modulation Input Connector

The Model 686 instrument series has two or four input connectors to receive an external analog signal that is used as modulating source. When the selected Run Mode is “Modulation” and the source is “External” the instrument will use this signal to modulate the specified carrier waveform. The connector type is a SMA.

External Modulating Input	Description
Connector	SMA on the Front Panel
Number of connectors	2 in 2-channels models or 4 in 4-channels models
Input Impedance	50 Ohm
Input Voltage Range	± 0.5 V for all modulations

Reference Clock Input Connector

The Model 686 series instrument can use an external clock source to generate the sampling clock frequency.

The connector type is a SMA.

Reference Clock Input	Value
Connector type	1 SMA on the Rear Panel
Input impedance	50 Ohm, AC coupled
Input Frequency range	5 MHz to 500 MHz

Reference Clock Output Connector

This connector outputs the internal 10MHz (100MHz Optional) reference clock used to synthesize the DAC sampling clock. If the clock source is set as internal the reference clock output provides a 10 MHz (100 MHz Optional) clock signal otherwise it is disabled.

The connector type is a SMA.

Reference Clock Output	Value
Connector	1 SMA on the Rear Panel
Output impedance	50Ω, AC coupled
Output Frequency	10MHz (100 MHz optional)

External Clock Input Connector

This connector input gives to the User the possibility to feed directly a sampling clock for the System.

This Clock bypasses the internal Clock Synthesizer of the instrument.

The connector type is a SMA.

External Clock Input	Value
Connector type	1 SMA on the Rear Panel
Input impedance	50 Ohm, AC coupled
Input Frequency range	321.5 MHz, 625 MHz, 1.25 GHz or 2.5 GHz (selectable)

Sync Clock Output Connector

This connector outputs a divided clock generated from sampling clock of the instrument. The resulting frequency can be selected by the User from a list of all possible values.

The connector type is a SMA.

Sync Clock Output	Value
Connector type	1 SMA on the Rear Panel
Input impedance	50 Ohm, AC coupled
Input Frequency range	20Ghz / M where $M = 40 \cdot 2^N$ and $N = 0, 1, 2, \dots, 7$

Digital Output connector

Important Note: this connector is not used by AFG application.

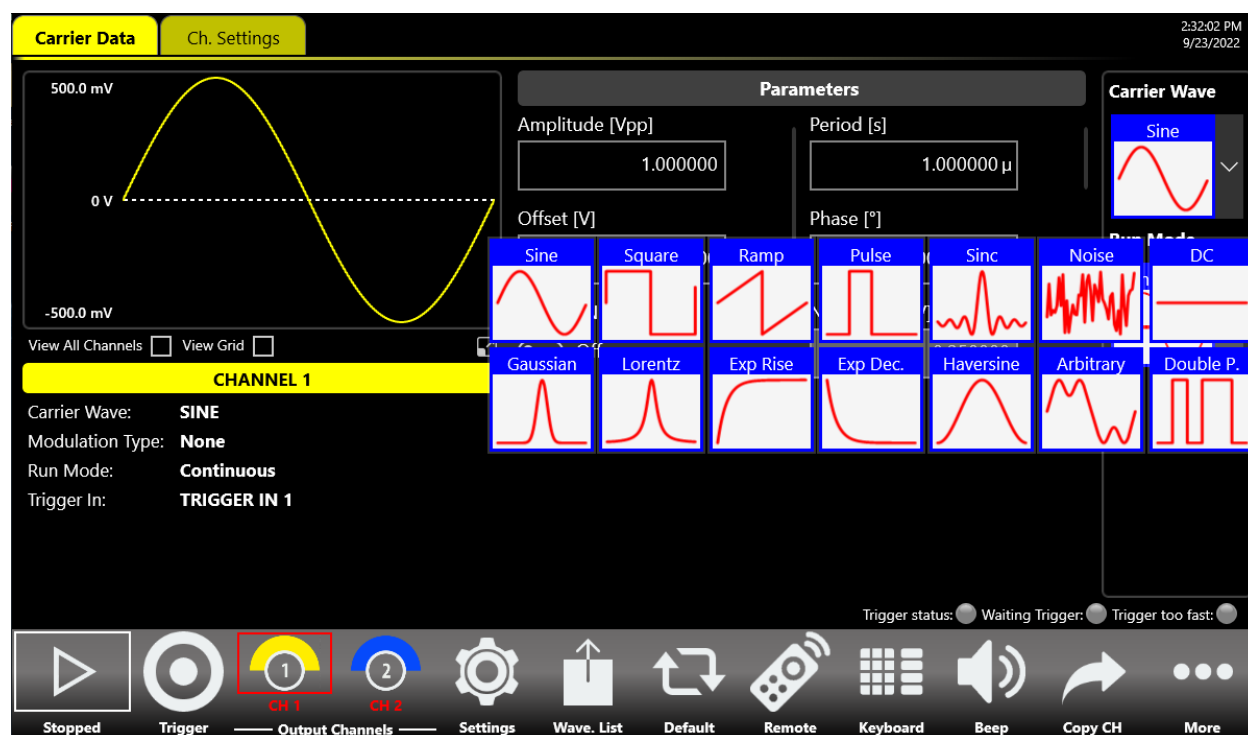
Sync In / Sync Out Connectors

The purpose of those connectors is to connect and synchronize together multiple instruments: up to 4 instruments can be linked together.

Predefined Waveforms

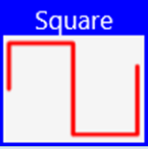
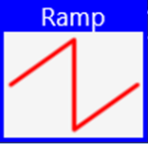
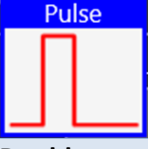
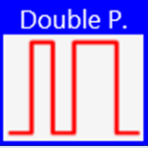
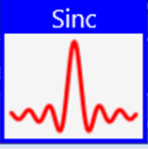
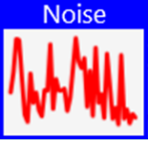
The Simple AFG application provides 13 predefined functions each of them described by its own set of parameters. It is also available the **Arbitrary** waveform that allows to load a waveform from a file or from remote.

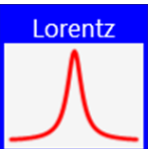
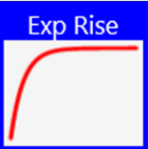
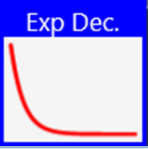
Touching the “Carrier Wave” button a dropdown menu opens where it is possible to select the waveform that is used as carrier waveform.



List and parameters of predefined waveforms

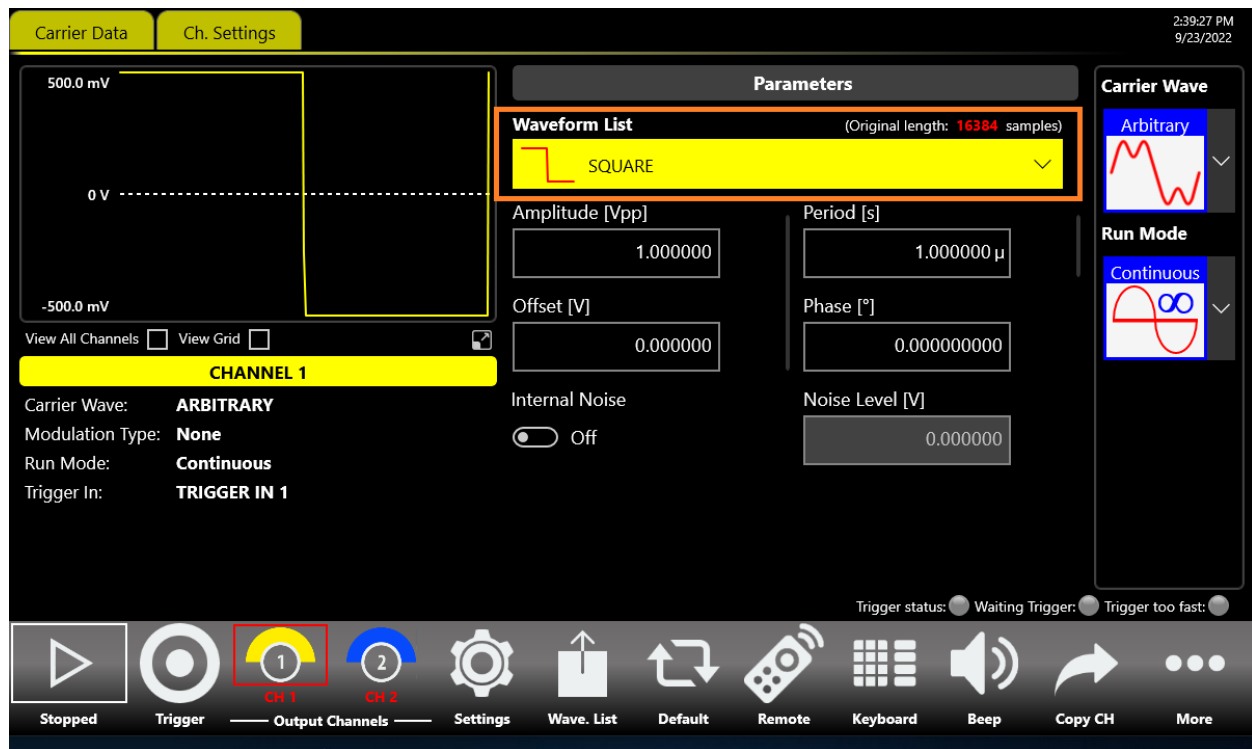
The following table shows the available waveforms, the parameters that define each of them and the possible combination of run mode and waveforms. The Continuous Run Mode has been omitted in the table because it is available for all waveforms.

Waveform	Parameters	AM, FM, PM, PSK, FSK	PWM	Sweep	Burst
Sine 	Amplitude, Offset, Frequency, Phase	✓		✓	✓
Square 	Amplitude, Offset, Frequency, Phase	Only AM		✓	✓
Ramp 	Amplitude, Offset, Frequency, Phase, Symmetry	✓		✓	✓
Pulse 	Amplitude, Offset, Frequency, Phase, Duty Cycle, Rising Edge, Falling Edge		✓		✓
Double Pulse 	Common: Offset, Frequency, Phase. Pulse 1 / Pulse 2: Amplitude P1/2, Offset P1/2, Rising Edge P1/2, Falling Edge P1/2, Width P1/2, Delay P1/2 Note: Delay2 is a delta delay relative to the end of the first pulse				✓
Sinc 	Amplitude, Offset, Frequency, Phase	✓		✓	✓
Noise 	Noise level (Amplitude), Offset				✓

Waveform	Parameters	AM, FM, PM, PSK, FSK	PWM	Sweep	Burst
DC Level 	Offset				
Gaussian 	Amplitude, Offset, Frequency, Phase	✓		✓	✓
Lorentz 	Amplitude, Offset, Frequency, Phase	✓		✓	✓
Exponential Rise 	Amplitude, Offset, Frequency, Phase	✓		✓	✓
Exponential Decrease 	Amplitude, Offset, Frequency, Phase	✓		✓	✓
Haversine 	Amplitude, Offset, Frequency, Phase	✓		✓	✓
Arbitrary 	Amplitude, Offset, Frequency, Phase	✓		✓	✓

Note: Consult the instrument datasheet to find out the specifications of frequency range for each waveform.

You can assign the “**Arbitrary**” waveform using the combo box highlighted in figure below (refer to “Waveform List” description paragraph):



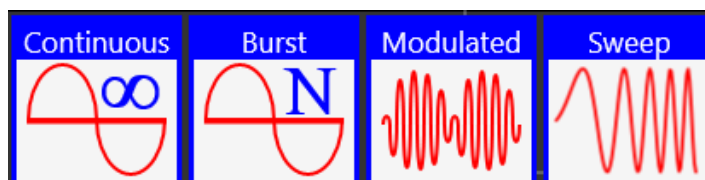
Note: When Arbitrary waveform is selected, the amplitude and offset of the original waveform are lost because the waveform is normalized. Anyway, the amplitude and offset of the normalized waveform can be modified as for any predefined waveforms.

Note: For Single-Ended models, in Stopped State, the instrument stops at the Baseline Offset of the selected channel for all the waveforms except for:

- Pulse Waveform – Stopped State: Low Level[V] if the polarity is positive or High Level[V] if the polarity is negative.
- Double Pulse Waveform – The stopped state is the Offset[V] parameter.

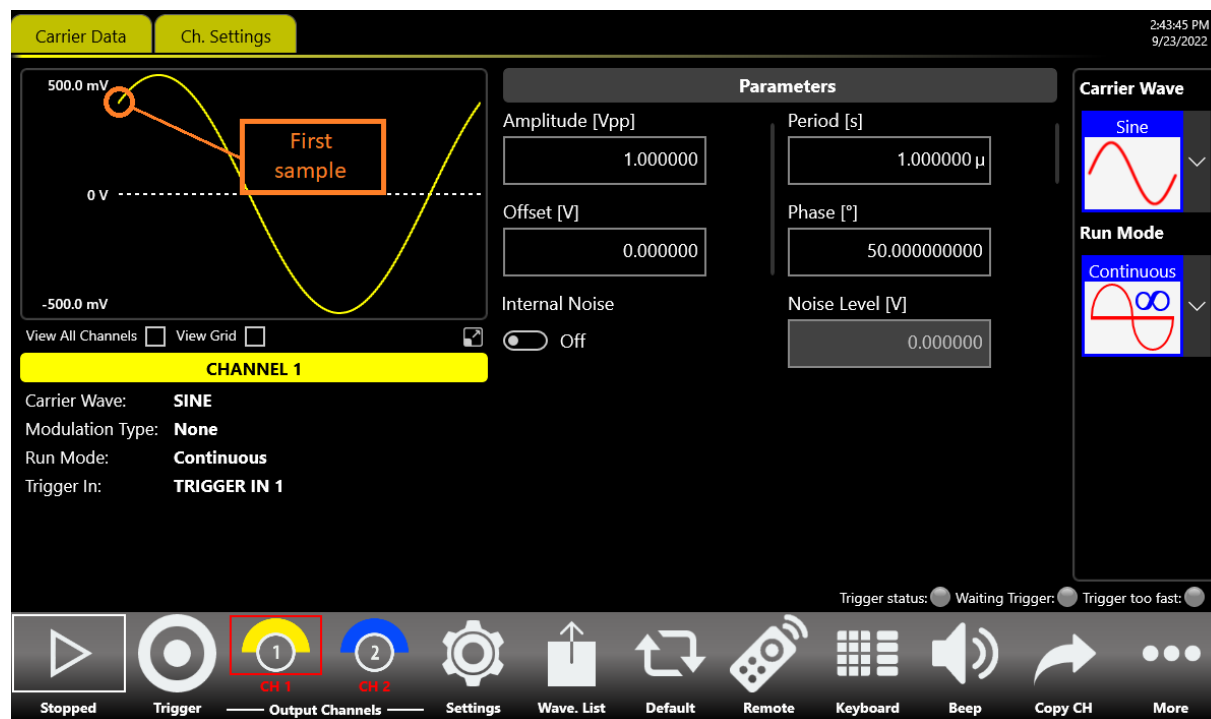
Run Mode

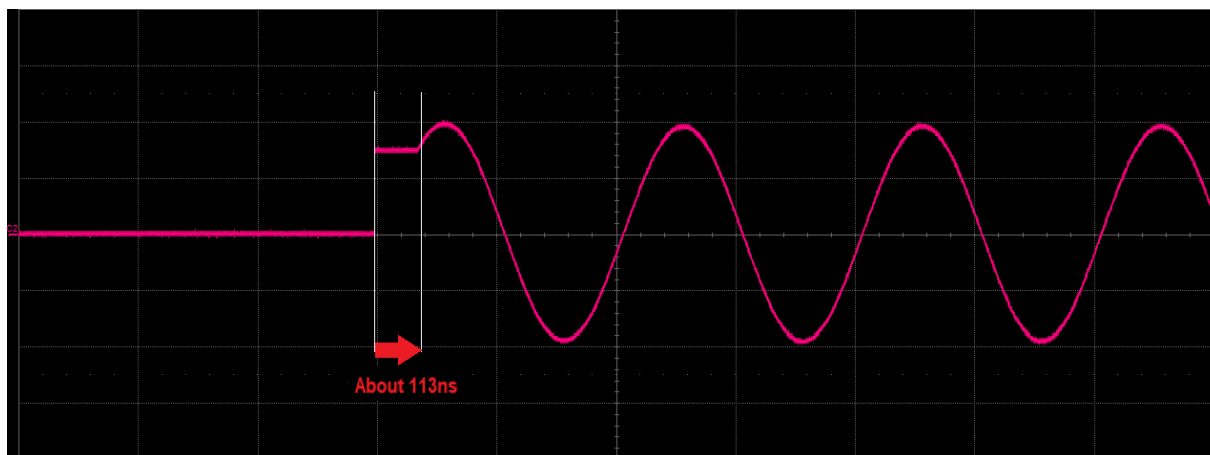
On the Carrier tab pressing the “Run Mode” button a menu opens showing all possible choices for the Run Mode. If “Modulation”, “Sweep” or “Burst” is selected the software moves directly on the secondary tab that takes the name of the selected Run Mode.



Note: after enabling a channel and pressing the Running button (see Command Tool Area section), the generation of the corresponding waveform will start keeping the first sample of the waveform constant for about 113ns.

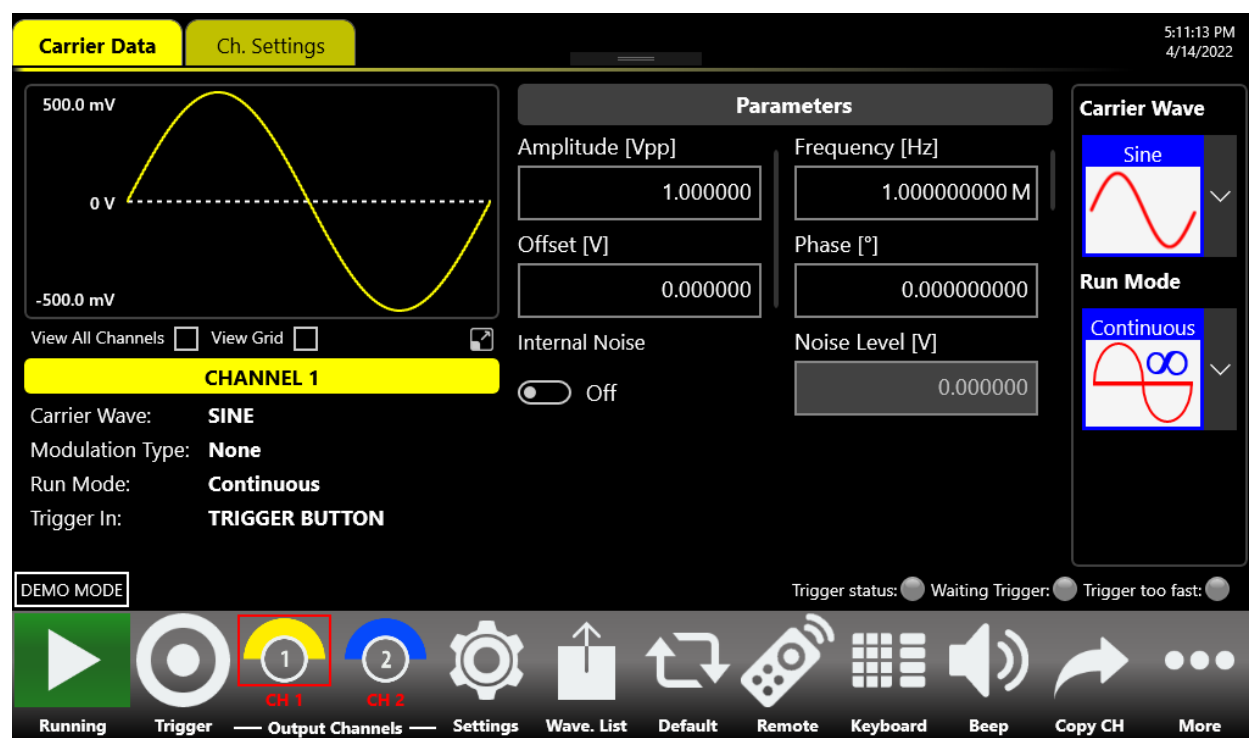
For example, in the picture below is represented the initial part of the generation of a sine wave with 50 deg of phase in continuous mode.





Continuous

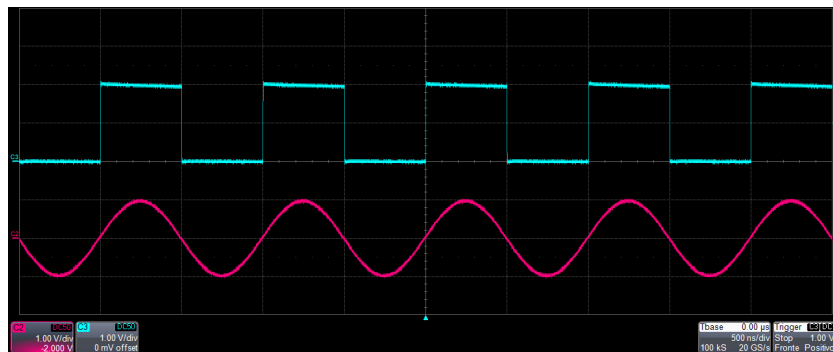
In the Continuous mode when the Run/Stop button is pressed the waveform is reproduced continuously until the Run/Stop button is pressed again or Waveform / Run Mode is changed.



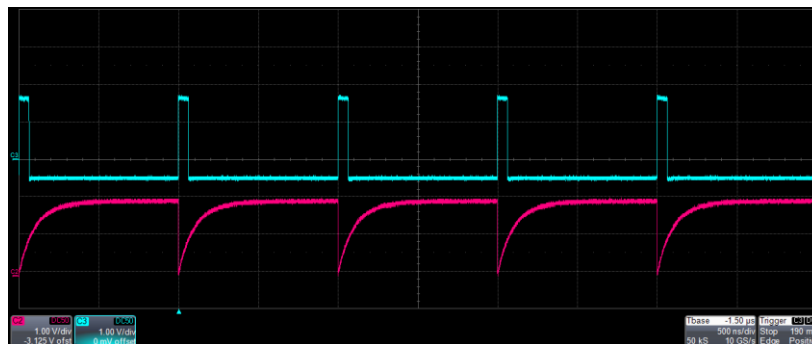
Marker Out behaviour in Continuous Run Mode

In Continuous mode, the **Marker Out** generates a pulse with a duty cycle of about 50% (Automatic width mode) or a User predefined width (Manual width mode) at the beginning of each period. The Marker have the same frequency of the carrier waveform until it is below 125MHz. Over 125MHz the Marker Out frequency is divided by an integer value as described in the following:

- Marker out frequency = Carrier frequency / 1 when: Carrier frequency < 125 MHz
- Marker out frequency = Carrier frequency / 2 when: 125 MHz ≤ Carrier frequency < 250 MHz
- Marker out frequency = Carrier frequency / 4 when: 250 MHz ≤ Carrier frequency < 500 MHz
- Marker out frequency = Carrier frequency / 8 when: 500 MHz ≤ Carrier frequency < 1 GHz
- Marker out frequency = Carrier frequency / 16 when: 1 GHz ≤ Carrier frequency < 2 GHz
- Marker out frequency = Carrier frequency / 32 when: 2 GHz ≤ Carrier frequency < 4 GHz
- Marker out frequency = Carrier frequency / 64 when: 4 GHz ≤ Carrier frequency ≤ 6.5 GHz



Marker out (blue, top) synchronous with the analog signal (red, bottom) in Automatic width mode



Marker out (blue, top) synchronous with the analog signal (red, bottom) in Manual width mode

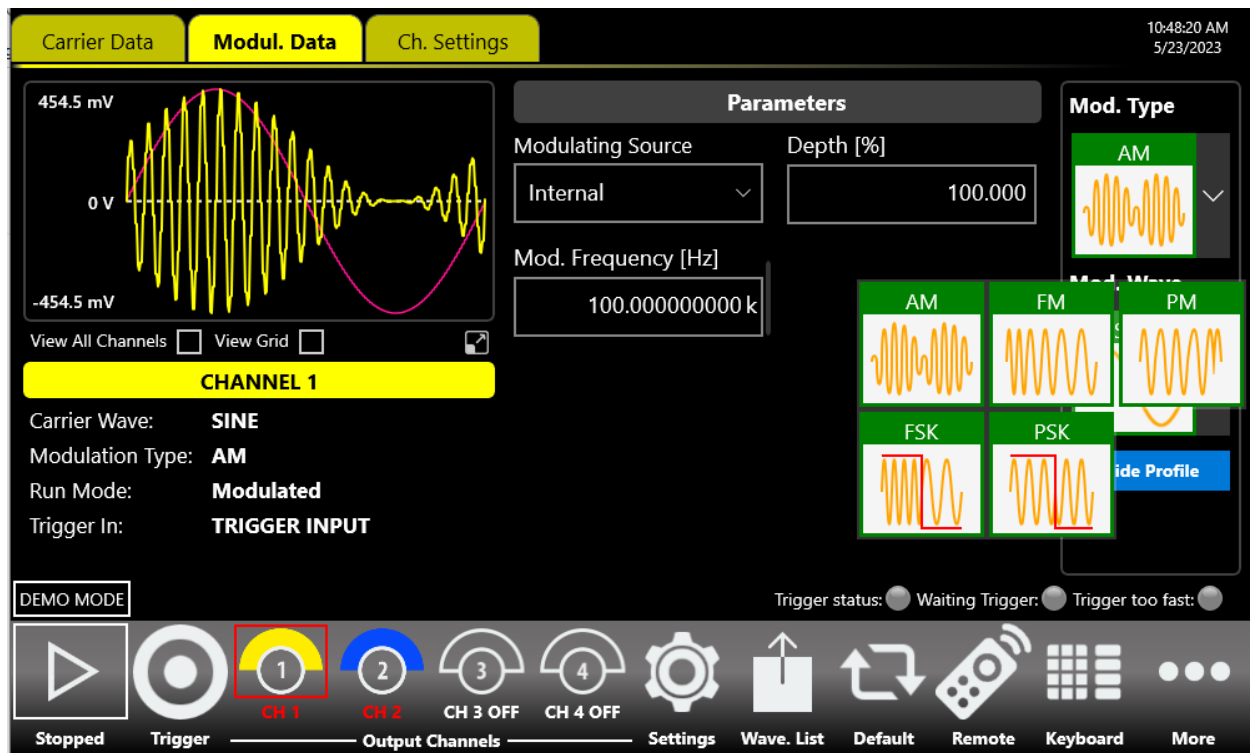
Note: for the Noise and DC waveforms, the **Marker Out** is unavailable in continuous mode.

Modulation

In this Run Mode, it is possible to modulate a carrier waveform with a modulation law that can be another waveform or an external signal. All waveforms except Noise, DC level and Double Pulse support the Modulation Run Mode.

The AM modulation is available only for the Square waveform.

Touching the “Mod. Type” button the modulation type menu opens as shown in the picture below.



With the button **Hide Profile** you can view and hide the waveform modulation profile on the graph

Hide Profile

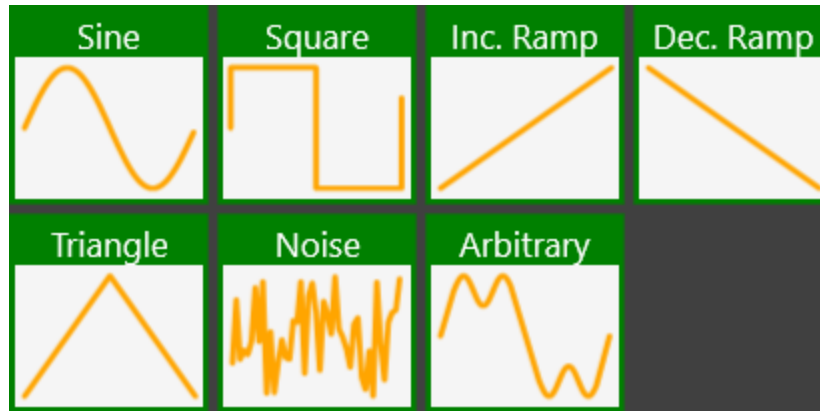
The modulation types are:

- **Amplitude Modulation (AM)**
- **Frequency Modulation (FM)**
- **Phase Modulation (PM)**
- **Frequency Shift Keying (FSK)**
- **Phase Shift Keying (PSK)**
- **Pulse Width Modulation (PWM)**

The PWM modulation is the only modulation supported by the Pulse waveform.

The types of modulation are explained in detail in the following sections of this chapter.

Touching the “Mod.Wave” button the modulation type menu opens. The possibility to choose the shape is only available for AM, FM, PM and PWM modulations.



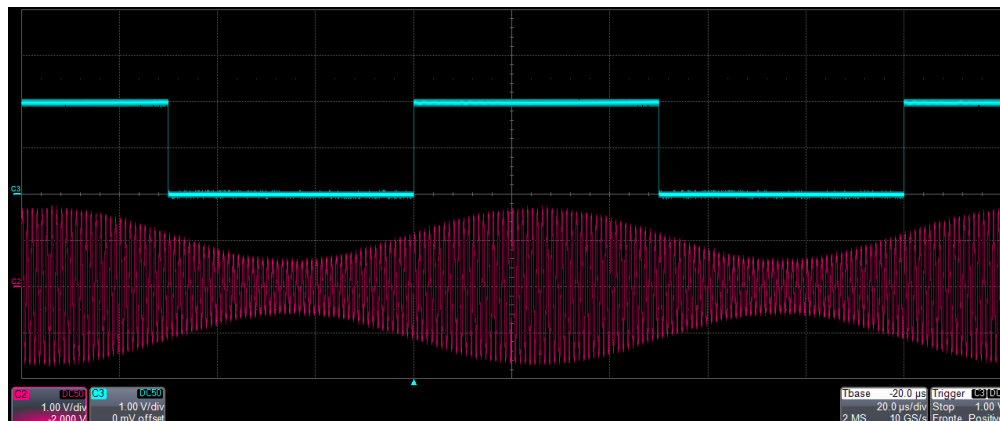
The modulation **Shapes** (when available) can be:

- **Sine**
- **Square**
- **Triangle**
- **Increase Ramp**
- **Decrease Ramp**
- **Noise**
- **Arbitrary**: it allows to load a Modulating Waveform from a waveform present in the Waveform List. By default, the “Arbitrary” waveform is a sine function. As Carrier Arbitrary you can select the waveform by combo box Waveform List.

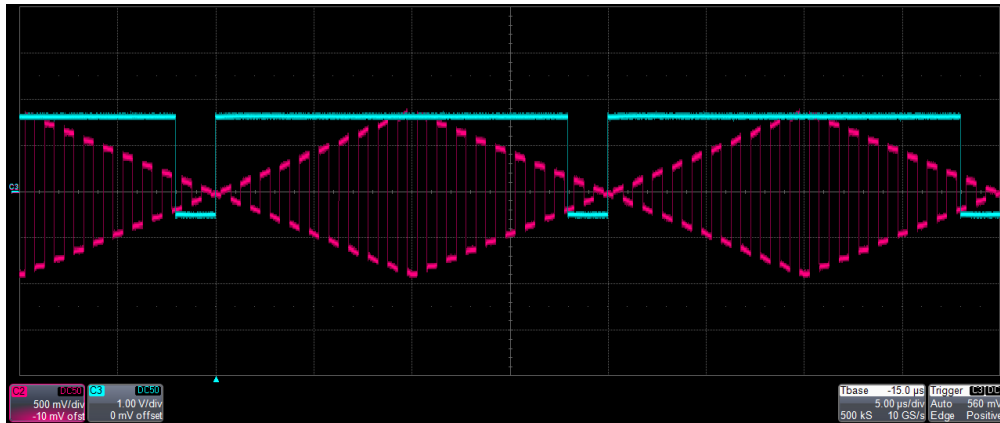
Marker Out behaviour in Modulation Run Mode

If the modulating source is **Internal**, the **Marker Out** generates a square wave synchronous with the modulating waveform. The leading edge of this pulse is positioned at the beginning of the modulating waveform.

If the modulating source is **External**, the **Marker Out** is disabled.



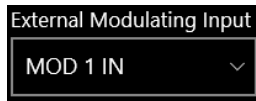
Marker out (blue, top) synchronous with the modulating waveform of the AM (red, bottom) in Automatic width mode



Marker out (blue, top) synchronous with the modulating waveform of the AM (red, bottom) in Manual width mode

Modulation General Parameter

- **Mod. Frequency [Hz]:** it defines the modulating frequency. It can vary between 500uHz and 80MHz.
- **Mod. Source:** the source can be **Internal** or **External**.
If **Internal** is selected it is possible to select a modulation type.
If **External** is selected the instrument accepts a modulating signal from one of the available **MOD IN** SMA input connector. Use the drop-down menu of **External Modulating Input** parameter to choose the right connector.



Modulation Types and associated parameters

- **Amplitude Modulation (AM):** the amplitude of the carrier waveform is modulated following the modulating wave shape. It is available for all functions except for the Pulse, Double Pulse, DC level and Noise.
The parameter **Depth [%]** controls the modulation depth between 0% and 120%.
- **Frequency Modulation (FM):** the frequency of the carrier waveform is modulated following the modulating shape. It is available for all functions except for the Pulse, Double Pulse, Square, DC level and Noise.
The parameter **Freq. Deviation [Hz]** defines the frequency deviation with respect to the carrier frequency. The Deviation is between 0 Hz and the maximum frequency that satisfies the following 2 relationships:
 - $\text{Carrier Frequency} - \text{Deviation} > 0 \text{ Hz}$
 - $\text{Carrier Frequency} + \text{Deviation} \leq \text{Maximum Carrier Frequency}$
 Where the Maximum Frequency depends on the selected carrier.

For example, for a sine function at 200 MHz the Deviation must be below 200MHz.

- **Phase Modulation (PM):** the phase of the carrier waveform is modulated following the modulating shape. It is available for all functions except for the Pulse, Double Pulse, Square, DC level and Noise.

The parameter **Phase Deviation [deg]** specifies the maximum phase deviation of the carrier waveform. It can vary in the range 0 to 360 degrees.

- **Frequency Shift Keying (FSK):** this modulation is a 2 level FSK. The carrier frequency switches between the initial carrier frequency and the initial carrier frequency + **Hop Frequency [Hz]**. It is available for all functions except for the Pulse, Double Pulse, Square, DC level and Noise.

Note: the maximum Hop Frequency depends on the selected carrier frequency and it must satisfy the following conditions:

- $Carrier\ Frequency + Hop\ Frequency \leq Maximum\ Carrier\ Frequency$
Where the Maximum Frequency depends on the selected carrier.

- **Phase Shift Keying (PSK):** this modulation is a 2 level PSK. The carrier phase is shifted by the value of **Hop Phase [deg]**. It is available for all functions except for the Pulse, Double Pulse, Square, DC level and Noise.

The hop phase is between 0 and 360 deg.

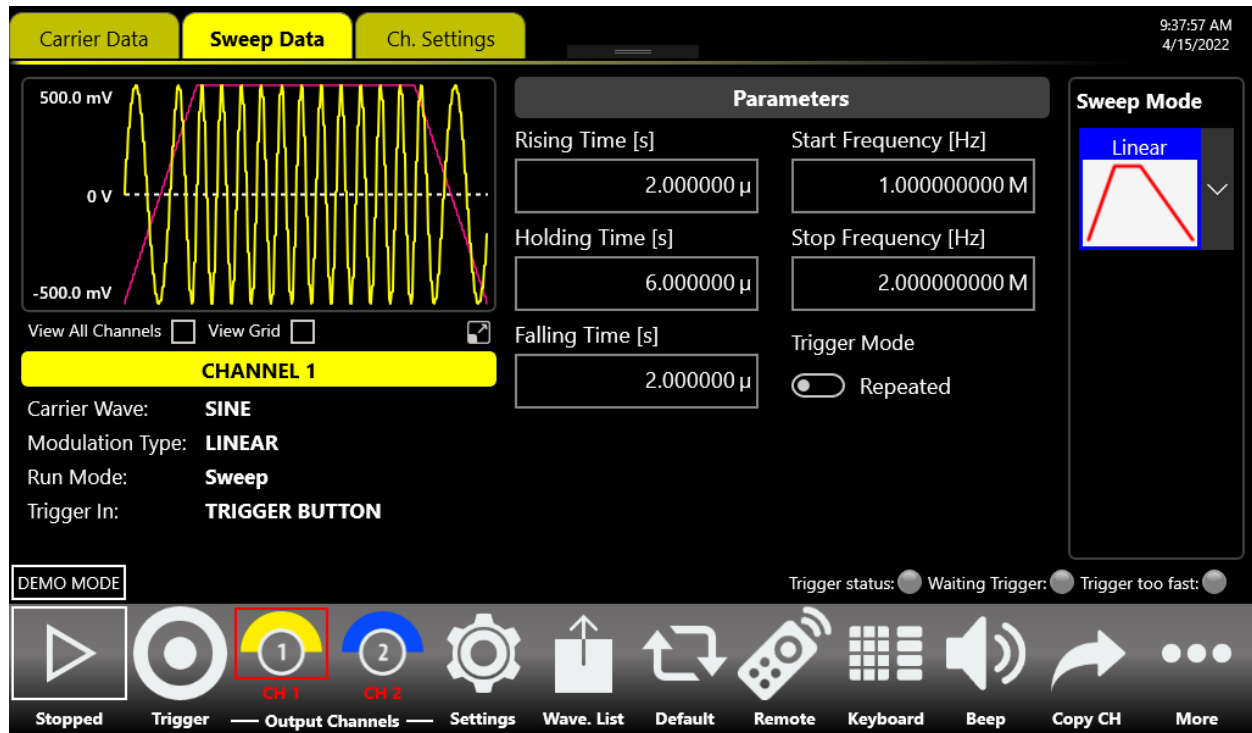
- **Pulse Width Modulation (PWM):** this modulation is available only for the Pulse waveform. It modulates the width of the Pulse of the quantity defined in the **PWM Deviation [%]** parameter that defines the maximum increase and decrease of the Duty Cycle in percentage. This deviation has to meet the following conditions:

- $PWM\ deviation\ [\%] \leq duty\ cycle\ [\%] - \frac{Rising\ Edge[s]}{0.8} * \frac{100[\%]}{Period\ [s]} - \epsilon$
- $PWM\ deviation\ [\%] \leq 100[\%] - duty\ cycle\ [\%] - \frac{Falling\ Edge[s]}{0.8} * \frac{100[\%]}{Period\ [s]} - \epsilon$

Where ϵ is a small margin to avoid that the duty cycle reaches the 0 % or the 100 % and the factor "0.8" takes into account that the edges are defined as 10-90%.

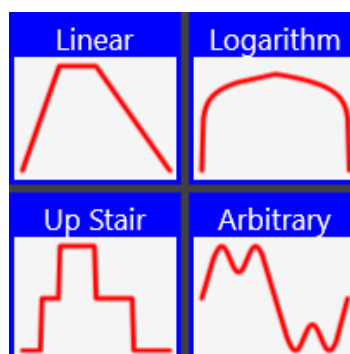
Sweep

The Sweep mode modulates the waveform frequency following a law that can be **Linear**, **Logarithm**, **Upstair** or **Arbitrary** (or User Defined). The **Arbitrary** selection gives the possibility to load the sweep profile from a waveform present in the Waveform List. The Sweep is available for all function except for Pulse, Double Pulse, Noise and DC level.



Sweep Mode

Touching the Sweep Mode button, a menu opens that gives the possibility to choose a sweep profile among the following:



- **Linear:** the frequency increases and decreases linearly.
- **Logarithm:** the frequency increases and decreases following a logarithmic function.
- **Upstair:** the frequency varies step by step. The number of steps is selectable through the **Step Number [N]** parameter.
- **Arbitrary:** allows to load a sweep profile from a waveform present in the Waveform List. By default, the “Arbitrary” waveform is a sine function. As Carrier Arbitrary you can select the waveform by combo box Waveform List.

Parameters

- **Rising Time [s]:** controls the time to increase the frequency from the Start Frequency up to the Stop Frequency.
- **Holding Time [s]:** defines the time that the frequency keeps the Stop Frequency.
- **Falling Time [s]:** controls the time to decrease the frequency from the Stop Frequency back to the Start Frequency.

Note: the time parameters must meet the following conditions:

$$Rising\ Time + Holding\ Time + Falling\ Time \leq 2000\ s$$

- **Step Number:** selects the number of frequency steps of the Upstair Sweep mode.
- **Start Frequency [Hz]:** selects the initial sweep frequency.
- **Stop Frequency [Hz]:** selects the final sweep frequency.

Note: the time to execute the sweep is the sum of Rising, Holding and Falling Time. When the Arbitrary mode is selected these 3 parameters are not meaningful and the time to execute corresponds to **Sweep Period [s]** value.

Sweep Period [s]

1.000 μ

Sweep Frequency [Hz]

1.000000000000 M

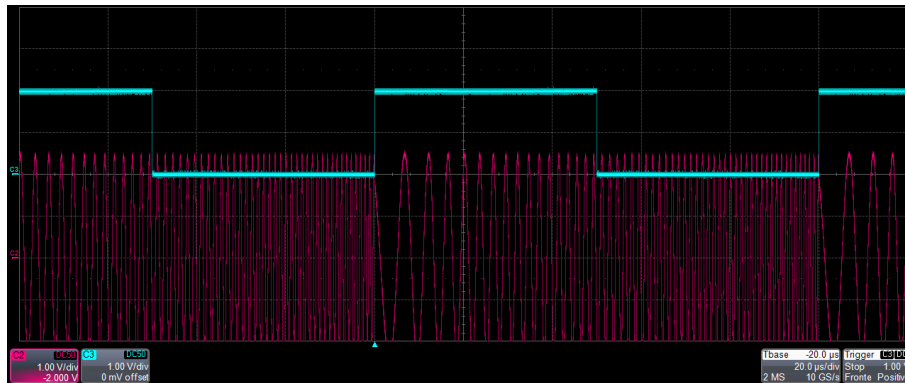
Sweep Trigger Mode

- **Repeat:** the instrument starts when the Run/Stop button is pressed and repeats the sweep continuously. This mode influences the maximum value of the Marker Skew parameter which will be 100s.
- **Triggered:** When the Run/Stop button is pressed the instrument waits for a Trigger signal. When the trigger is detected, it generates the sweep profile then stops waiting for a new Trigger. During the wait for a Trigger the instrument continues to generate the waveform with its frequency equal to the start frequency.

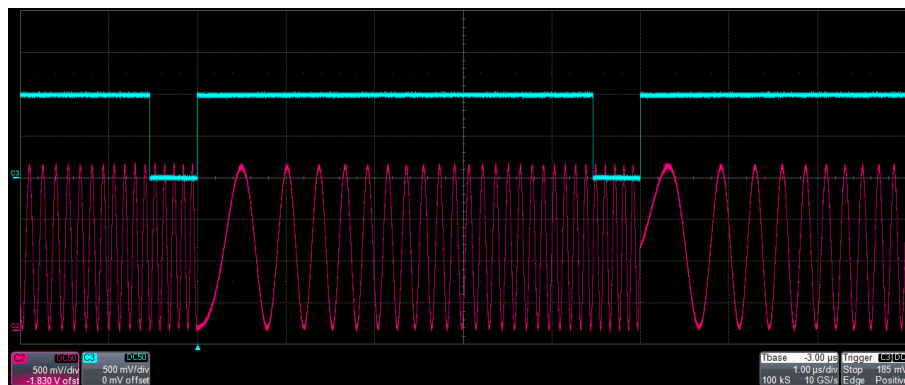
In this mode only automatic mode is available for marker's width parameter and the maximum Marker skew is 1.8us.

Marker Out behaviour in Sweep Run Mode

In this Run Mode the **Marker Out** generates a square wave with the rising edge placed at the beginning of each sweep and the marker's period is equal to the sum of Rising Time, Holding Time and Falling time parameters. In Sweep Triggered Mode, the Marker Width Mode is always set to Automatic.



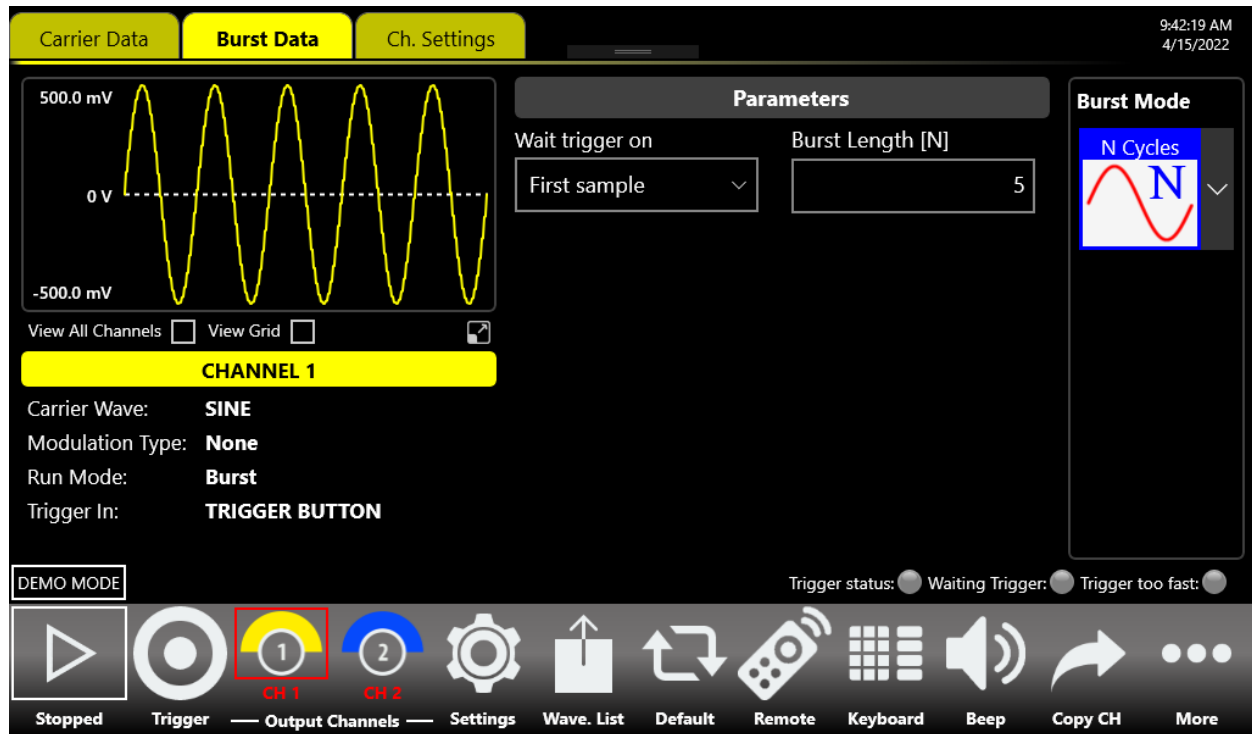
Marker out (blue, top) synchronous with the sweep (red, bottom) in Automatic width mode



Marker out (blue, top) synchronous with the sweep (red, bottom) in Manual width mode (only for Repeated mode)

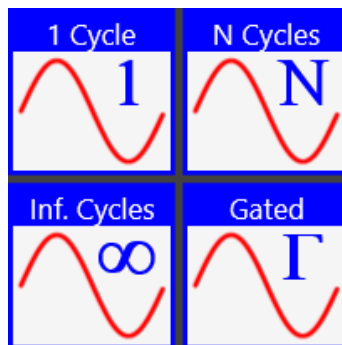
Burst

In Burst mode a waveform is repeated a predefined number of times or until the Trigger signal is at High Level depending on the selected Burst Type. This Run Mode is available for all waveforms except the DC level.



Burst Mode

The burst mode allows to define the behaviour of the instrument after the Trigger reception. Touching the “Mode” button the following menu opens and it is possible to select the burst type.



- **1 Cycles:** the instrument waits for a Trigger. When the trigger is detected, it generates one time the carrier waveform then it stops and waits for the next Trigger.

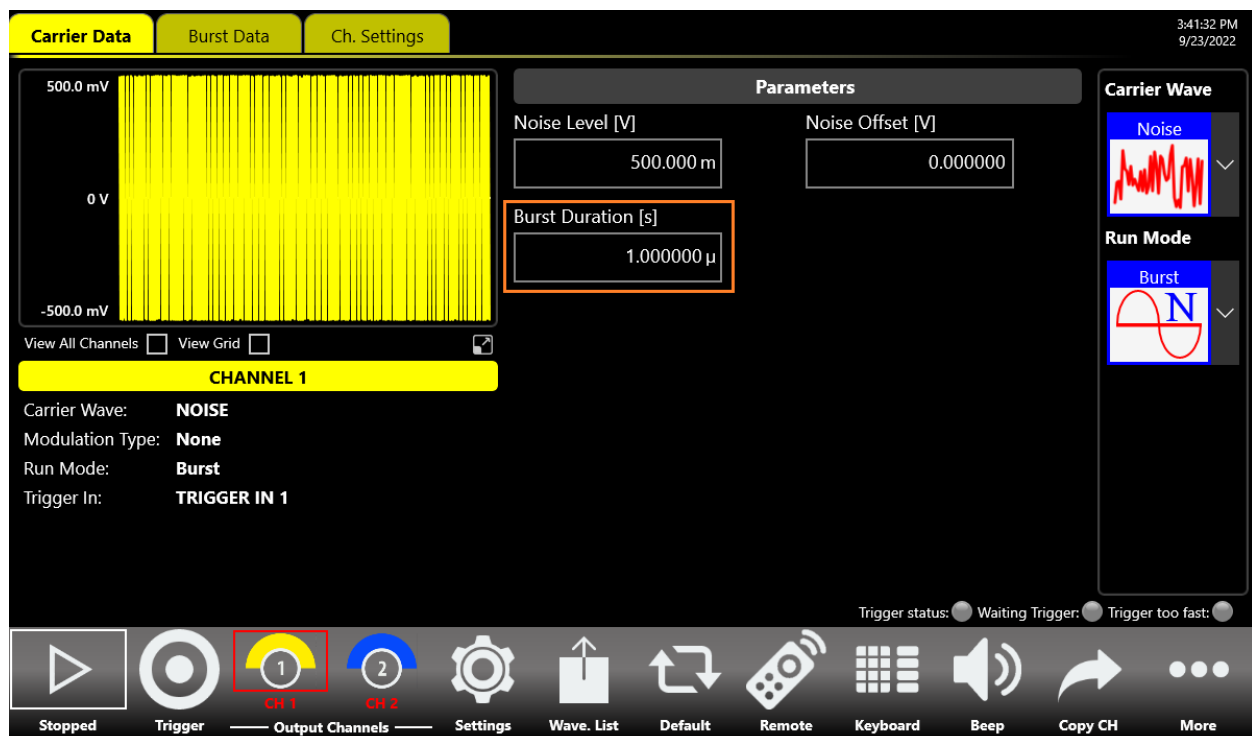
- **N-Cycles:** the instrument waits for a Trigger. When the trigger is detected it generates N times the carrier waveform then it stops and waits for the next Trigger. The number of N cycles to generate is defined by the **Number of Cycles** parameter.
- **Inf-Cycles:** this mode is similar to the previous one, but after the Trigger the generation starts until the Run/Stop Button is pressed.
- **Gated:** the waveform is generated only when the Trigger is “true”. When the trigger returns to “false” the instrument terminates the current burst sequence then it returns waiting for the next trigger.

In Gated mode, if the source of trigger is external, a trigger condition is “true” when it crosses the selected Threshold with the selected Edge, otherwise it’s “false” when it crosses the selected Threshold with the opposite Edge; the ‘Both’ option for the Edge parameter is meaningless in this modality and it will be disabled.

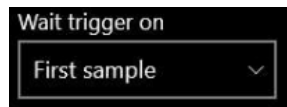
For more information about previous parameters see the Trigger In section.

If the source of trigger is Button, a trigger condition is “true” when the Trigger button is held press, otherwise it is “false” when you release it.

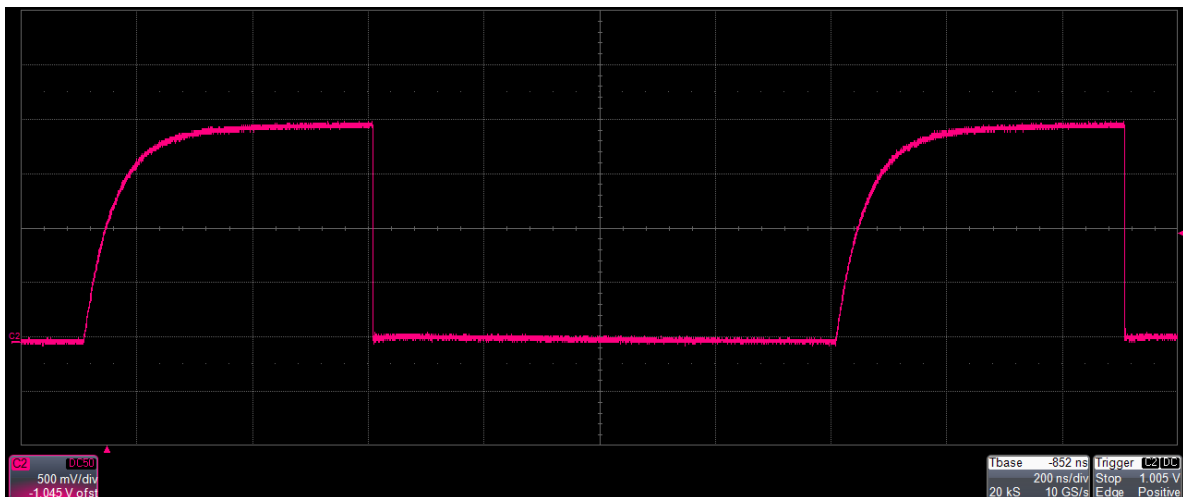
Note: if the Noise is selected as carrier and the channel is in 1-Cycle Burst or N-Cycles Burst mode, the **Duration Burst [s]** control will appear in Carrier Setting and the user can specify the duration of a single burst generation.



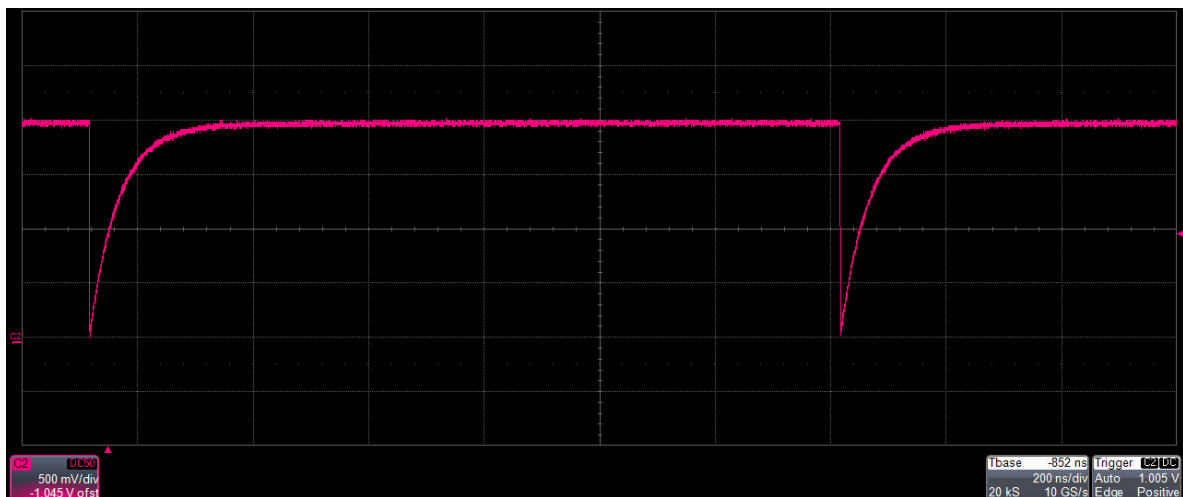
Note: after the execution of the each burst the Output voltage level keeps the Last Sample voltage (equal to the First sample) of the carrier waveform. This is true except for Exponential Rise, Exponential Decrease and Arbitrary Carrier where the **Wait Trigger On** parameter is enabled:



In this case the user can set which voltage level will be generated between one burst and another: the carrier's first or last sample.



Exponential Rise Carrier, 1-Cycle Burst Mode, Wait Trigger On First Sample



Exponential Rise Carrier, 1-Cycle Burst Mode, Wait Trigger On Last Sample

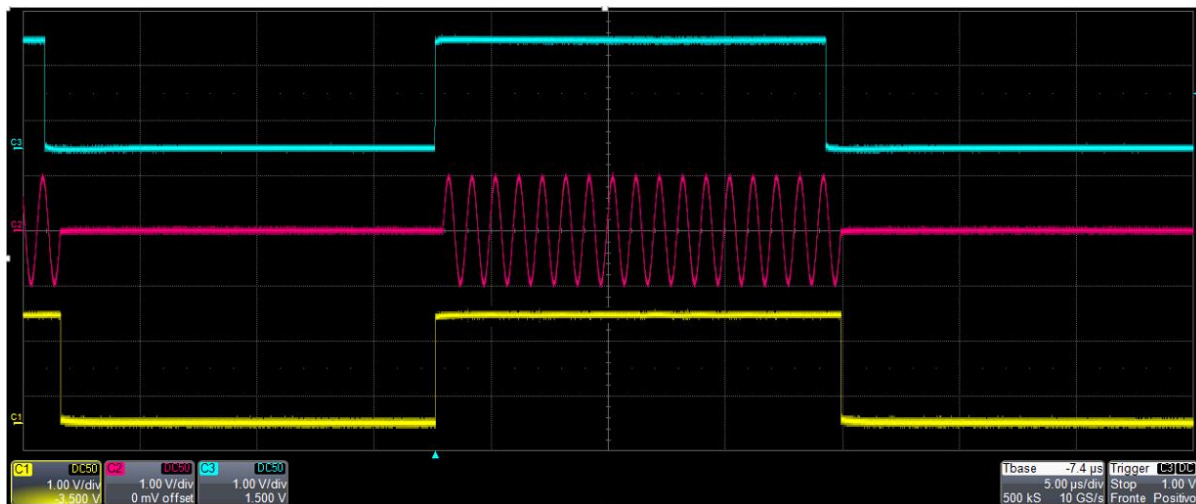
Marker Out behaviour in Burst Run Mode

In this Run Mode, the **Marker Out** generates a pulse with a duration equal to the duration of the burst sequence or to the gate time duration (time when the Trigger signal is at High level).

So the marker's width parameter is meaningless (the Marker Width Mode is always set to Automatic) while the maximum marker's skew is 1.8us.

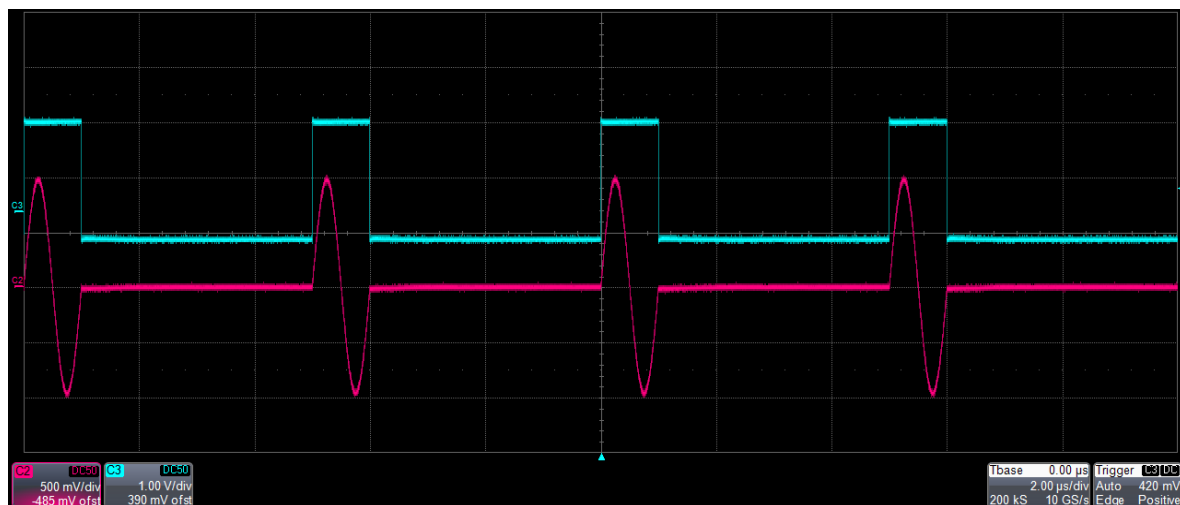
Es.1: in the following picture the gated mode operation is shown. From top to bottom there are:

- C3 (blue trace) is the Trigger signal source from the external Trigger In 1 connector.
- C2 (red trace) is the generated signal.
- C1 (yellow trace) is the marker out.



Burst Gated with Trigger from external Trigger In 1

Es.2: in the following picture the 1-Cycle Burst mode operation is shown. A sine wave carrier (frequency = 1Mhz) is generated using the internal timer (interval = 5 us).

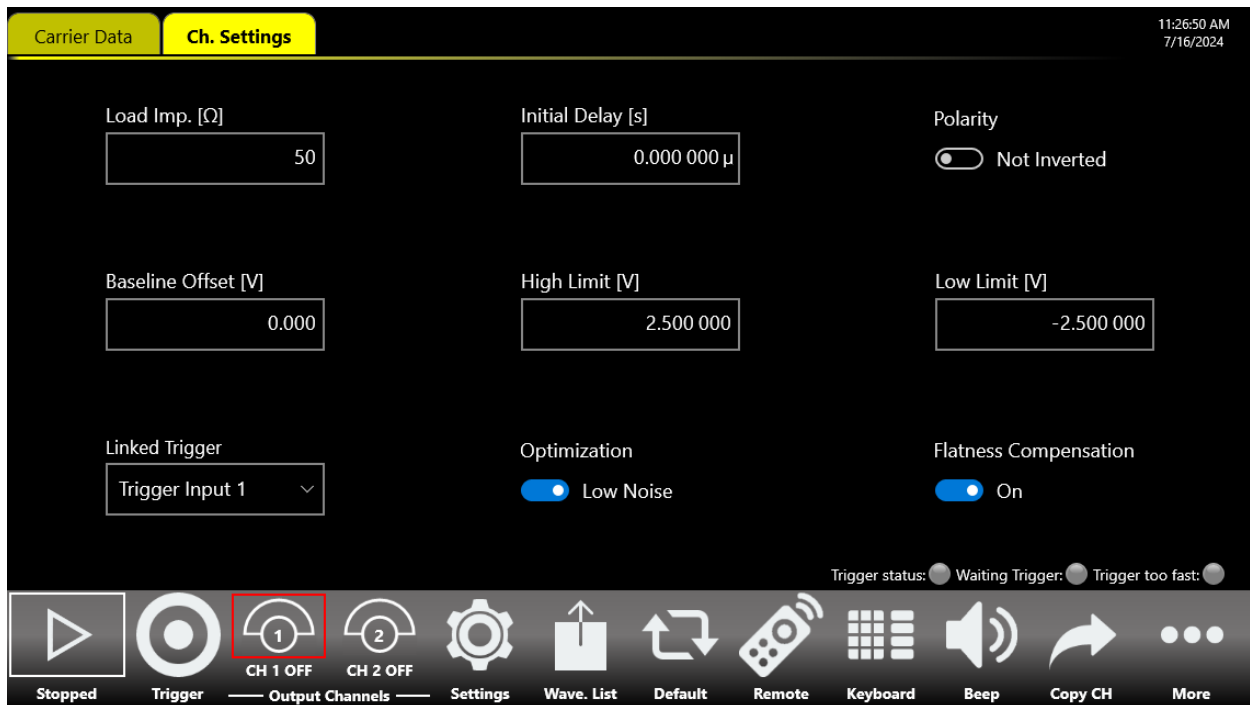


Marker out (blue, top) synchronous with 1-Cycle Burst (red, bottom)

Channels and Device Setting

Channels Settings

Touching the Ch. Setting Tab **Ch. Settings** or slide left right until you reach the setting page.



The parameters present in Channel settings are described in the table below:

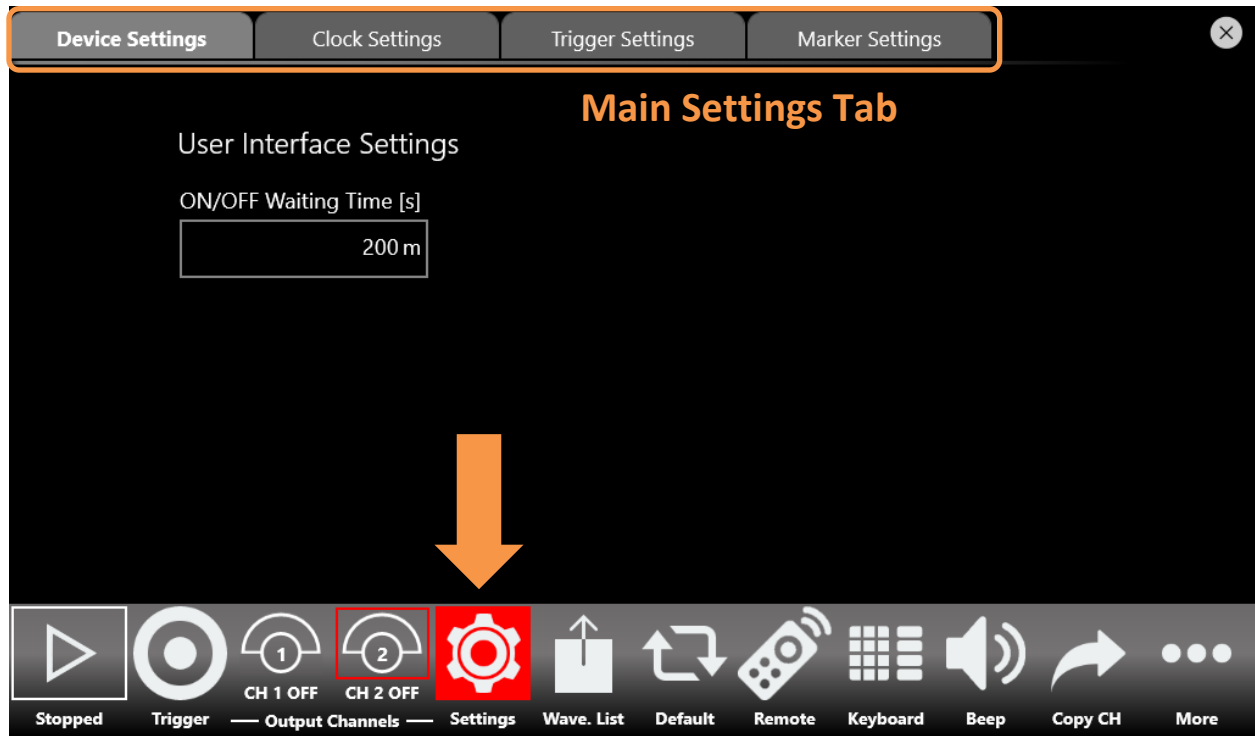
Channel Setting	Description
Initial Delay	Set the Initial Delay of the selected channel. The delay range is 0s to 8.5s.
Load Imp. [Ω]	The instrument applies the appropriate scaling to the output waveform to get the right amplitude on the defined Load expressed in Ohm. The impedance range is 1 Ohm to 1 MOhm.
Baseline Offset[V]	Defines the DC offset value added to the output signal respect to the ground level. The range is between -2.5 V to 2.5 V on 50 Ohm load and it depends on the selected load impedance (i.e. -5V to 5V into open load). Note: This parameter is available only for the instruments with single-ended outputs.

Vocm [V]	Controls the common mode voltage of the differential channel. The range is between -2 V to 2 V on 50 Ohm load and it depends on the selected load impedance. Note: This parameter is available only for the instruments with differential outputs.
Polarity	This control allows you to invert the polarity of the output signal.
High Limit [V]	Sets the maximum voltage that the channel can generate. This limit is verified during the generation, but it doesn't take into account the Baseline Offset or Vocm level. The constraint High Limit > Low Limit must be met. This feature can be used to ensure that the load will not be damaged. Note: The High Limit doesn't force the voltages parameter settings to stay below the limit, but during the generation the part of the waveform that exceeds the limit will be clamped at the High Limit.
Low Limit [V]	Sets the maximum voltage that the channel can generate. This limit is verified during the generation, but it doesn't take into account the Baseline Offset or Vocm level. The constraint Low Limit < High Limit must be met. This feature can be used to ensure that the load will not be damaged. Note: The Low Limit doesn't force the voltages parameter settings to stay above the limit, but during the generation the part of the waveform that exceeds the limit will be clamped at the Low Limit.
Linked Trigger	Sets the Trigger In source. It's possible choose between two trigger-in sources for 2-channel models or four trigger-in sources for 4-channel models (Trigger Inputs connectors). In any case the external trigger-in signal will be evaluated only if <i>Trigger Source</i> parameter is set to TRIGGER IN and <i>Run Mode</i> is set to Triggered Sweep or Burst.
Optimization	Every digital to analogue converter is affected by a series of systematic errors (INL, DNL, timing errors...) which give rise to a non-linearity in the spectrum of the generated output signal. • <i>High Linearity Mode</i>: it's possible to reduce the Noise Spectral Density of the output channel for a better linearity. • <i>Low Noise Mode</i>: is useful for reducing noise when the generated signal has a low frequency or when it consists of constant components.
Flatness Compensation	If enabled, it reduces the amplitude impairments of the signal generated in the high frequency ranges; if disabled it improves the spectral purity of the output signal.

Settings button



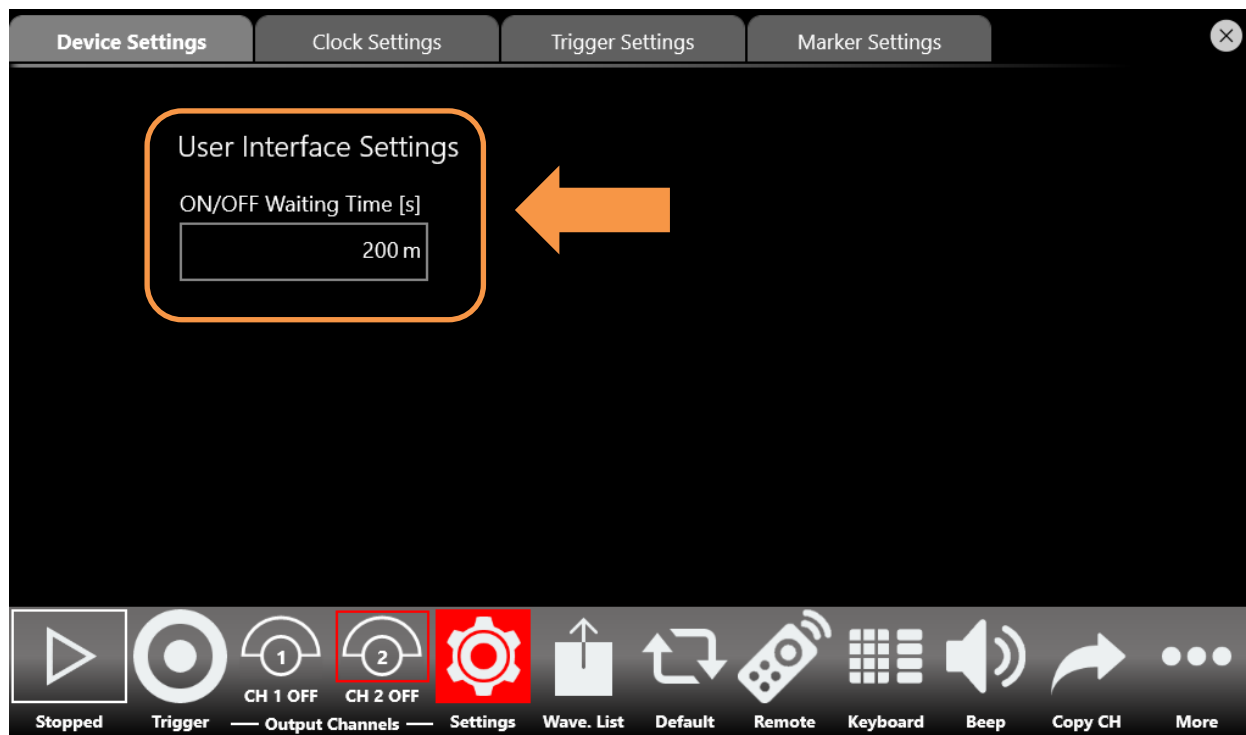
Touch the “Settings” button to open the dedicated **Main Setting Tabs** page relative to the, **Device Settings**, **Clock Settings**, **Trigger Settings** and **Marker Settings** tab.



Device Settings tab

The **Device Settings** page contains some parameters valid for the device.

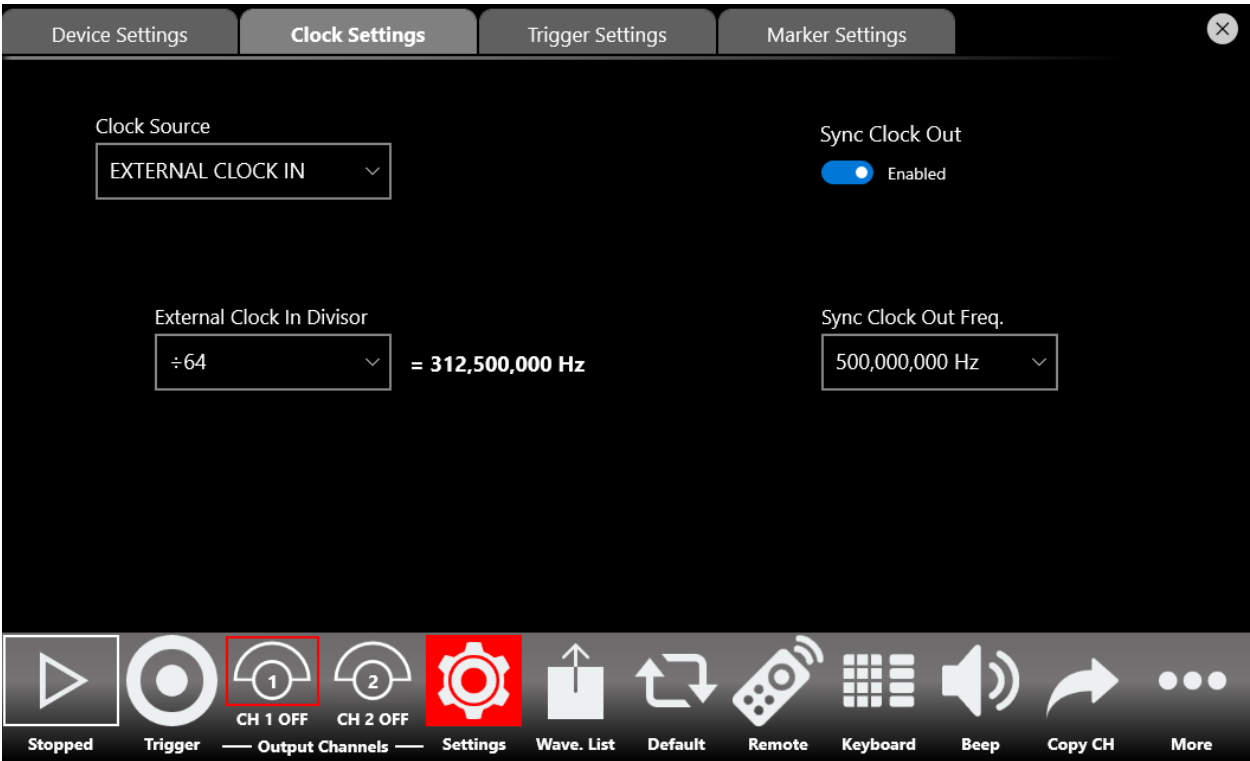
In this tab is present a UI setting: ON/OFF Waiting Time[s].



Device Setting	Description
ON/OFF Waiting Time[s]	This parameter allows the user to set how long they must hold down the Channel button to turn the channel's output ON or OFF. The range of the ON/OFF Waiting Time is 0s to 2s. The default value is 200 ms. Usage Example: Turn ON the Channel 1 

Clock Settings tab

The **Clock Settings** parameters are located in the Clock Settings tabs.



Timing Setting	Description
Clock Source	If Internal Clock is selected, the internal DAC sampling clock is synthesized using a 10Mhz reference clock generated internally. If Reference Clock In is selected, the DAC sampling clock is synthesized using the clock provided externally to the Ref.Clock In SMA connector. In this case the Ref Clk Frequency [Hz] control will appear and the user must specify the Reference Clock frequency[Hz]. The frequency range of this clock is: 5 Mhz to 500 Mhz. If External Clock In is selected, the internal clock synthesizer is bypassed and the clock signal provided at the External Clock Input SMA connector will feed directly the sampling clock for the system. In this case the External Clock In Freq. control will appear to define the frequency applied to the External Clock Input signal (four values are available: 312.5 MHz, 625 MHz, 1.25 GHz or 2.5 GHz).
Sync Clock Out	It enables or disables the external Sync Clock Output. This clock can be used to provide a trigger input signal synchronous with the system clock

in order to avoid the Trigger In to analog Out delay from jittering and to reduce the Trigger In to analog Out latency.

When the Sync Clock Output is enabled the **Sync Clock Out Freq** control will appear and the user can choose Sync Output clock frequency from a list of all the possible values.

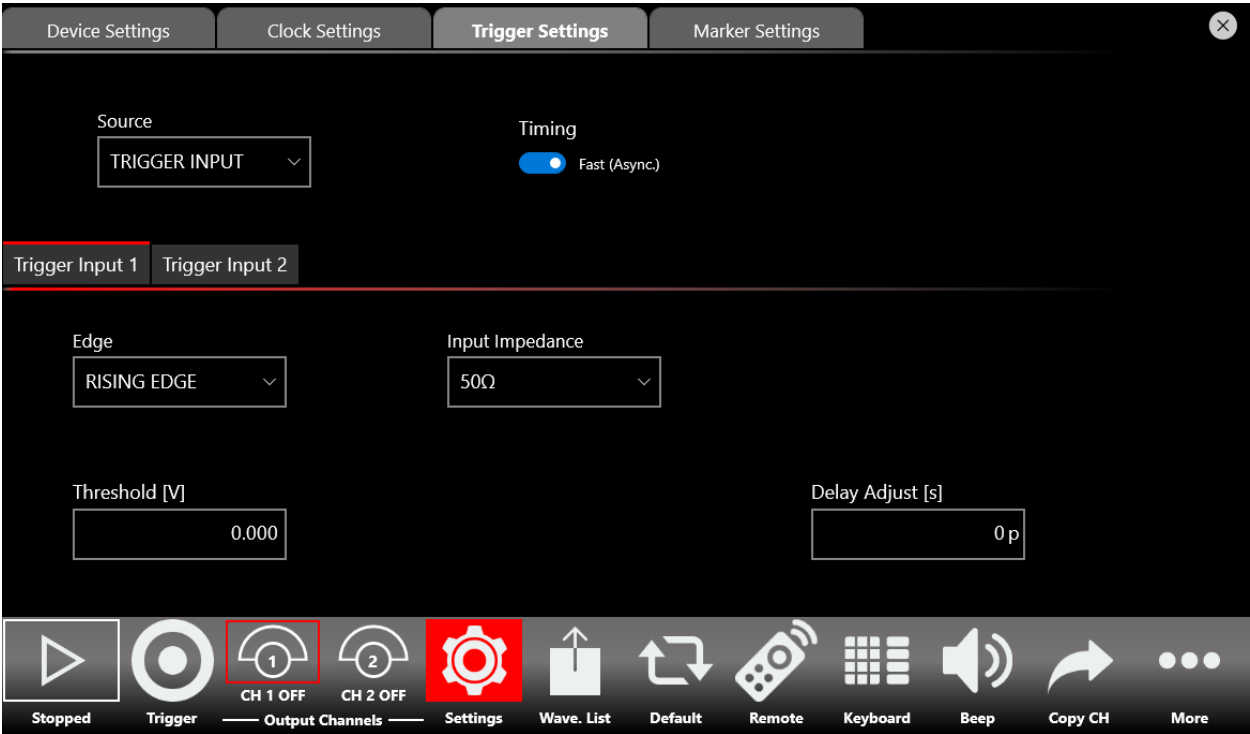
Note: Sync Clock Output can be enabled even if the instrument is in Stopped mode. In the same way you can set the Synch Clock Out Frequency but pay attention that the reported frequency is linked to the Sampling Clock rate. If the sampling rate is changed in stopped mode, the frequency of the clock out will be consistent with that displayed in the parameter only after having sent the instrument in Running mode.

Trigger Settings tab

The Trigger In settings parameters are located in the **Trigger Settings** tab. You can set both internal trigger sources (such as a timer, physical button...) and external trigger sources.

While the Source and Timing parameters are common for all Triggers Inputs, the remaining ones can be set independently for each of the Trigger Input signal.

Depending on the number of channels of the model, 2 or 4 Trigger Inputs are available: the 686-2C models have two Trigger Inputs: In 1 and In 2 while the 686-4C models have four Trigger Inputs: In 1, In 2, In 3 and In 4 (the corresponding connectors are located in the front panel of the instruments).



Trigger In Settings: 2 Channels Model

Trigger In Setting	Description
Source	Trigger Button: the Trigger event is provided to the instrument by pressing the Trigger button on the keyboard or the Trigger button on the menu toolbar or by issuing a trigger by Remote Command. Timer: the Trigger signal is internally generated by a Timer. The Timer count interval is set by Timer Interval [s] textbox. Trigger Input: a Trigger event is generated by the signal applied externally to the Trigger In SMA connector when it crosses the selected Threshold with the selected Slope. Up to two or four trigger-in sources can be set. Their respective threshold and slope values can be defined in the relative textbox and slider. The association between an analog output channel and

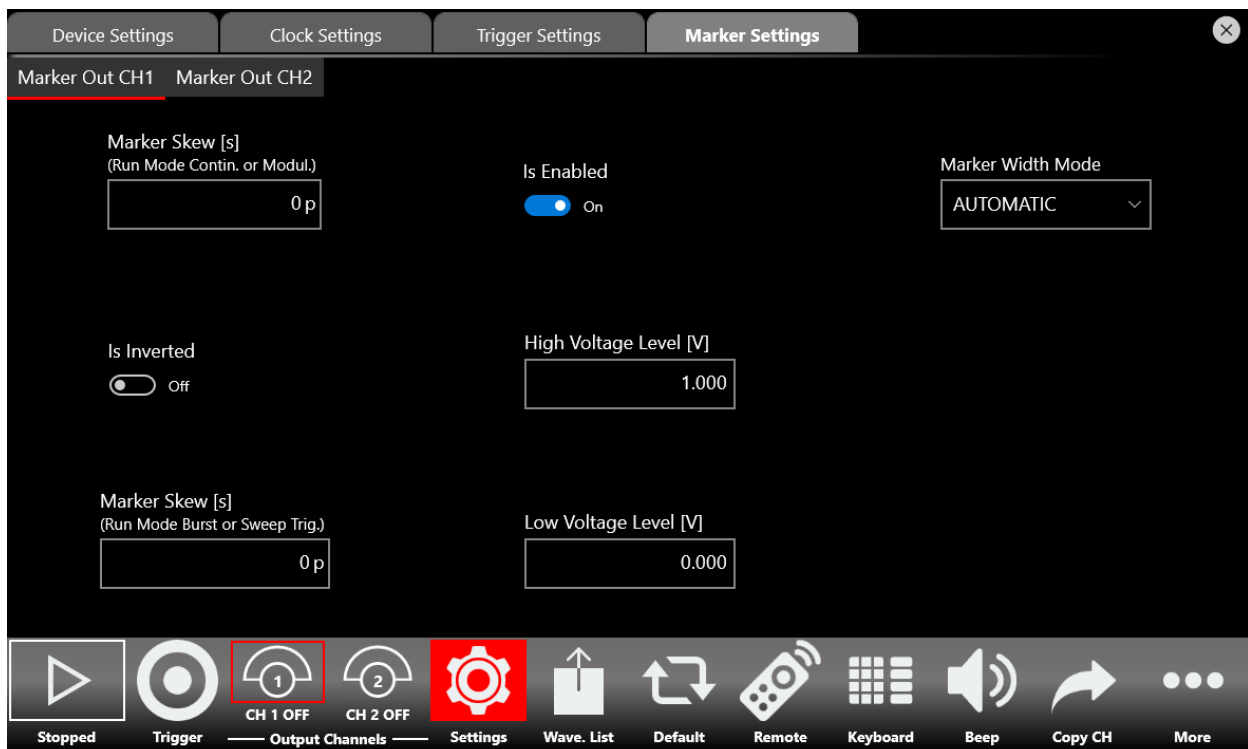
	the trigger in signal takes place thanks to <i>Linked Trigger</i> parameter in Ch. Settings. Note: the Trigger buttons and the Trigger from remote command are always active, independently from the selected Trigger Source.
Timing	When <i>Slow (Sync.)</i> is selected the Trigger input signal is supposed to be asynchronous with the system clock. In this case a hardware time measurement circuit is enabled to keep the Trigger In to Out jitter as low as possible. When <i>Fast (Async.)</i> is selected the Trigger input signal is expected to be synchronous with the Sync Clock Out (and so with the system clock as well). In this case the time measurement circuit is disabled so that the Trigger In to Out delay is lower. Note: on a 4-channels model, when the Timing is set to <i>Slow (Sync.)</i> mode, the only two Trigger Inputs available are Trigger Input 1 and Trigger Input 2.
Timer Interval [s]	Set the timer count interval. It has effect only when the selecting trigger Source is Timer. It has effect only when the selecting Source is <i>Timer</i>.
Edge	When <i>Rising Edge</i> is selected the trigger is detected when the signal on the "Trigger In" SMA connector crosses the threshold from low to high. The <i>Falling Edge</i> option is the opposite. "Both Edges" means that Trigger is sensitive to both edges of the signals. It has effect only when the selecting Source is <i>Trigger Input</i>. Note: If at least one of the channels, to which a specific Trigger In is linked to, is in Burst Gated as Run Mode, then the 'Both Edges' option will be disabled.
Threshold [V]	It is the threshold that the external signal applied to the Trigger In connector must cross to issue a Trigger event to the instrument. It has effect only when the selecting Source is <i>Trigger Input</i>.
Input Impedance	It selects the Trigger In 1 connector impedance: 1 kOhm or terminated into 50 Ohm.
Delay Adjust [s]	When the Timing is set to <i>Fast (Async.)</i> then the Trigger In signal is evaluated on the rising/falling edge of the Sync Clock Out. In order to optimize the timing margins, it is possible to apply a delay to the Trigger Input signal. The Delay Adjust parameter specifies the delay applied to the Trigger Input signal. The range of the delay is 0ps to 2418ps. The resolution of the delay is 78ps.

Marker Settings

In the **Marker Settings** page, it is possible to define the behaviour and parameters of the Marker Out signals (the corresponding connectors are located in the front panel of the instruments).

The 686-2C models have two Marker Out: Marker Out CH1, Marker Out CH2. The 686-4C models have four Marker Outs: Marker Out CH1, Marker Out CH2, Marker Out CH3, Marker Out CH4. Each Marker Out has its own set of parameters. It is possible to switch among these set of parameters by clicking on the Marker Out CH1, ..., the Marker Out Ch4 icons located in the Marker Out Main Setting Bar.

Each marker refers to its respective output channel.



Marker Out Settings: 2 Channels Model

The available parameters for the markers out are:

Marker Out Setting	Description
High Voltage Level [V]	Specifies the Marker high level Voltage. The Voltage Level range is -0.5 V to 1.65 V on 50 Ohm load (it doubles on open load).
Low Voltage Level [V]	Specifies the Marker Low level Voltage. The Voltage Level range is -0.5 to 1.65 V on 50 Ohm load (it doubles on open load).

Is Enabled	Enables or disables the Marker Out. When the Marker Out is disabled it is forced to 0V. Note: using some Waveform Carriers or setting specific Run Modes it may happen that the marker has no meaning. In that case the state button will be disabled.
Marker Width Mode	Allows the User to choose between Automatic Width mode (Marker out width is automatically adjusted by the AFG Software) and Manual Width mode (Marker Out width is fixed to User defined value). This parameter has effect only for not Triggered modes.
Marker Width [s]	Specifies the width of the marker out when Manual width mode is selected. The range is from 500 ps up to (Marker Out Period - 2.1ns).
Marker Skew [s] (Run Mode Contin. or Modul.)	It sets the delay between the marker and the analog channel. This parameter is valid only if the run Continuous Mode or Modulated Mode or Sweep Repeated Mode is been selected for the associated channel to the Marker out. The skew range is 0s to 100s.
Marker Skew [s] (Run Mode Burst or Sweep Trig.)	It sets the delay between the marker and the analog channel. This parameter is valid only if the run Sweep Triggered Mode or Burst Mode is been selected for the associated channel to the Marker out. The skew range is 0s to 1.8us.
Is Inverted	It sets the polarity of the marker out signal.

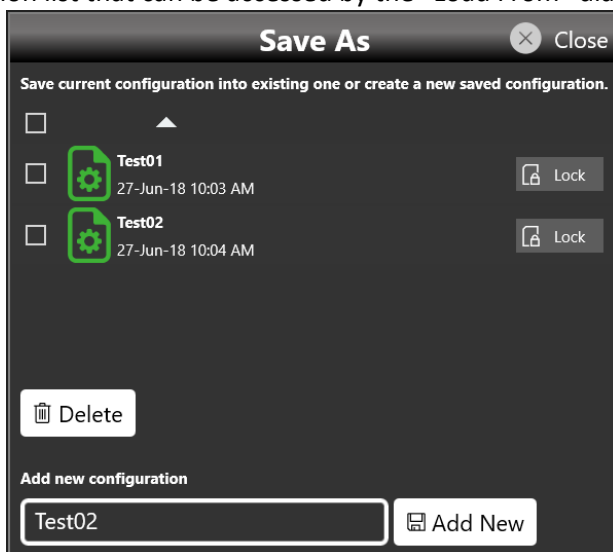
Main Command Button Description

Save As...

A configuration (all the parameter values relating to the currently state of the instrument) can be saved



by means of the “Save As” button that will open the following dialog box. The configuration will be saved in the configuration list that can be accessed by the “Load From” dialog box:



In this page you can add a new configuration entry or overwrite an existing one. To create a new configuration entry, it is necessary to write a name in the text box in the bottom of the page and click on “Add_New”.

Export Configuration



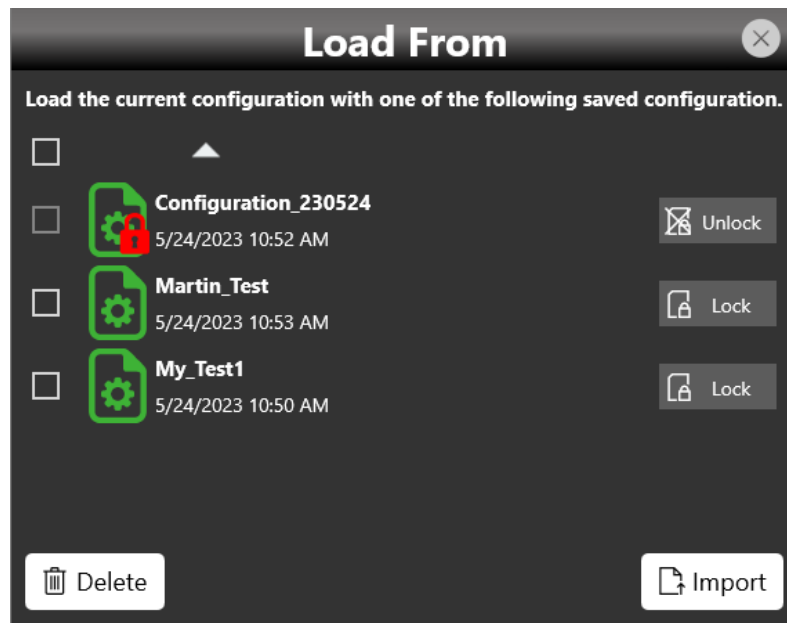
If you touch the Export Configuration button a proprietary binary .zip file relative to the current configuration will be exported. The exported file can be used to share configurations between different users or instruments.

Load From...



Touching the “Load From” button in the “More” menu, a page will be opened that shows the list of all the saved and imported configurations. If you select an existing Configuration, you will load all the settings into the instrument.

In the “Load From” page it is also possible to manage the configuration list: you can delete, import or lock a configuration. When a configuration is locked it can’t be deleted or overwritten.

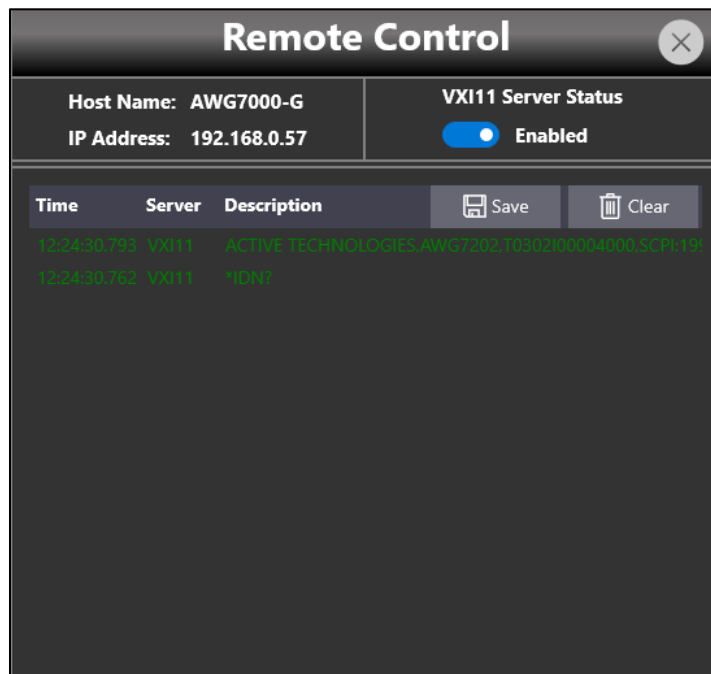


If you touch the Import Configuration button you can import the configuration file that comes from a different machine or from a different user. The imported configuration will be inserted in the “Load From” list.

Remote Control



The “Remote” button located in the Command Bar opens the page of the SCPI server. In that page there is the list of all the commands received by the SCPI server and its replies. If the text of the command is displayed in **green** it means that the command is correct and it has been accepted by the server. If the text of the command is displayed in **red** it means that the command is wrong and it hasn't been accepted by the server.



In the top of the page the Host Name and the IP Address of the instrument are shown. The slider on the right side of the page allows to enable or disable the SCPI server. It is enabled by default.

Remote Desktop Connection

The following credentials must be used when connecting to the instrument by a remote desktop connection:

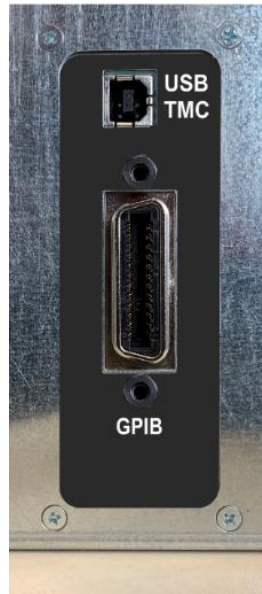
Computer Name: AWG7000

User Name: awg7000

Password: 1234

GPIB and USBTMC (with – GPIBUSB option only)

While the VXI-11 server is always available to remotely control all the parameters of the instrument, with GP-IB / USB-TMC option the Model 686 series is provided with a GPIB electronic interface and a USBTMC port (refer to the Programmer manual for a complete description of all instrument setting commands and data transfer commands).

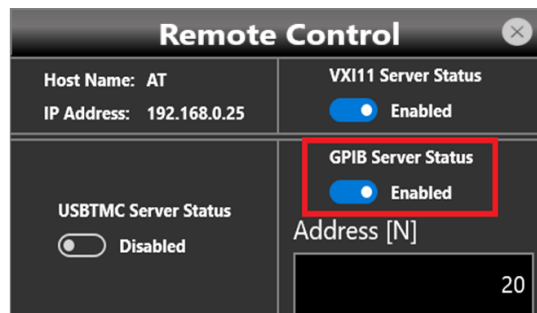


GPIB and USBTMC Connector (optional)

GPIB control

Follow these instructions to setting up the instrument for remote communication using the GPIB (General Purpose Interface Bus) interface:

- Connect one side of a GPIB cable to the GPIB port of the 686 (on the rear panel), and your GPIB controller on the other side.
- In “Remote Control Panel” window verify that GPIB Server Status is Enabled; if not, press the button to enable it.

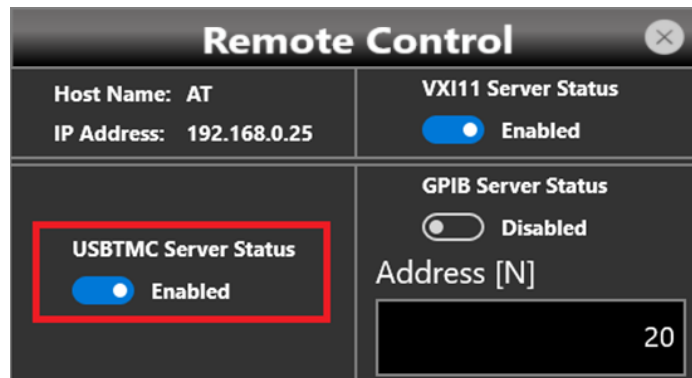


- Set the GPIB address. A unique device address must be assigned to each device on the bus. The default setting for the GPIB configuration is Address [N] 20. If you need to change the GPIB address set it in Address [N] parameter.

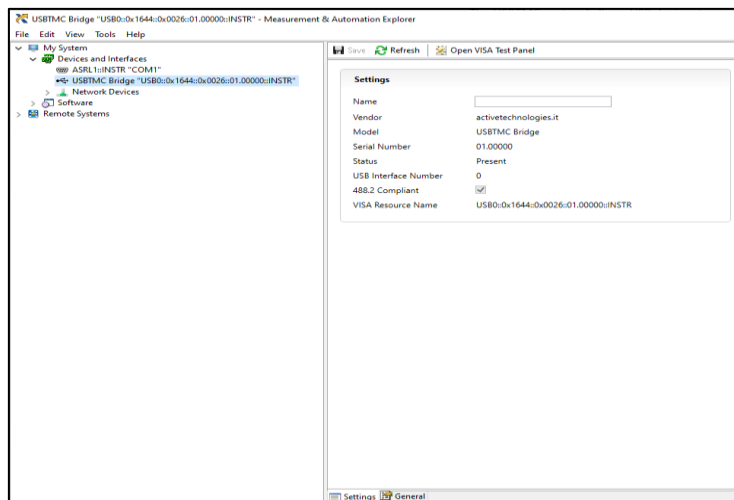
USBTMC control

The USBTMC protocol allows USB devices to communicate using IEEE488 style messages. This lets you run the SCPI commands using USB hardware:

- Connect an appropriate USB cable (A MALE - B MALE) between the USBTMC port of the 686 (on the rear panel) and your Client-PC.
- In Remote Control Panel verify that USBTMC Server Status is Enabled; if not, press the button to enable it.



- If you are using NI-VISA software, launch the NI-MAX tool on the Client-PC. After a short time a USBTMC device will be recognize.

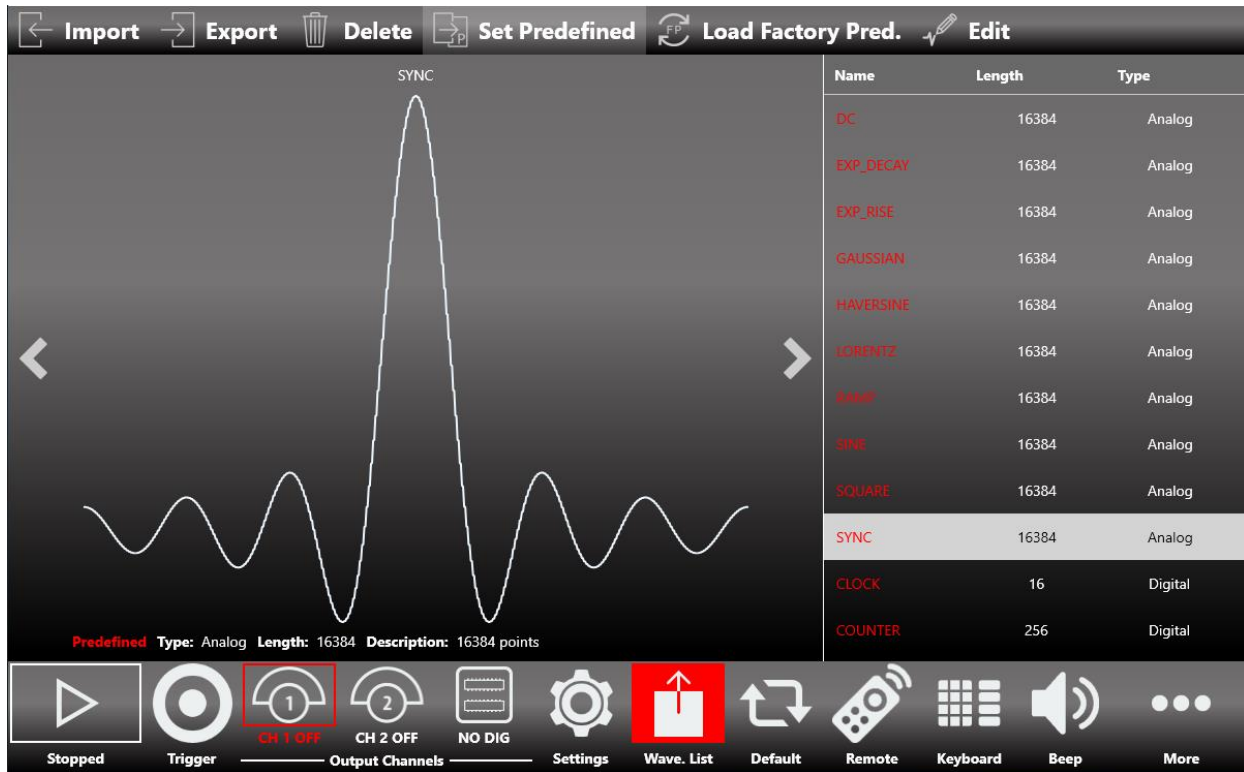


- Now you can send the SCPI commands to the 686 resource using the NI Visa Test Panel.

Waveform List



By pressing the  button the Waveform List page will open showing all the waveforms available in the current configuration.



The Model 686 series contains by default a set of Factory Predefined Waveforms.

The Predefined Waveforms are the ones in red color on the list, the imported waveforms are the ones in gray.

Please note that:

- You can create your own set of Predefined Waveform by promoting waveforms in the list to Predefined ones.
- You can delete a Predefined waveform with the exception of SINE and DC waveforms.
- It is possible to restore the Factory Predefined waveforms by pressing the



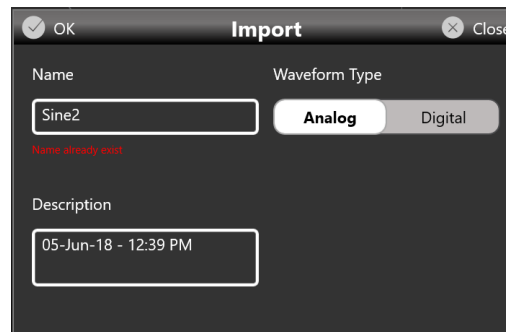
button. All the waveforms previously imported will not be deleted.

How to import a waveform from a file

Import button  allow you to import data from a file to create a new waveform.

The supported file formats are:

- .txt – New line (\n) separated text file (one column only with no header)
 - .zip – Compressed file in binary proprietary format
 - .trc – LeCroy oscilloscope binary file format
 - .bin – Binary file. The software will use two bytes for each sample (little endian format)
1. Press the import button and the Windows File Browser will open, select the file you would like to import. The Import page will open.
 2. In the Import dialog, the Name and Description fields will be automatically filled with default values. The imported waveform is normalized.
 3. Press OK to confirm or Close to cancel the operation.



How to export a waveform to a file

- Select a waveform on the waveform list.
- Press the  **Export** button.
- The exported waveform will be stored in a proprietary binary .zip file format that can be shared with other instruments running the same application.
- You can export also the Predefined waveforms.
Please note this special case: if you export a Predefined waveform and then you try to import it again on the list, it will be imported as a standard waveform.

How to promote a waveform to a Predefined

Please note that when a configuration is loaded only all Predefined waveforms and those directly used at the time of saving will appear in the Waveform list.

Before saving a configuration it's necessary to promote an imported waveform as Predefined if the user want to keep it in the Waveform List.

- Select an imported waveform on the waveform list.
- Press the  **Set Predefined** button.
- The waveform will appear on the list in red color to show that it has been promoted to Predefined.

How to edit a Waveform

- Prerequisites: "Waveform Editor" software installed.
- Select a waveform on the waveform list.
- Press the  **Edit** button to launch the "Waveform Editor".
- Please refer to the "Waveform Editor" user manual for a complete explanation about editing and creating waveforms.

How to create a new Waveform

- Prerequisites: "Waveform Editor" software installed.
- Press the  button in the **More...** menu to launch the "Waveform Editor".
- Please refer to the "Waveform Editor" user manual for a complete explanation about editing and creating waveforms.

Channel Coupling

In electronics design and testing, you sometimes want to have two synchronized clock signals that are related by a frequency ratio; one clock needs to maintain a certain frequency ratio with the other clock. Or perhaps, you want to simulate an amplifier with an offset; the amplifier output needs to maintain a defined offset from the input amplitude.

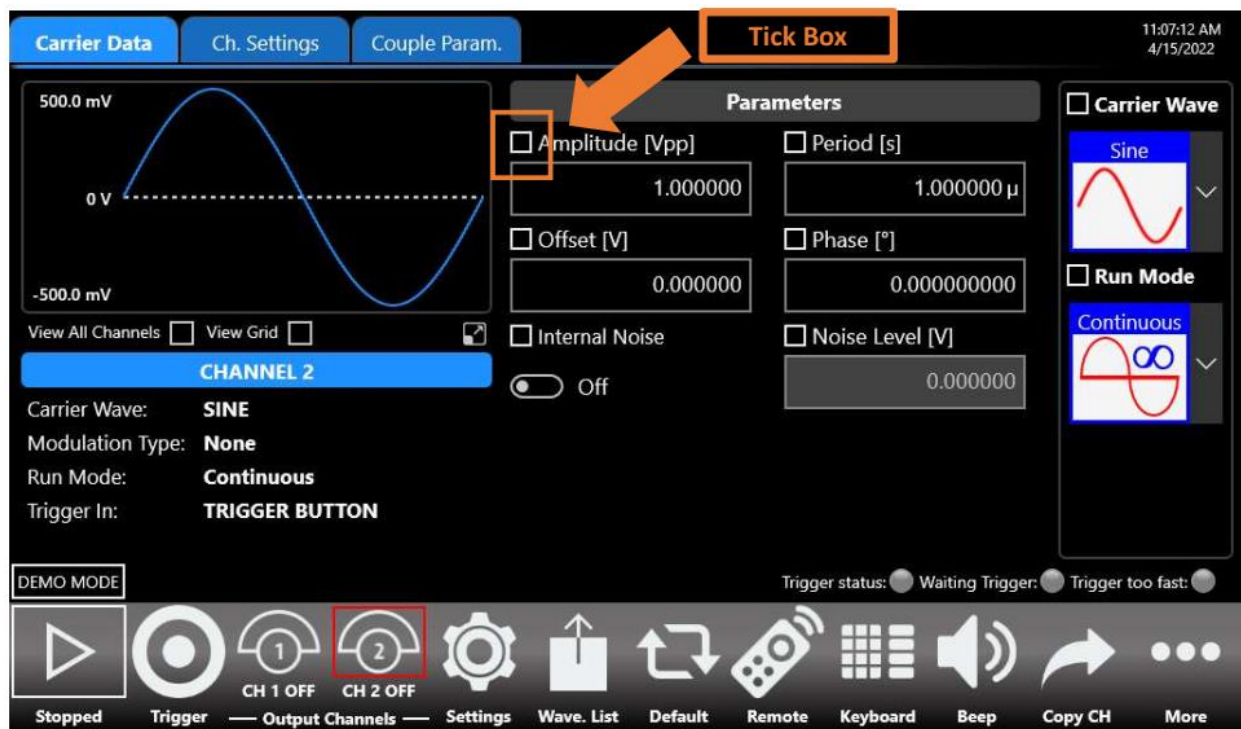
In **Channel Uncoupled** mode the behavior of the signal generated by a specific channel depends only on the parameters set on the channel itself, while it is independent of any parameter set on the other channels of the instrument.

In **Channel Coupled** mode the user can choose which Channel N parameters are related to the respective Channel 1 (reference channel) parameters. If this modality is enabled, near all Channel N (where $N > 1$) parameters will appear a tick box, where the user can choose whether this should be coupled or not to the corresponding Channel 1 (CH1) parameter.

The coupled parameters will change in real time following the changes of Channel 1 parameters.

Some parameters are already coupled by default when this mode is selected.

The Operating Mode "Channel Coupled" is not allowed between channels with different Output Format. (e.g: the Symmetry parameter of channel CH2, where a Ramp is selected as carrier, cannot be coupled if the carrier of CH1 is a Sine waveform, because for this one the parameter doesn't exist).

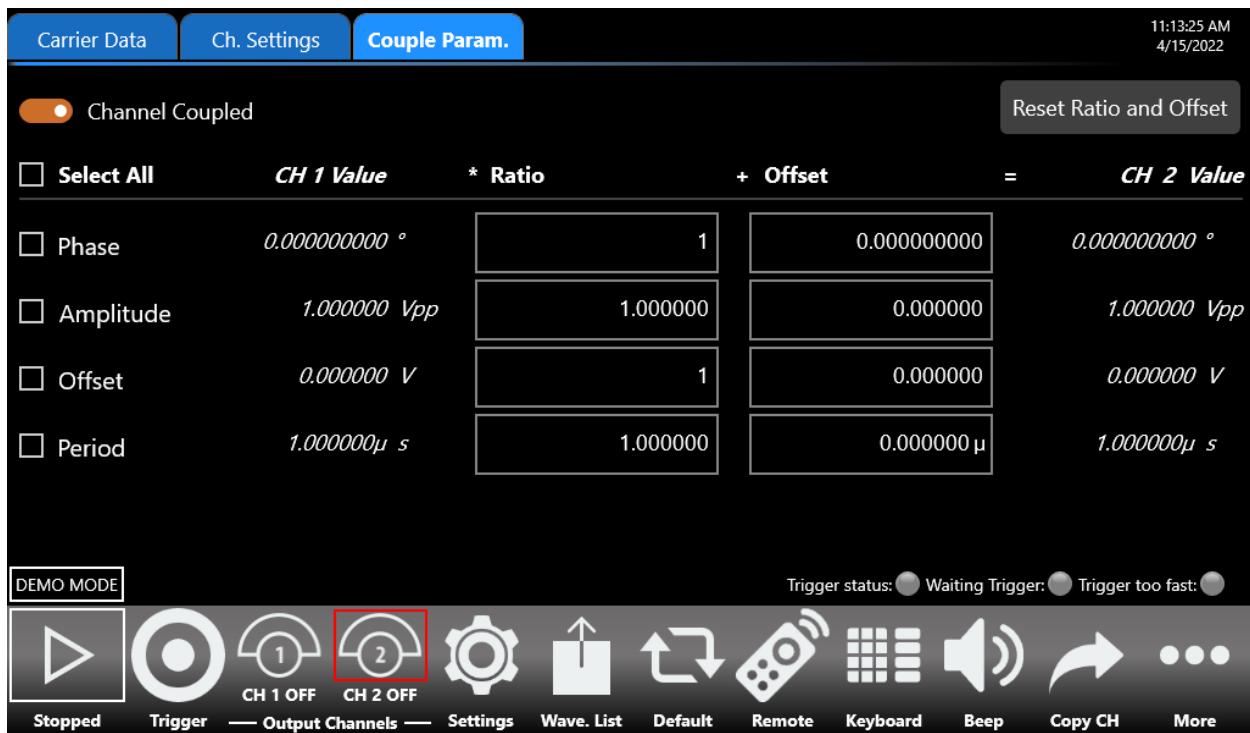


The Channel Coupling section allows you to specify that Channel N parameter like frequency, amplitude, offset etc. is related to Channel 1 parameter by a ratio (multiplying) and an offset (adding).

The equation that specifies the relation between the Channel N parameter to the Channel 1 parameter is the following:

$$\text{Ch[N] Parameter} = \text{CH1 Parameter} * \text{Ratio} + \text{Offset}$$

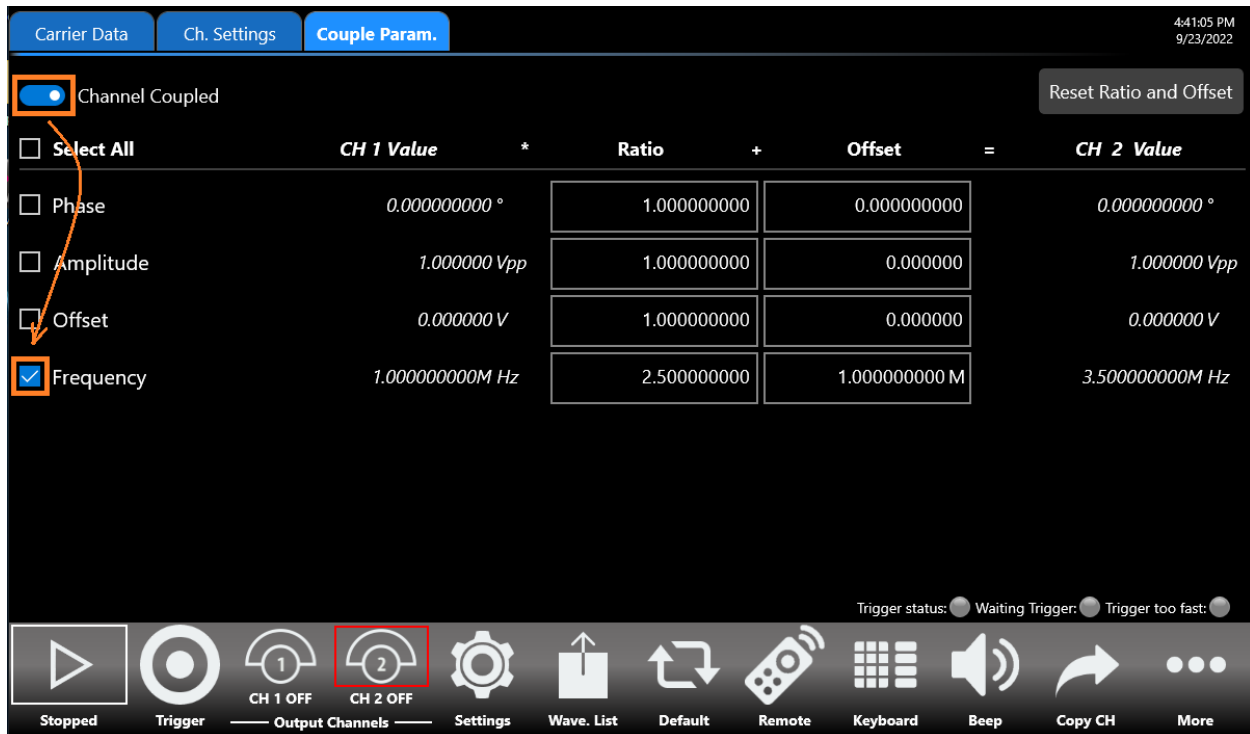
The coupled parameters will change in real time following the changes of Channel 1 parameters to maintain the specified ratio and offset. You can couple multiple parameters together.



- Swipe up or down to select which parameters to couple to the Channel 1.
- **Toggle Channel Coupled:** enables or disables the channel coupling.
- **Reset Ratio and Offset:** resets the Ratio and Offset parameters to their default values (1 and 0).
- **Select All:** selects/deselects all the available coupling parameters on the selected channel.
- Press the parameter checkbox to select/deselect the single coupling parameter.
- When waveforms have different carrier shapes, only the common parameters will be available on this page.
- Ratio or Offset auto limit their ranges in base of the linked parameter value set.

Example: Frequency Coupling

1. Suppose that the channel 1 has a frequency of 1 MHz.
2. In Channel 2 Couple Param. Tab **Couple Param.** enable Channel Coupled and check the Frequency parameter



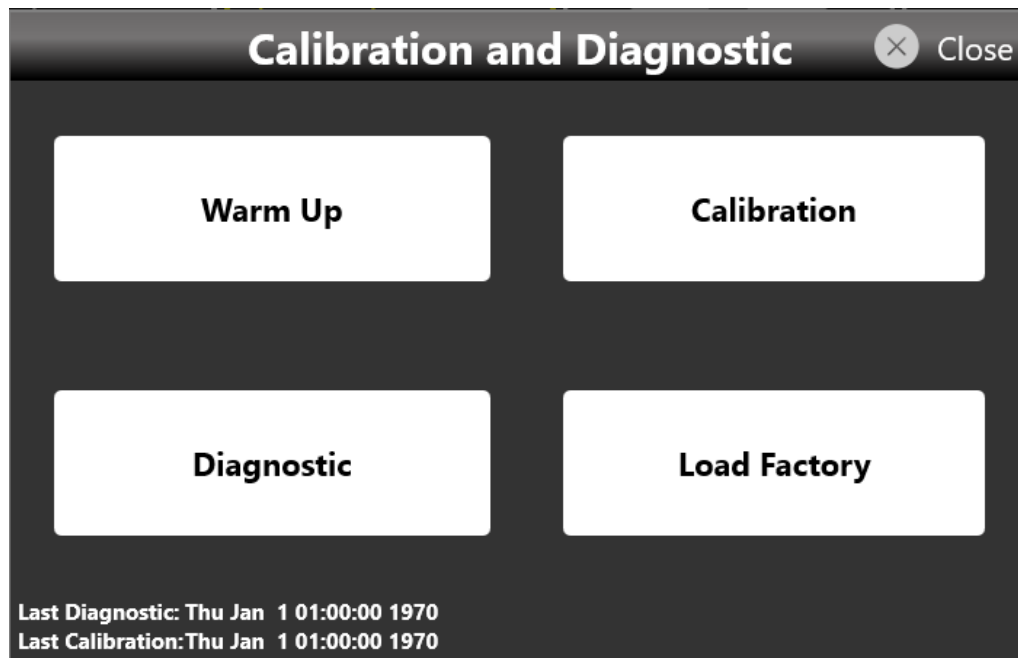
3. Set Ratio to 2.5 and Offset to 1M
4. The Channel 2 frequency will be 3.5 MHz

Now the channel 1 and the channel 2 frequency parameters are coupled and the Frequency parameter textbox in the Channel 2 Carrier page will be disabled.

On the contrary, the Channel 1 frequency parameter will be enabled and if the frequency is changed to 5 MHz then the channel 2 frequency will automatically change to 13.5MHz.

Calibration and Diagnostic

The calibration button  in the More... menu  opens the Calibration and Diagnostic page.



Below a description of the actions executed by pressing the buttons located in the page:

- *Warm up* button: touching this button will start the warm up instrument procedure that will take 30 minutes. The elapsed time is shown. The procedure can be stopped by touching the Stop button located at the bottom right of the Warm Up page
- *Calibration button*: touching this button will start the self-calibration of the instrument. The logs of the procedure are displayed in a text box that can be saved at the end of the procedure for further analysis.
- *Diagnostic* button: touching this button will start the self-diagnostic of the instrument. The logs of the procedure are displayed in a text box that can be saved at the end of the procedure for further analysis.
- *Load Factory* button: touching this button will load the factory calibration parameters.

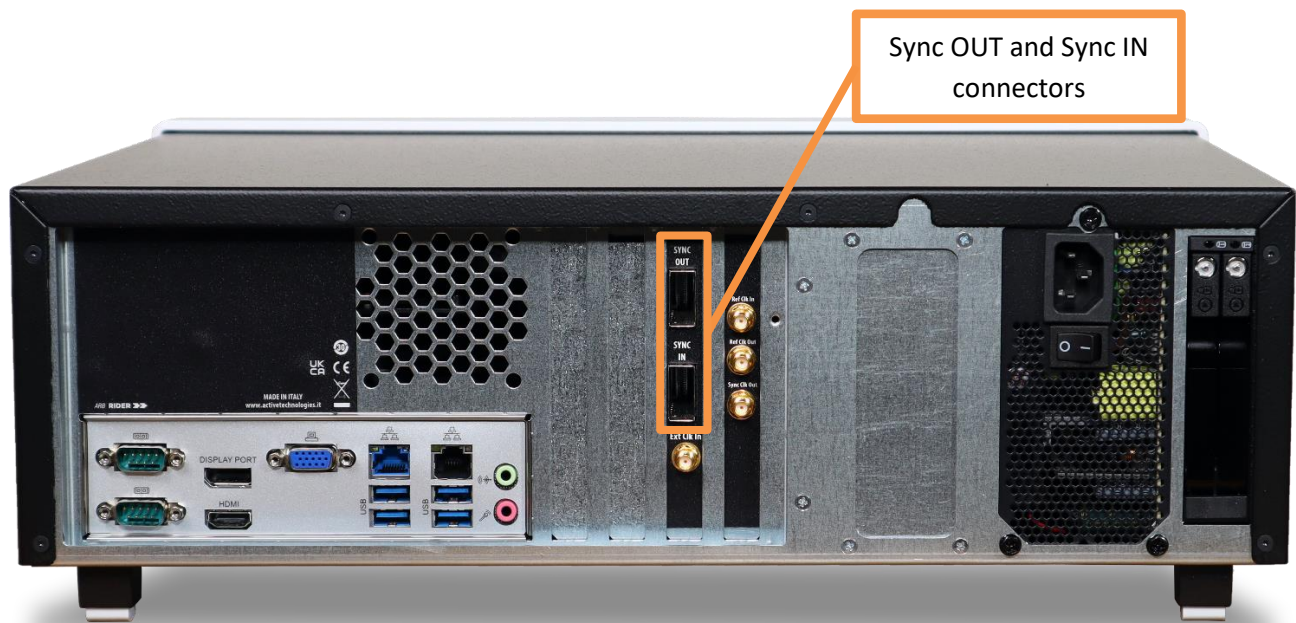
Multi-Instrument System

In a Multi-Instrument configuration, a Master device can control every triggering and timing setting in order to synchronize its operation with that of other Slave devices.

You can connect up to four Model 686 units to have a system with up to 16 synchronized analog channels.

In order to set up a multi-Instrument system the following steps are required:

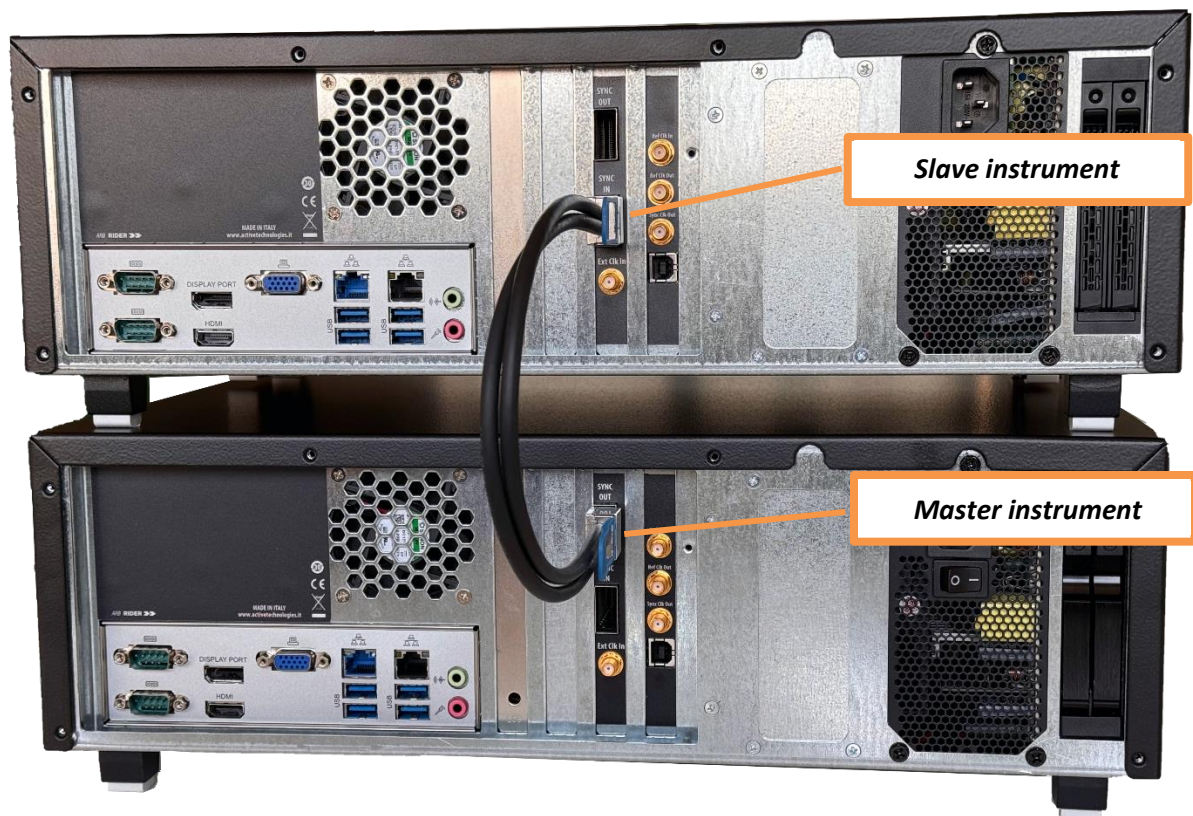
- Turn off the instruments;
- Select the instrument that you want to use as Master. The other units will be considered as Slaves.
- Using the 686-SYNC cable, connect the Master **Sync Out** connector to the Slave **Sync In** connector located on the rear panel of the instruments. Then connect the Sync Out of the slave to the Sync In of the next slave device and so on up to the last slave device.



Once all the instruments have been connected, turn them on and launch the Simple AFG Application in all the instruments.

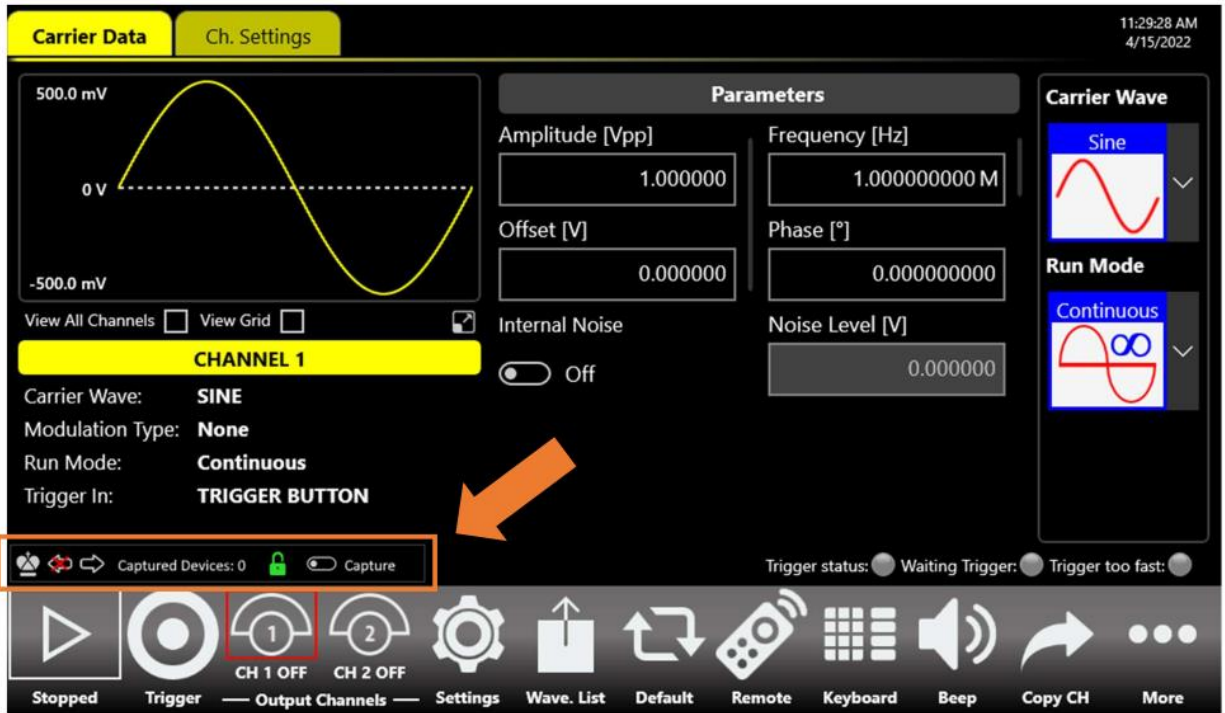
Please note that:

- Turn off the instruments before connecting or disconnecting 686-SYNC cables
- The external clock source and external trigger input are available in Master device only.

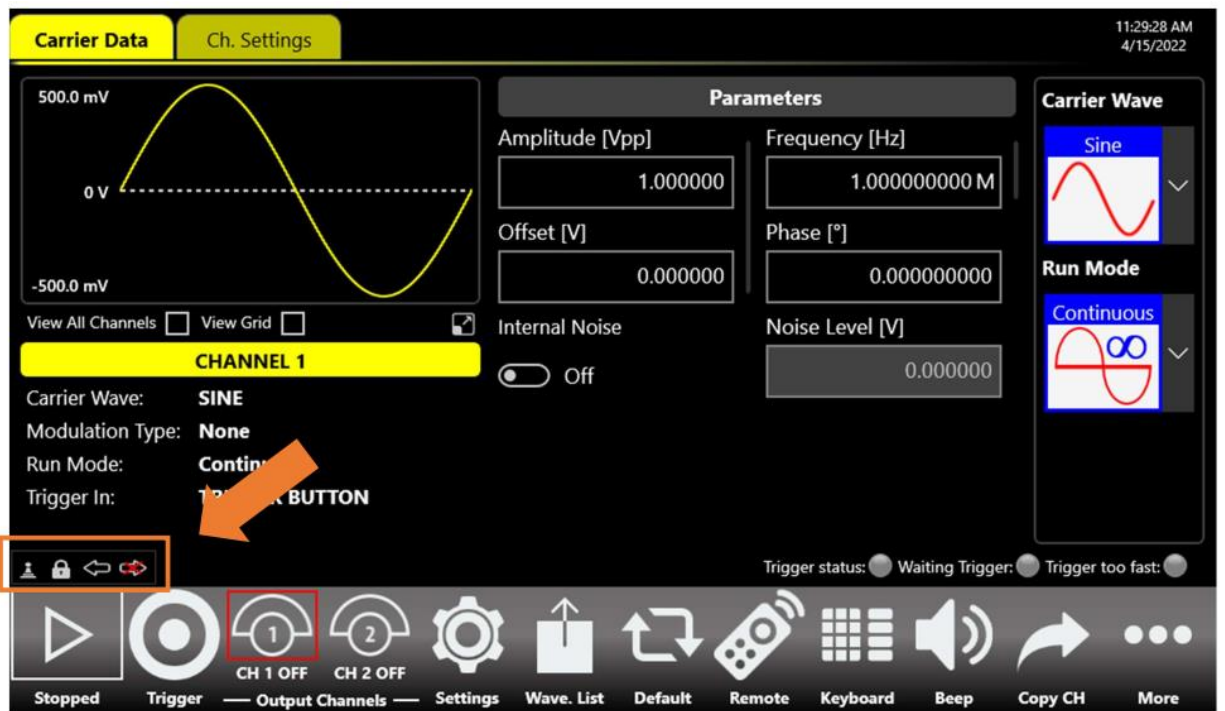


The following steps describe the operations that you should perform to set up a multi-Instrument system and start the generation of two devices.

1. On Master and Slave units, launch the Simple AFG Application.
2. On the Master unit (the one with the Sync OUT port connected) will appear a new Master Multi-instrument bar:



- On the Slave unit (the one with the Sync IN port connected) will appear a new Slave Multi-instrument bar:



- Slide the Capture switch on the Master Multi-Instrument bar:



- The lock icon will change to show that a device has been captured:

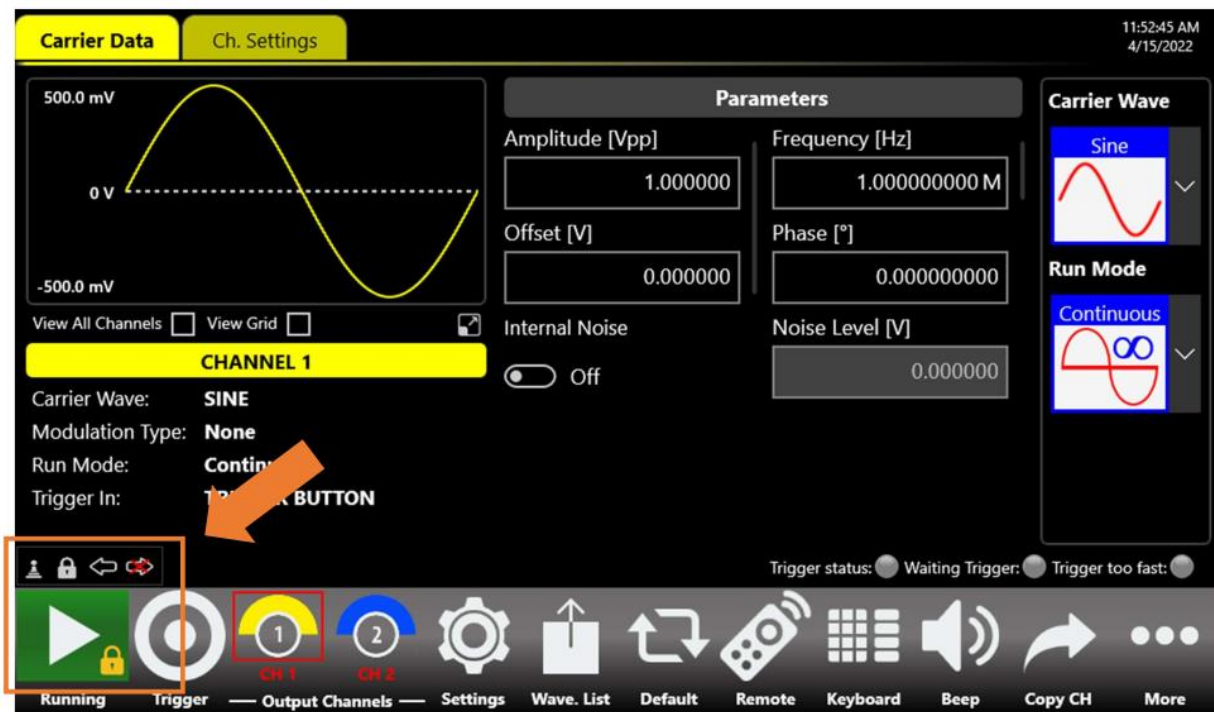


The number of captured devices is now 1.

- On the slave device also the lock icon will change to show that the instrument has been captured:



- Press Start on the Master device. Both the master and the slave instruments will start synchronously. A lock symbol on the start button of the captured instrument indicates that it is controlled by the master device:

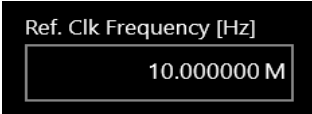


- To stop the generation, press the Stop button on the master device;

9. To unlink the instruments, slide the Release switch on the master device. In this way the two devices can be independently controlled by their respective interfaces.

Behavior of a Multi-instrument system

As already specified, the setting of triggering and timings is managed by the instrument identified as master. The table below summarizes the behavior of Multi-instrument system depending on the settings that involve the synchronization. All the parameters not shown can be set independently between the various instruments at the discretion of the user.

Parameters, Settings and Commands	Behavior on instruments
Clock Source and its parameters:   	After selecting the clock source and connecting the eventual external source on the master instrument, this will be propagated to all the other instruments in the system to synchronize them with each other.
Start/Stop button/SCPI Commands:  - AWGControl:RUN, - AWGControl:STOP	Enabled only on the master instrument which propagates them to all slave instruments. Pressing the button on a slave instrument has no effect. The commands have effects only on master instrument which propagates them to all slave instruments; if launched on a slave instrument they return an error.
Trigger Button/ SCPI Command:  *TRG	Enabled only on the master instrument which propagates it to all slave instruments. Pressing the button or launching the command on a slave instrument has no effect.

Timer: Source TIMER Timer Interval [s] 1.000000000 m	The Timer Interval value of the Timer present on the master is automatically set on all the slave instruments. After that each instrument will consider or not the Timer event depending on the run mode (for example in Continuous Mode the Timer is meaningless).
Trigger In 1/2/3/4 Source and its parameters: Source TRIGGER INPUT	The Trigger In 1 and Trigger In 2 (and Trigger In 3 and Trigger In 4 only in 4-channel models) can be detected and propagated to the slaves only by the master instrument. Therefore it is necessary to connect the trigger sources on the master and set its parameters (Threshold, Edge,...) on it.

Master Multi-Instrument Bar

The Multi-Instrument toolbar will appear on the Master device when it detects the connection with other Model 686 units via the 686-SYNC cable:



The description of these symbols is summarized in the following table:

	It notifies that the instrument is the Master one. It is the first device of the chain and it can control the execution of all the connected slave devices.
	It notifies that no other device has been found backward in the device chain. It will appear only in the master instrument.
	It notifies that a slave device has been found forward in the device chain.

	It notifies that the master instrument has not captured the slave instruments connected. By sliding the Capture button, the master instrument will capture and control the connected slave devices.
	It shows the number of devices that are controlled by the master instrument after a Capture event.
	It notifies that the master instrument has captured the slave devices, so it can control the execution of whole the instruments chain. By sliding the Release button, the master instrument will release the control of the connected slave devices.

Slave Multi-Instrument Toolbar

A new Multi-Instrument toolbar will appear on the Slave device when it detects the connection with other Model 686 units via the 686-SYNC cable:



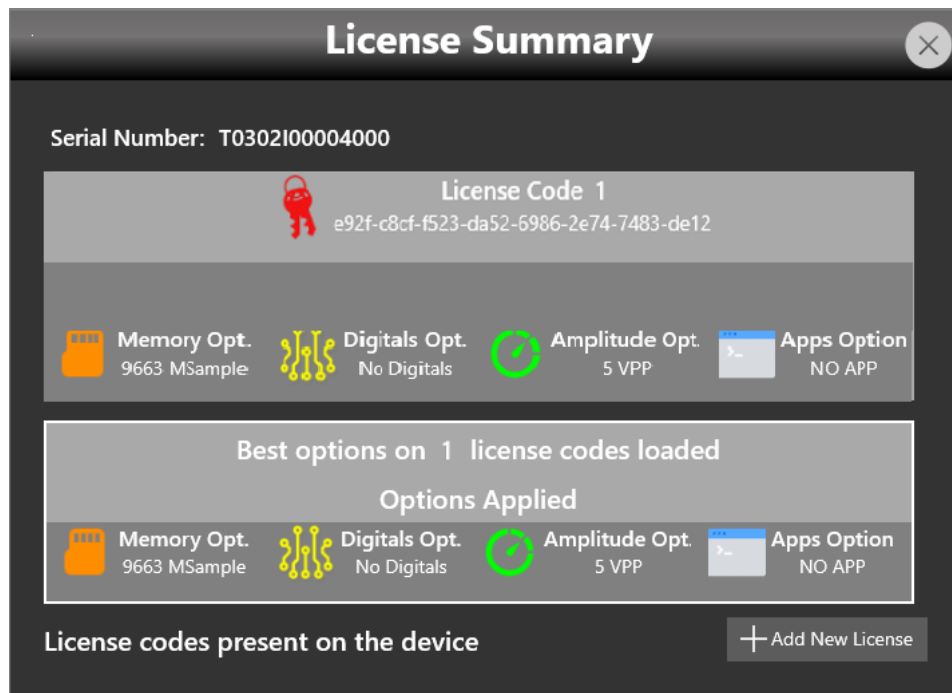
The description of these symbols is summarized in the following table:

	It notifies that the instrument is a slave device and, when captured, it can be controlled by the master unit.
	It notifies that the instrument has not been captured by the master device.
	It notifies that the instrument has been captured and will be controlled by the master device.
	It notifies that another device has been found backward in the device chain. It could be the master unit or another slave unit.
	It notifies that another slave device has been found forward on the chain.
	It notifies that no other slave device have been found forward, so this instrument is the last one in the chain.

License



The license button  in the More.. menu  opens the License page to manage the license options.



Memory and Amplitude Option are predefined Options; otherwise touching the Add New License button, it is possible to enter a new licence key to enable any of the following features:

686-2C models:

- a. Apps Option: - DPG (Serial Pattern Generator application license).

Model 686-4C models:

- a. Digitals Option: - 8/16/32 Digitals output;
- b. Apps Option: - DPG (Serial Pattern Generator application license).

To get the licence key please contact your distributor sales representative.

Certifications

Berkeley Nucleonics Corporation certifies compliance to the following standards as of the time of publication. Please see the EC Declaration of Conformity document shipped with your product for current certifications.

EMC Compliance

EC DECLARATION OF CONFORMITY - EMC

The instrument meets intent of EC Directive 2014/30/EU for Electromagnetic Compatibility. Compliance was demonstrated to the following specifications listed in the Official Journal of the European Communities:

EN 61326-1:2013, EN 61326-2-1:2013 EMC requirements for electrical equipment for measurement, control, and laboratory use. ¹

Electromagnetic Emissions:

EN 55011:2010, Radiated and Conducted Emissions Group 1, Class A ^{2 3}

EN 61000-3-2/A2:2009 Harmonic Current Emissions, Class A

EN 61000-3-3:2008 Voltage Fluctuations and Flickers, Pst = 1

Electromagnetic Immunity:

EN 61000-4-2:2009 Electrostatic Discharge, 4 kV contact, 8 kV air, 4 kV vertical/horizontal coupling planes ⁴

EN 61000-4-3/A2:2010 RF Radiated Electromagnetic Field, 3 V/m, 80-1000 MHz; 3 V/m, 1400 MHz - 2 GHz; 1 V/m, 2 GHz - 2.7 GHz

EN 61000-4-4/A1:2010 Electrical Fast Transient/Burst, 1 kV on power supply lines, 0.5 kV on I/O signal data and control lines ⁴

EN 61000-4-5:2006 Power Line Surge, 1 kV AC Mains, L-N, L-PE, N-PE ⁴

EN 61000-4-6:2009 RF Conducted Electromagnetic Field, 3 Vrms, 0.15 MHz - 80 MHz

EN 61000-4-11:2004 Mains Dips and Interruptions, 0%/1 cycle, 70%/25 cycles, 0%/250 cycles ^{4 5}

¹ To ensure compliance with all applicable EMC standards, use high-quality shielded interface cables.

² Emissions which exceed the levels required by this standard may occur when the instrument is connected to a test object.

³ This product is intended for use in nonresidential areas only. Use in residential areas may cause electromagnetic interference.

⁴ Meets Performance Criteria "B" limits of the respective standard: during the disturbance, product undergoes a temporary degradation or loss of function or performance which is self-recoverable.

⁵ Performance Criteria "C" applied for 70%/25

Safety Compliance

EC DECLARATION OF CONFORMITY – LOW VOLTAGE

The instrument meets intent of EC Directive 2014/35/EU for Product Safety. Compliance was demonstrated to the following specifications as listed in the Official Journal of the European Communities:

EN 61010-1:2010 Safety requirements for electrical equipment for measurement, control, and laboratory use – Part 1: General requirements

EN 61010-2:030:2010 Safety requirements for electrical equipment for measurement, control, and laboratory use – Part 2-030: Particular requirements for testing and measuring circuits

The design of the instrument has been verified to conform to the following limits put forth by these standards:

- Mains Supply Connector: Overvoltage Category II, instrument intended to be supplied from the building wiring at utilization points (socket outlets and similar).
- Measuring Circuit Terminals: No rated measurement category. Terminals not intended to be connected directly to the mains supply.
- Unit: Pollution Degree 2, operating environment where normally only dry, non-conductive pollution occurs. Temporary conductivity caused by condensation should be expected.



Environmental Compliance

END-OF-LIFE HANDLING

The instrument is marked with this symbol to indicate that it complies with the applicable European Union requirements to Directives 2012/19/EU and 2013/56/EU on Waste Electrical and Electronic Equipment (WEEE) and Batteries.

The instrument is subject to disposal and recycling regulations that vary by country and region. Many countries prohibit the disposal of waste electronic equipment in standard waste receptacles.

RESTRICTION OF HAZARDOUS SUBSTANCES (RoHS)

This instrument and its accessories conform to the 2011/65/EU RoHS2 Directive.



Contact Us

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Model 686 Simple AFG User Manual

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